Exploring Quantum Properties of Light-Element Xenes: DFT Studies of the Electronic Structure and Optical Response for Boron and Beryllium Allotropes

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A thesis submitted in fulfilment of
the requirements for the degree of
Doctor of Philosophy

School of Physics
The University of Sydney
Australia

27 June 2026

Abstract

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List of Publications

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Acknowledgements

Thank everyone.

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Chapter 1

Introduction

Some text

Chapter 2

Fundamentals I: Many-Body Quantum Physics and Density Functional Theory

We start our discussion of the fundamentals in this chapter. Here, we introduce the many-body quantum framework behind density functional theory. The ionic structure creates an external potential, and the interacting electrons in this potential determine the electronic properties of the materials studied here. In principle, the many-body wave function Ψ contains the full electronic information of the system. However, as Ψ depends on all electronic coordinates, direct calculation becomes impractical for realistic materials. We therefore proceed from the many-body wave function Ψ to the electron density $n(\vec{r})$, and then construct practical density functionals.

This chapter partly draws on my notes from the Coursera course *Density Functional Theory*, offered by École Polytechnique and taught by Francesco Sottile and Lucia Reining [1].

We develop this chapter as a gradual reduction of the electronic problem. In section 2.1, we begin with quantum time evolution and introduce the many-body hamiltonian. This microscopic description leads to expectation values, which we examine in section 2.2 to motivate the necessity for reduced descriptions. We then introduce density functionals and establish the Hohenberg–Kohn theorem in section 2.3. With the electron density $n(\vec{r})$ as the central variable, we construct the Kohn–Sham framework in section 2.4. Finally, section 2.5 demonstrates practical exchange-correlation approximations and brings the formal theory to the working form of density functional theory.

2.1 A story starts with time evolution quantum mechanics

Imagine that we are watching the evolution of electrons and atoms as a microscopic performance. You blink, and suddenly the stage configuration has shifted. Something has changed. But how do we describe this evolution in physical language?

In classical physics, we started with Newton's laws to describe motion. Later, Lagrangian and Hamiltonian mechanics provided a different framework for describing physics from the perspective of degrees of freedom, symmetries, and conservation laws. The twentieth century then brought the quantum revolution, establishing quantum mechanics as the fundamental framework for describing microscopic physics. In this new framework, the Schrödinger equation plays the central role.

2.1.1 Time-dependent Schrödinger equation

In quantum mechanics, the state of a physical system at a given instant is represented by a vector $|\psi(t)\rangle$. In a chosen representation, the same state can be expressed as a wave function. This state vector encapsulates all information required to predict the statistics of measurement outcomes at time t.

Next, we switch to the coordinate representation. We define the wave function as the projection of the state onto the position eigenbasis:

$$\psi(\vec{r}, t) \coloneqq \langle \vec{r} | \psi(t) \rangle \tag{2.1}$$

here, $|\vec{r}\rangle$ is an eigenstate of the position operator $\hat{\vec{r}}$.

The time evolution of the state is determined by the hamiltonian operator $\hat{H}$. It corresponds to the total energy of the system. For a closed system with a time-independent hamiltonian, the expectation value of the energy is conserved. This evolution is described by the time-dependent Schrödinger equation [2]:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle \tag{2.2}$$

here, the left-hand side gives the time derivative of the state vector; the right-hand side shows that this change is generated by the action of the hamiltonian operator.

Suppose we know the state vector at an initial time $t = t_0$. This elegant equation then uniquely determines its future state and, in turn, the probabilities of measurement outcomes.

2.1.2 Time evolution operator

Suppose that the hamiltonian is independent of time, so that it does not depend explicitly on t. In this case, the Schrödinger equation admits a formal solution. Given a reference state vector $|\psi(t_0)\rangle$, the state at time t can be expressed with the time-evolution operator $\hat{U}(t, t_0)$:

$$|\psi(t)\rangle = \hat{U}(t, t_0)\,|\psi(t_0)\rangle \tag{2.3}$$

Subsequently, we introduce the explicit exponential representation of the time-evolution operator:

$$\hat{U}(t, t_0) = \mathrm{e}^{-\frac{\mathrm{i}}{\hbar}\hat{H}(t-t_0)} \tag{2.4}$$

With the time difference $\tau = t - t_0$, we can express the time-evolution as a single-parameter family $\hat{U}(\tau)$:

$$\hat{U}(\tau) = \mathrm{e}^{-\frac{\mathrm{i}}{\hbar}\hat{H}\tau} \tag{2.5}$$

If we stipulate $t_0 = 0$, this expression becomes:

$$\hat{U}(t) = \mathrm{e}^{-\frac{\mathrm{i}}{\hbar}\hat{H}t} \tag{2.6}$$

We use $\hat{U}(t, t_0)$ to propagate the state forward in time while preserving its norm. Since $\hat{H}$ is Hermitian, $\hat{U}(t, t_0)$ is unitary. This property can be described as:

$$\hat{U}^\dagger(t, t_0)\,\hat{U}(t, t_0) = \hat{U}(t, t_0)\,\hat{U}^\dagger(t, t_0) = \hat{I} \tag{2.7}$$

Starting from the unitarity of $\hat{U}(t, t_0)$, quantum time evolution is reversible. Here, we denote the evolution from time t back to the reference time t_0 as $\hat{U}(t_0, t)$:

$$\hat{U}^\dagger(t, t_0) = \hat{U}^{-1}(t, t_0) = \hat{U}(t_0, t) \tag{2.8}$$

In the present time-independent case, we can further show that the Hermitian conjugate corresponds to evolving backward in time. In terms of the time difference τ, this relation can be written as:

$$\hat{U}^\dagger(\tau) = \hat{U}^{-1}(\tau) = \hat{U}(-\tau) \tag{2.9}$$

Accordingly, if we choose $t_0 = 0$, this expression becomes:

$$\hat{U}^\dagger(t) = \hat{U}^{-1}(t) = \hat{U}(-t) \tag{2.10}$$

The two-time evolution operator $\hat{U}(t, t_0)$ maps a quantum state from the reference time t_0 to time t. By construction, successive time evolutions must be consistent with each other. This requirement leads to the composition law:

$$\hat{U}(t_2, t_0) = \hat{U}(t_2, t_1)\hat{U}(t_1, t_0) \tag{2.11}$$

In the time-independent case, $\hat{U}(t, t_0)$ depends only on the time difference $\tau = t - t_0$. The operators $\hat{U}(\tau)$ therefore form a one-parameter unitary group:

$$\hat{U}(\tau_2)\,\hat{U}(\tau_1) = \hat{U}(\tau_2 + \tau_1) \tag{2.12}$$

In particular, when we choose $t_0 = 0$, the one-parameter composition law becomes:

$$\hat{U}(t_1 + t_2) = \hat{U}(t_1)\,\hat{U}(t_2) \tag{2.13}$$

For the one-parameter family $\hat{U}(\tau)$, the identity element, $\hat{I}$, corresponds to $\tau = 0$:

$$\hat{U}(0) = \hat{I} \tag{2.14}$$

In this sense, these relations mean that time translation can be composed consistently, and that the evolution can always be undone by evolving backward.

Unitarity also guarantees probability conservation:

$$\begin{aligned}\langle\psi(t)|\psi(t)\rangle &= \langle\psi(t_0)|\,\hat{U}^\dagger(t,t_0)\,\hat{U}(t,t_0)\,|\psi(t_0)\rangle \\ &= \langle\psi(t_0)|\,\hat{I}\,|\psi(t_0)\rangle \\ &= \langle\psi(t_0)|\psi(t_0)\rangle\end{aligned} \tag{2.15}$$

If the state is normalized at time t_0, then $\langle\psi(t_0)|\psi(t_0)\rangle = 1$, and the state remains normalized for all later times.

This is why we can view $\hat{U}(t,t_0)$ as the quantum analogue of a time-translation operator. It moves the state through Hilbert space without changing its total probability.

Finally, for a small time step Δt, we return to the exponential form:

$$\hat{U}(\Delta t) = \mathrm{e}^{-\frac{\mathrm{i}}{\hbar}\hat{H}\Delta t} \tag{2.16}$$

and obtain the short-time expansion:

$$\hat{U}(\Delta t) \approx \hat{I} - \frac{\mathrm{i}}{\hbar}\hat{H}\Delta t + \mathcal{O}(\Delta t^2) \tag{2.17}$$

Acting on $|\psi(t)\rangle$ and taking the limit $\Delta t \to 0$ then recovers the differential form of the time-dependent Schrödinger equation.

We now consider the more general case in which the hamiltonian depends explicitly on time and is denoted by $\hat{H}(t)$. The state is still propagated by the evolution operator $\hat{U}(t_1,t_0)$ as in equation (2.3).

In this case, its corresponding time-evolution operator can be written as a time-ordered exponential [3]:

$$\hat{U}(t_1,t_0) = \hat{T}\exp\left(-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t_1}\hat{H}(t)\mathrm{d}t\right) \tag{2.18}$$

here, $\hat{T}$ denotes the time-ordering operator. It orders products of $\hat{H}(t)$ such that operators at later times appear to the left. This is required when the hamiltonians at different times do not commute:

$$[\hat{H}(t), \hat{H}(t')] \neq 0 \tag{2.19}$$

Equivalently, the hamiltonian operators evaluated at earlier times act first on the state vector.

2.1.3 Many-body Schrödinger equation

Up to now, we have mainly discussed a single-particle quantum system described by the time-dependent Schrödinger equation (2.2).

In realistic situations, quantum systems often involve many interacting particles. For instance, electrons in a material evolve under both external fields and their mutual Coulomb interactions. We therefore extend the description to a nonrelativistic many-body system including electrons, nuclei, and their interactions.

To describe this larger ensemble, we generalize our single-particle state $|\psi(t)\rangle$ to a many-body state $|\Psi(t)\rangle$, an element of a larger Hilbert space constructed by the tensor product of individual particle spaces:

$$|\Psi(t)\rangle \in \mathbb{H}_1 \otimes \mathbb{H}_2 \otimes \cdots \otimes \mathbb{H}_N = \mathbb{H} \tag{2.20}$$

For identical fermions, the physical many-body states belong to the antisymmetric subspace of $\mathbb{H}$.

For example, in a two-particle system, the total Hilbert space is $\mathbb{H} = \mathbb{H}_1 \otimes \mathbb{H}_2$. It is worth noting that not every state $|\Psi\rangle$ can be factored as $|\psi_1\rangle \otimes |\psi_2\rangle$, because many physically relevant states are entangled. This means that their correlations cannot be described by individual particle states alone. In the coordinate representation, we express the wave function as:

$$\Psi(x_1, x_2, \cdots, x_N, t) = \langle x_1, x_2, \cdots, x_N | \Psi(t) \rangle \tag{2.21}$$

where each x_i includes both spatial and spin degrees of freedom.

The dynamics are governed by a many-body hamiltonian operator $\hat{H}$ and the time evolution is still described by the Schrödinger equation (2.2).

We work in Hartree atomic units unless stated otherwise. In this unit system, $\hbar = 1$, $m_\mathrm{e} = 1$, $e = 1$, and $4\pi\varepsilon_0 = 1$. As a result, energies are measured in Hartree and lengths in the Bohr radius, and the nuclear masses M_I are also given in units of the electron mass.

We adopt i labels the electrons and I labels nuclei. Subsequently, we explicitly express each physical contribution to the many-body hamiltonian as a separate operator [4]:

$$\text{Electron kinetic energy:} \qquad \hat{T}_\mathrm{e} = -\frac{1}{2}\sum_i \nabla_i^2 \tag{2.22}$$

$$\text{Electron-nucleus Coulomb attraction:} \qquad \hat{V}_\mathrm{eN} = -\sum_{i,I} \frac{Z_I}{|\vec{r}_i - \vec{R}_I|} \tag{2.23}$$

$$\text{Electron-electron Coulomb repulsion:} \qquad \hat{V}_\mathrm{ee} = \sum_{i<j} \frac{1}{|\vec{r}_i - \vec{r}_j|} \tag{2.24}$$

$$\text{Nuclear kinetic energy:} \qquad \hat{T}_\mathrm{N} = -\sum_I \frac{1}{2M_I}\nabla_I^2 \tag{2.25}$$

$$\text{Nucleus-nucleus Coulomb repulsion:} \qquad \hat{V}_\mathrm{NN} = \sum_{I<J} \frac{Z_I\, Z_J}{|\vec{R}_I - \vec{R}_J|} \tag{2.26}$$

Summing all these terms, we arrive at the full many-body hamiltonian:

$$\hat{H} = -\frac{1}{2}\sum_i \nabla_i^2 - \sum_{i,I} \frac{Z_I}{|\vec{r}_i - \vec{R}_I|} + \sum_{i<j} \frac{1}{|\vec{r}_i - \vec{r}_j|} - \sum_I \frac{\nabla_I^2}{2M_I} + \sum_{I<J} \frac{Z_I\, Z_J}{|\vec{R}_I - \vec{R}_J|} \tag{2.27}$$

For a given nuclear configuration $\{\vec{R}_I\}$, we regard the hamiltonian as $\hat{H}(\{\vec{r}_i\}; \{\vec{R}_I\})$, where the nuclear coordinates enter as external parameters.

Considering the parameters in this hamiltonian, we observe that the many-body wave functions for each state λ depend explicitly on the $3N$ spatial coordinates of the electrons and their spin degrees of freedom σ_i. For the fixed nuclear positions, the stationary electronic states satisfy the eigenvalue equation:

$$\hat{H}\Psi_\lambda(x_1, x_2, \cdots, x_N) = E_\lambda \Psi_\lambda(x_1, x_2, \cdots, x_N) \tag{2.28}$$

here, N denotes the number of electrons, and $x_i \coloneqq (\vec{r}_i, \sigma_i)$.

In practice, due to the significantly greater mass of nuclei compared to electrons, we commonly adopt the clamped-nuclei approximation within the Born-Oppenheimer framework. For fixed nuclear positions, the nucleus-nucleus repulsion contributes only an additive constant, $\hat{V}_{\mathrm{NN}}(\{\vec{R}_I\})$. The corresponding hamiltonian can then be rewritten as:

$$\begin{aligned} \hat{H} &= -\frac{1}{2}\sum_i \nabla_i^2 + \sum_{i<j} \frac{1}{|\vec{r}_i - \vec{r}_j|} + \sum_i v_{\mathrm{ext}}(\hat{\vec{r}}_i) + \hat{V}_{\mathrm{NN}}(\{\vec{R}_I\}) \\ &= -\frac{1}{2}\sum_i \nabla_i^2 + \sum_{i<j} \frac{1}{|\vec{r}_i - \vec{r}_j|} + \sum_i v_{\mathrm{ext}}(\hat{\vec{r}}_i) + E_{\mathrm{NN}} \end{aligned} \tag{2.29}$$

here, $E_{\mathrm{NN}} \coloneqq \hat{V}_{\mathrm{NN}}(\{\vec{R}_I\})$.

At this stage, it is convenient to collect all one-body terms acting on the electrons into an external potential. We define the external potential as:

$$v_{\mathrm{ext}}(\vec{r}) \coloneqq -\sum_I \frac{Z_I}{|\vec{r} - \vec{R}_I|} \tag{2.30}$$

For fixed nuclear positions $\{\vec{R}_I\}$, the electron-nucleus Coulomb attraction term $\hat{V}_{\mathrm{eN}}$ can be rewritten in the one-body form:

$$\hat{V}_{\mathrm{eN}} = \sum_i v_{\mathrm{ext}}(\hat{\vec{r}}_i) \tag{2.31}$$

In what follows, we denote the corresponding operator by $\hat{V}_{\mathrm{ext}} = \sum_i v_{\mathrm{ext}}(\hat{\vec{r}}_i)$.

More generally, $v_{\mathrm{ext}}(\vec{r})$ may also include any additional applied one-body fields, so that all system-specific information enters through v_{ext}.

After truncating each one-particle space to a finite basis of size d, the many-body Hilbert space dimension scales as:

$$\dim(\mathbb{H}) \sim d^N \tag{2.32}$$

This complexity is often called the curse of dimensionality [5]. It motivates approximate methods such as Hartree–Fock theory and density functional theory [6, 7]. We will return to these methods in later sections.

When electron interactions are completely disregarded, the problem simplifies to N independent single-electron problems. If we further ignore fermionic antisymmetry, the many-body wave function reduces to a simple product of single-particle wave functions:

$$\Psi(x_1, x_2, \cdots, x_N) = \psi_{n_1}(x_1)\psi_{n_2}(x_2)\cdots\psi_{n_N}(x_N) \tag{2.33}$$

here, $\lambda = \{n_1, n_2, \cdots, n_N\}$ labels the states.

In the coordinate representation, we can represent the antisymmetric wave function as a Slater determinant:

$$\Psi = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\vec{r}_1, \sigma_1) & \psi_1(\vec{r}_2, \sigma_2) & \cdots & \psi_1(\vec{r}_N, \sigma_N) \\ \psi_2(\vec{r}_1, \sigma_1) & \psi_2(\vec{r}_2, \sigma_2) & \cdots & \psi_2(\vec{r}_N, \sigma_N) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_N(\vec{r}_1, \sigma_1) & \psi_N(\vec{r}_2, \sigma_2) & \cdots & \psi_N(\vec{r}_N, \sigma_N) \end{vmatrix} \tag{2.34}$$

here, the antisymmetric N-electron wave function Ψ is constructed from a set of single-particle spin-orbital wave functions $\{\psi_i(\vec{r}, \sigma)\}$.

Finally, we can describe the full time-evolution of the many-body wave function in coordinate space by the time-dependent Schrödinger equation (2.2):

$$i\frac{\partial}{\partial t}\Psi(x_1, x_2, \cdots, x_N, t) = \hat{H}\,\Psi(x_1, x_2, \cdots, x_N, t) \tag{2.35}$$

2.1.4 From Schrödinger picture to Heisenberg picture

Up to now, we have adopted the Schrödinger picture to describe the time evolution of quantum systems. In this picture, states carry all the time dependence, while operators remain fixed. Specifically, the quantum state $|\psi(t)\rangle$ evolves in time according to the Schrödinger equation (2.2).

For a given observable represented by a Hermitian (self-adjoint) operator $\hat{\Omega}$, its expectation value is calculated as:

$$\langle\hat{\Omega}\rangle = \langle\psi(t)|\hat{\Omega}|\psi(t)\rangle \tag{2.36}$$

In the standard formulation of quantum mechanics, observables represent measurable physical quantities, so their expectation values are real. This is why observables are taken to be Hermitian operators, which ensures real eigenvalues and guarantees that $\langle\hat{\Omega}\rangle \in \mathbb{R}$.

To set up the transformation, we return to the Schrödinger picture. Recall (2.3) and the definition of the time-evolution operator in (2.4). With $\hat{U}(t, t_0)$, the two pictures are related as follows.

In the Heisenberg picture, the state at the reference time t_0 is taken to be time-independent:

$$|\psi_{\mathrm{H}}\rangle = |\psi(t_0)\rangle \tag{2.37}$$

We keep the state fixed and shift the time dependence to the operators. For any Schrödinger picture operator $\hat{\Omega}$, we can define its Heisenberg counterpart $\hat{\Omega}_{\mathrm{H}}(t)$ as:

$$\hat{\Omega}_{\mathrm{H}}(t) = \hat{U}^{\dagger}(t, t_0)\,\hat{\Omega}\,\hat{U}(t, t_0) = \mathrm{e}^{\frac{\mathrm{i}}{\hbar}\hat{H}(t-t_0)}\,\hat{\Omega}\,\mathrm{e}^{-\frac{\mathrm{i}}{\hbar}\hat{H}(t-t_0)} \tag{2.38}$$

This definition transfers the time dependence from the state to the operator.

With these definitions, the expectation value is unchanged from Schrödinger to the Heisenberg picture. Applying the expression of time-evolution operator(2.3), the following relation demonstrates the equivalence of the two pictures:

$$\begin{aligned}\langle\hat{\Omega}\rangle\,(t) &= \langle\psi(t)|\,\hat{\Omega}\,|\psi(t)\rangle \\ &= \langle\psi(t_0)|\,\hat{U}^{\dagger}(t, t_0)\,\hat{\Omega}\,\hat{U}(t, t_0)\,|\psi(t_0)\rangle \\ &= \langle\psi_{\mathrm{H}}|\,\hat{\Omega}_{\mathrm{H}}(t)\,|\psi_{\mathrm{H}}\rangle\end{aligned} \tag{2.39}$$

For a time-independent hamiltonian and an operator without explicit time dependence in the Schrödinger picture, differentiating (2.38) gives the Heisenberg equation of motion [2]:

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\Omega}_{\mathrm{H}}(t) = \frac{\mathrm{i}}{\hbar}\left[\hat{H}, \hat{\Omega}_{\mathrm{H}}(t)\right] \tag{2.40}$$

This form is particularly convenient for discussing conserved quantities and symmetries, since it expresses the dynamics directly at the operator level.

2.1.5 From Heisenberg picture to the continuity equation

Having seen that any local one-body observable can be written as an integral over a single function, we now introduce that central quantity. The one-body density contains all the information needed to compute local one-body expectation values and serves as the fundamental variable in density functional theory [7, 8, 9]. In what follows, we review its precise definition:

$$n(\vec{r}) = \langle\psi|\ \hat{n}(\vec{r})\,|\psi\rangle = \langle\psi|\sum_i \delta(\vec{r} - \hat{\vec{r}}_i)\,|\psi\rangle \tag{2.41}$$

This defines the electron density at point $\vec{r}$. Physically, $n(\vec{r})\mathrm{d}^3r$ is the expected number of electrons in the volume element d^3r around $\vec{r}$. In density functional theory, $n(\vec{r})$ serves as the fundamental variable replacing the full many-body wave function.

It can be shown that the ground-state density $n(\vec{r})$ uniquely determines the external potential up to an additive constant, and hence all ground-state observables can be written as functionals of $n(\vec{r})$. This dramatic reduction comes from a $3N$-dimensional wave function to a three-dimensional density, which lies at the heart of density functional theory, making formerly intractable many-body problems computationally feasible.

The time-dependent density-functional theory relies on the continuity equation in the Heisenberg picture. We first define the particle density operator in the Schrödinger-picture as:

$$\hat{n}(\vec{r}) = \sum_i \delta(\vec{r} - \hat{\vec{r}}_i) \tag{2.42}$$

and its corresponding Heisenberg-picture counterpart is defined by the unitary time evolution as:

$$\hat{n}_{\mathrm{H}}(\vec{r},t) = \hat{U}^{\dagger}(t)\,\hat{n}(\vec{r})\,\hat{U}(t) \tag{2.43}$$

Its time derivative follows from the Heisenberg equation of motion,

$$\frac{\partial}{\partial t}\hat{n}_{\mathrm{H}}(\vec{r},t) = \frac{\mathrm{i}}{\hbar}\left[\hat{H}, \hat{n}_{\mathrm{H}}(\vec{r},t)\right] \tag{2.44}$$

Next, we split the hamiltonian into kinetic and potential parts:

$$\hat{H} = \hat{T}_{\mathrm{e}} + \hat{V}_{\mathrm{ext}} + \hat{V}_{\mathrm{ee}} \tag{2.45}$$

One shows that both $\hat{V}_{\mathrm{ext}}$ and $\hat{V}_{\mathrm{ee}}$ commute with $\hat{n}(\vec{r})$:

$$\left[\hat{V}_{\mathrm{ext}} + \hat{V}_{\mathrm{ee}},\, \hat{n}_{\mathrm{H}}(\vec{r},t)\right] = 0 \tag{2.46}$$

They are diagonal in the coordinate basis, so only the kinetic term contributes:

$$\frac{\partial}{\partial t}\hat{n}_{\mathrm{H}} = \frac{\mathrm{i}}{\hbar}\left[\hat{T}_{\mathrm{e}}, \hat{n}_{\mathrm{H}}\right] \tag{2.47}$$

The Heisenberg-picture current operator can now be introduced:

$$\hat{\vec{j}}_{\mathrm{H}}(\vec{r},t) = \frac{1}{2m_{\mathrm{e}}}\sum_i \left[\hat{\vec{p}}_i\,\delta(\vec{r}-\hat{\vec{r}}_i) + \delta(\vec{r}-\hat{\vec{r}}_i)\,\hat{\vec{p}}_i\right]_{\mathrm{H}} \tag{2.48}$$

here, $\hat{\vec{p}}_i$ is the momentum operator of particle i. A direct commutator calculation then yields

$$\frac{\mathrm{i}}{\hbar}\left[\hat{T}_{\mathrm{e}}, \hat{n}_{\mathrm{H}}(\vec{r},t)\right] = -\nabla\cdot\hat{\vec{j}}_{\mathrm{H}}(\vec{r},t) \tag{2.49}$$

Putting these results into equation (2.44) gives the operator-level continuity equation:

$$\frac{\partial}{\partial t}\,\hat{n}_{\mathrm{H}}(\vec{r},t) + \nabla\cdot\hat{\vec{j}}_{\mathrm{H}}(\vec{r},t) = 0 \tag{2.50}$$

Taking expectation values in any Heisenberg-picture state $|\Psi_{\mathrm{H}}\rangle$ then yields the familiar continuity equation for the physical density $n(\vec{r},t) = \langle\Psi_{\mathrm{H}}|\,\hat{n}_{\mathrm{H}}(\vec{r},t)\,|\Psi_{\mathrm{H}}\rangle$ and current $\vec{j}(\vec{r},t) = \langle\Psi_{\mathrm{H}}|\,\hat{\vec{j}}_{\mathrm{H}}(\vec{r},t)\,|\Psi_{\mathrm{H}}\rangle$:

$$\frac{\partial}{\partial t}\, n(\vec{r},t) + \nabla\cdot\vec{j}(\vec{r},t) = 0 \tag{2.51}$$

This exact conservation law provides a fundamental constraint on the time evolution of the density and current. It also serves as one of the basic structural ingredients of time-dependent density functional theory, where the many-body dynamics are reformulated in terms of the time-dependent density $n(\vec{r},t)$.

2.2 Toward the many-body problem

In fact, a many-body wave function explicitly depends on all particle coordinates, making it practically impossible to calculate and store exactly due to the exponential increase in complexity with particle number.

Nevertheless, what we ultimately aim to describe are measurable properties. These include observables like position, momentum, energy, or spin. We can extract them from the wave function through expectation values:

$$\langle \hat{\Omega} \rangle = \langle \Psi | \, \hat{\Omega} \, | \Psi \rangle \tag{2.52}$$

2.2.1 An introduction to grand canonical ensemble

Although the present chapter mainly focuses on the zero-temperature ground-state formalism, it is useful to briefly record the grand-canonical framework because it provides the natural extension to finite temperature and varying particle number [10].

Before proceeding further, it is essential to introduce the grand canonical ensemble, as it naturally accommodates quantum many-body systems exchanging particles and energy with external reservoirs. This treatment is particularly crucial in density functional theory calculations for realistic materials, where fluctuations in particle number and thermal effects are common.

In the grand canonical ensemble, the expectation value of any observable operator $\hat{\Omega}$ is computed as follows:

$$\langle \hat{\Omega} \rangle = \frac{1}{\mathcal{Z}} \sum_{\alpha} \mathrm{e}^{-\beta(E_\alpha - \mu N_\alpha)} \, \langle \Psi_\alpha | \, \hat{\Omega} \, | \Psi_\alpha \rangle \tag{2.53}$$

here, the summation runs over all quantum states labeled by α, and Ψ_α represents the many-body quantum states of the system, characterized by their energies E_α and particle numbers N_α. The inverse temperature β is defined as $\beta = 1/(k_B T)$, with T being the temperature and k_B the Boltzmann constant. The chemical potential

μ controls particle exchange between the system and its reservoir. Therefore, the partition function $\mathcal{Z}$ serves as a normalization factor and is defined as:

$$\mathcal{Z} = \sum_{\alpha} \mathrm{e}^{-\beta(E_\alpha - \mu N_\alpha)} \tag{2.54}$$

In quantum many-body theory and calculation, employing the grand canonical ensemble allows systematic inclusion of particle number fluctuations and thermal equilibrium effects, providing a robust and realistic statistical-mechanical description essential for accurate modeling of complex quantum materials.

In what follows, we mainly focus on the ground-state setting: zero temperature and fixed particle number; while the grand-canonical form is noted for completeness and later extensions.

2.2.2 From wave function to practical functionals

Let us stay at zero temperature and fixed particle number N. Consider the operator for the measurement $\sum_{i,j,\cdots} \hat{\Omega}\left(x_i, x_j, \cdots\right)$, and the ground-state N-body wave function $\Psi(x_1, x_2, \cdots, x_N)$. By inserting resolutions of the identity in the coordinate basis for all particles, $I = \int \mathrm{d}x \, |x\rangle \langle x|$ [11], we can represent the expectation value as:

$$\begin{aligned} \langle \hat{\Omega} \rangle &= \langle \Psi | \, \hat{\Omega} \, | \Psi \rangle \\ &= \iint \cdots \int \mathrm{d}x_1 \mathrm{d}x_2 \cdots \mathrm{d}x_N \, \Psi^*(x_1, x_2, \cdots, x_N) \sum_{i,j,\cdots} \hat{\Omega}\left(x_i, x_j, \cdots\right) \Psi(x_1, x_2, \cdots, x_N) \end{aligned} \tag{2.55}$$

Therefore, we may regard the expectation value of the observable as a functional of the many-body wave function Ψ, and write

$$\Omega[\Psi] = \langle \Psi | \, \hat{\Omega} \, | \Psi \rangle \tag{2.56}$$

here, the square brackets $[\Psi]$ (rather than parentheses) emphasize that Ω is a functional; it is a mapping from the entire wave function Ψ to a scalar.

A functional simply assigns a number to a function. In our case of equation (2.55), its form is a known integral. However, evaluating it requires the exact many-body wave function, which we generally cannot compute or store. Moreover, integration

acts as an averaging operation that smooths out microscopic fluctuations, so only the coarse-grained features of the integrand contribute to the result.

Consequently, we can discard irrelevant details of the many-body wave function and work with simplified approximations tailored to specific observables. In practice, we often use approximate wave functions: methods such as Hartree–Fock, configuration interaction, and quantum Monte Carlo follow this strategy by retaining the dominant components of the wave function while approximating or sampling the rest.

A more radical question is whether we need to work with explicit wave functions at all, or whether the expectation value in equation (2.55) can be reformulated entirely in terms of other, more tractable quantities.

To address this question, it is natural to start from the energy, since the ground-state is determined variationally through the hamiltonian expectation value.

In the case of this approximation, for a given hamiltonian $\langle\Psi|\,\hat{H}\,|\Psi\rangle$, we can regard it as depending on the form of the kinetic energy operator $\langle\Psi|\,\hat{T}\,|\Psi\rangle$, external potential operator $\langle\Psi|\,\hat{V}_{\text{ext}}\,|\Psi\rangle$, and the interaction operator $\langle\Psi|\,\hat{V}_{\text{int}}\,|\Psi\rangle$.

To specify the quantum dynamics and interactions of our system within the Born–Oppenheimer approximation, we work at the fixed nuclear coordinates $\{\vec{R}_I\}$ and write the clamped nuclei hamiltonian as:

$$\hat{H}_{\text{BO}}(\{\vec{R}_I\}) = -\frac{1}{2}\sum_i \nabla_i^2 + \sum_i v_{\text{ext}}(\hat{\vec{r}}_i) + \sum_{i<j}\frac{1}{|\vec{r}_i - \vec{r}_j|} + E_{\text{NN}}(\{\vec{R}_I\}) \tag{2.57}$$

In the present electronic many-body problem within the Born–Oppenheimer approximation, the nuclear coordinates are clamped, and the nuclear-nuclear repulsion becomes a negligible constant. We can identify the interaction operator as the electron-electron Coulomb repulsion (2.24), and the external potential acting on the electrons is the attractive electron-nucleus interaction, which can be written as a sum of one-body terms (2.31):

$$\begin{aligned}\hat{V}_{\text{int}} &\coloneqq \hat{V}_{\text{ee}} = \sum_{i<j}\frac{1}{|\vec{r}_i - \vec{r}_j|}\\ \hat{V}_{\text{ext}} &\coloneqq \hat{V}_{\text{eN}} = \sum_i v_{\text{ext}}(\hat{\vec{r}}_i)\end{aligned} \tag{2.58}$$

Using the expression for the electron kinetic energy operator in equation (2.22), we arrive at:

$$\begin{aligned}\langle\Psi|\,\hat{H}\,|\Psi\rangle &= \langle\Psi|\,\hat{T}\,|\Psi\rangle + \langle\Psi|\,\hat{V}_{\text{ext}}\,|\Psi\rangle + \langle\Psi|\,\hat{V}_{\text{int}}\,|\Psi\rangle \\ &= \langle\Psi| - \frac{1}{2}\sum_i \nabla_i^2\,|\Psi\rangle + \langle\Psi|\sum_i v_{\text{ext}}(\vec{r}_i)\,|\Psi\rangle + \langle\Psi|\sum_{i<j}\frac{1}{|\vec{r}_i-\vec{r}_j|}\,|\Psi\rangle\end{aligned} \tag{2.59}$$

Having treated both the kinetic energy operator and the interaction term as fixed, specifying the external potential $v_{\text{ext}}(\vec{r})$ uniquely determines the hamiltonian $\hat{H}$. For any normalized many-body state $|\Psi\rangle$, we define the energy expectation value as:

$$E \coloneqq \langle\Psi|\,\hat{H}\,|\Psi\rangle \tag{2.60}$$

When $|\Psi\rangle$ is chosen to be an eigenstate of $\hat{H}$, denoted by $|\Psi_n\rangle$, the corresponding eigenvalue E_n and eigenstate are determined by the Schrödinger equation:

$$\hat{H}\,|\Psi_n\rangle = E_n\,|\Psi_n\rangle \tag{2.61}$$

Accordingly, one may view the spectrum and the eigenstates as depending on the external potential through the hamiltonian:

$$v_{\text{ext}}(\vec{r}) \longmapsto \left\{|\Psi_n\left[v_{\text{ext}}(\vec{r})\right]\rangle\,, E_n\left[v_{\text{ext}}(\vec{r})\right]\right\}_n \tag{2.62}$$

here, $E_n\left[v_{\text{ext}}\right]$ denotes the energy eigenvalue associated with the stationary state $|\Psi_n\left[v_{\text{ext}}\right]\rangle$, namely:

$$E_n\left[v_{\text{ext}}(\vec{r})\right] \coloneqq \langle\Psi_n\left[v_{\text{ext}}(\vec{r})\right]|\,\hat{H}\,|\Psi_n\left[v_{\text{ext}}(\vec{r})\right]\rangle \tag{2.63}$$

The grand-canonical ensemble admits a compact operator formulation in terms of a trace over the many-body Hilbert space. Invoking the cyclic invariance of the trace, we obtain:

$$\langle\hat{\Omega}\rangle = \frac{1}{\mathcal{Z}}\,\text{Tr}\left[\mathrm{e}^{-\beta\left(\hat{H}-\mu\hat{N}\right)}\,\hat{\Omega}\right] \tag{2.64}$$

with the grand-canonical partition function:

$$\mathcal{Z} = \sum_{\alpha} \mathrm{e}^{-\beta(E_\alpha - \mu N_\alpha)} = \mathrm{Tr}\left[\mathrm{e}^{-\beta\left(\hat{H}-\mu\hat{N}\right)}\right] \tag{2.65}$$

Note that the sum over states in equation (2.53) is nothing but a trace over the many-body Hilbert space. By using the cyclic invariance of the trace, we can therefore recast both the expectation value and the partition function in the compact operator form:

$$\langle\hat{\Omega}\rangle = \frac{1}{\mathcal{Z}} \sum_{\alpha} \mathrm{e}^{-\beta(E_\alpha - \mu N_\alpha)} \langle\Psi_\alpha|\,\hat{\Omega}\,|\Psi_\alpha\rangle \tag{2.66}$$

Equivalently, by expanding the trace in the eigenbasis of $\hat{H}$, equation (2.64) reduces to the sum representation in equation (2.66).

In principle, all physical observables derive from the many-body Schrödinger equation (2.35), yet the exponential growth of the Hilbert-space dimension renders direct diagonalization intractable for realistic systems.

If the kinetic energy operator and Coulomb interaction are fixed, the ground-state total energy becomes a functional of the external potential we introduced in equation (2.62).

A central challenge is that evaluating the grand-canonical expectation in equation (2.66) remains highly nontrivial, as the trace involves high-dimensional integrals and non-commuting operators.

Our aim is to express the expectation values—which are, in principle, highly complex functionals—in terms of a single simple function, so that practical approximations become possible.

Next, recall the general expression for an observable in a many-body system: equation (2.55). If the observable is a local one-body operator, we have:

$$\begin{aligned}
\langle\hat{\Omega}_1\rangle &= \langle\Psi|\,\hat{\Omega}_1\,|\Psi\rangle \\
&= \iint\cdots\int \mathrm{d}x_1\mathrm{d}x_2\cdots\mathrm{d}x_N\; \Psi^*(x_1, x_2, \cdots, x_N) \sum_i \hat{\Omega}_1(x_i)\Psi(x_1, x_2, \cdots, x_N) \\
&= \int \mathrm{d}x_1 \Omega_1(x_1) N \int \mathrm{d}x_2\cdots\mathrm{d}x_N\; \Psi^*(x_1, x_2, \cdots, x_N)\Psi(x_1, x_2, \cdots, x_N)
\end{aligned} \tag{2.67}$$

We can find that this integral naturally factors into two components: the one-body operator kernel $\Omega_1(x_1)$, and the corresponding one-body density [12, 13]:

$$n(x_1) = N \int \mathrm{d}x_2 \cdots \mathrm{d}x_N \Psi^*(x_1, x_2, \cdots, x_N)\Psi(x_1, x_2, \cdots, x_N) \tag{2.68}$$

Accordingly, the ground-state expectation value of the one-body operator can be written as:

$$\langle \hat{\Omega}_1 \rangle = \int \mathrm{d}x_1 \, \Omega_1(x_1) \, n(x_1) \tag{2.69}$$

here, x denotes the one-particle coordinate. If x refers to the spatial coordinate only, we write $x = \vec{r}$ and denote the usual spatial density as $n(\vec{r})$. More generally, in many-body notation x denotes a combined space-spin coordinate, and we write $x = (\vec{r}, \sigma)$. In this case, $n(x) = n(\vec{r}, \sigma)$ is the spin-resolved one-body density.

In many condensed matter physics applications, we work with the spin-summed density. To recover the spin-summed spatial density, we write a discrete sum over the spin index:

$$n(\vec{r}) = \sum_{\sigma} n(\vec{r}, \sigma) \tag{2.70}$$

or use an integral notation for the spin index,

$$n(\vec{r}) = \int \mathrm{d}\sigma \, n(\vec{r}, \sigma) \tag{2.71}$$

Both notations express the same idea: the spatial density is obtained by summing over the spin degree of freedom.

Throughout this chapter, for simplicity, when we write $\int \mathrm{d}x$, it means $\sum_\sigma \int \mathrm{d}\vec{r}$. For spin-independent external potentials, we write $v_{\text{ext}}(x) = v_{\text{ext}}(\vec{r})$.

The density is also constrained by particle number conservation. For a spin-resolved density $n(x)$, we define the particle number functional as:

$$N[n] \coloneqq \int \mathrm{d}x \, n(x) \tag{2.72}$$

and in the ground-state setting with fixed particle number we require:

$$N[n] = N \tag{2.73}$$

Equivalently, for the spin-summed spatial density, the same condition takes the form $\int \mathrm{d}\vec{r}\, n(\vec{r}) = N$. This constraint will be enforced explicitly when we later vary an energy functional with respect to n.

We also note that not every non-negative function is an admissible density. In general, a candidate $n(x)$ must be compatible with an antisymmetric N-body wave function and the chosen external potential. In what follows, we keep these representability requirements implicit, and we work within the set of admissible densities, and equation (2.73) provides the particle number constraint.

The expectation value reduces to:

$$\langle \hat{\Omega}_1 \rangle = \int \mathrm{d}x_1\, \Omega_1(x_1) n(x_1) \tag{2.74}$$

Thus, rather than computing the full many-body integral, we need only determine the one-body density $n(x_1)$.

Next, consider the expectation value of the electron-electron Coulomb interaction energy:

$$\begin{aligned}\langle \hat{V}_{\mathrm{ee}} \rangle &= \iint \cdots \int \mathrm{d}x_1 \mathrm{d}x_2 \cdots \mathrm{d}x_N\, \Psi^*(x_1, x_2, \cdots, x_N) \sum_{i,j>i} \frac{1}{|\vec{r}_i - \vec{r}_j|} \Psi(x_1, x_2, \cdots, x_N) \\ &= \int \mathrm{d}x_1 \mathrm{d}x_2\, \frac{1}{2|\vec{r}_1 - \vec{r}_2|} N(N-1) \int \mathrm{d}x_3 \cdots \mathrm{d}x_N\, \Psi^*(x_1, x_2, \cdots, x_N) \Psi(x_1, x_2, \cdots, x_N)\end{aligned} \tag{2.75}$$

here, the two-body density naturally appears:

$$n_2(x_1, x_2) = N(N-1) \int \mathrm{d}x_3 \cdots \mathrm{d}x_N\, \Psi^*(x_1, x_2, \cdots, x_N) \Psi(x_1, x_2, \cdots, x_N) \tag{2.76}$$

So that the expectation value of the Coulomb interaction energy reduces to:

$$\langle \hat{V}_{\text{ee}} \rangle = \frac{1}{2} \int \mathrm{d}x_1 \mathrm{d}x_2 \, \frac{n_2(x_1, x_2)}{|\vec{r}_1 - \vec{r}_2|} \tag{2.77}$$

Similarly, the expectation value of the kinetic energy takes the form:

$$\begin{aligned} T &= -\iint \cdots \int \mathrm{d}x_1 \mathrm{d}x_2 \cdots \mathrm{d}x_N \, \Psi^*(x_1, x_2, \cdots, x_N) \sum_i \frac{\nabla_i^2}{2} \Psi(x_1, x_2, \cdots, x_N) \\ &= -N \iint \cdots \int \mathrm{d}x_1 \mathrm{d}x_2 \cdots \mathrm{d}x_N \, \Psi^*(x_1, x_2, \cdots, x_N) \frac{\nabla_1^2}{2} \Psi(x_1, x_2, \cdots, x_N) \\ &= -\frac{1}{2} \int \mathrm{d}x_1 \, \left[\nabla_{x_1'}^2 \, \rho(x_1, x_1') \right]_{x_1' = x_1} \end{aligned} \tag{2.78}$$

We can also separate this integral via the matrix of density:

$$\rho(x_1, x_1') = N \int \cdots \int \mathrm{d}x_2 \cdots \mathrm{d}x_N \, \Psi^*(x_1, x_2, \cdots, x_N) \Psi(x_1', x_2, \cdots, x_N) \tag{2.79}$$

and get the simplified expectation value of the kinetic energy operator:

$$T = -\frac{1}{2} \int \mathrm{d}x_1 \, \left[\nabla_{x_1'}^2 \, \rho(x_1, x_1') \right]_{x_1' = x_1} \tag{2.80}$$

We begin by insisting on a single functional variable Q such that every observable of interest $\hat{\Omega}_i$ can be written in the form $\hat{\Omega}_i[Q]$. Having identified Q, we then derive the explicit mappings $\hat{\Omega}_i[Q]$ for each target observable. Finally, we develop a practical algorithm or an approximation scheme to compute Q itself, thereby rendering all observables $\hat{\Omega}_i[Q]$ tractable.

Given a time-dependent many-body wave function $\Psi(x_1, x_2, \cdots, x_N; t)$. We denote the corresponding ground-state ($g \coloneqq 0$) many-body wave function as $\Psi_g(x_1, x_2, \cdots, x_N; t)$. We can also denote the excited states by an index α ($\alpha \geq 1$), as $\Psi_\alpha(x_1, x_2, \cdots, x_N; t)$.

Then, we can describe the ground-state one-body density:

$$n_g(x_1, t) = N \iint \cdots \int \mathrm{d}x_2 \, \mathrm{d}x_3 \cdots \mathrm{d}x_N \, \Psi_g^*(x_1, x_2, \cdots, x_N; t) \, \Psi_g(x_1, x_2, \cdots, x_N; t) \tag{2.81}$$

When this system makes transitions to excited states, the one-body transition density is obtained by integrating other coordinates:

$$f_n^{\mathrm{eh}}(x_1,t)=\iint\cdots\int \mathrm{d}x_2\,\mathrm{d}x_3\cdots\mathrm{d}x_N\,\Psi_g^*(x_1,x_2,\cdots,x_N;t)\,\Psi_n(x_1,x_2,\cdots,x_N;t) \tag{2.82}$$

Equivalently, the reduced integral reads:

$$f_n^{\mathrm{eh}}(x_1,t)=\int \Psi_g^*(\vec{x};t)\,\Psi_n(\vec{x};t)\prod_{j=2}^{N}\mathrm{d}x_j \tag{2.83}$$

This marginal integral represents the one-body transition density from the ground-state g to the n-th excited state.

In the presence of an oscillating electromagnetic field, we introduce the Bohr frequency:

$$\omega_{ng}=(E_n-E_g)/\hbar \tag{2.84}$$

The corresponding phase factor associated with the excitation energy is $\mathrm{e}^{-i(E_n-E_g)t}$. If one separates the trivial time phases, we obtain the time-dependent form of the stationary state wave functions for the ground and excited states:

$$\Psi_{g/n}(\vec{x};t)=\Phi_{g/n}(\vec{x})\,\mathrm{e}^{-iE_{g/n}t/\hbar} \tag{2.85}$$

Therefore, the one-body transition density leads to the following time-dependent form:

$$\begin{aligned} &f_n^{\mathrm{eh}}(x_1,t)\\ =&\mathrm{e}^{-i(E_n-E_g)t}\iint\cdots\int \mathrm{d}x_2\,\mathrm{d}x_3\cdots\mathrm{d}x_N\,\Psi_g^*(x_1,x_2,\cdots,x_N;t)\,\Psi_n(x_1,x_2,\cdots,x_N;t)\end{aligned} \tag{2.86}$$

here, the phase-independent part of the one-body transition density follows as:

$$\iint\cdots\int \mathrm{d}x_2\,\mathrm{d}x_3\cdots\mathrm{d}x_N\,\Psi_g^*(x_1,x_2,\cdots,x_N;t)\,\Psi_n(x_1,x_2,\cdots,x_N;t) \tag{2.87}$$

Subsequently, we recall a one-body operator $\hat{\Omega}^{(1)}$ with kernel $\Omega_1(x_1)$, the one-body density $n(x_1)$, the one-body density matrix $\rho(x_1, x_1')$, and the pair density $n_2(x_1, x_2)$. Then, the spatial parts are contained in x_1, x_2:

$$\begin{aligned} \langle \hat{\Omega}^{(1)} \rangle &= \int \mathrm{d}x_1 \, \Omega_1(x_1) \, n(x_1) \\ T &= -\frac{1}{2} \int \mathrm{d}x_1 \, \left[\nabla^2_{x_1'} \, \rho(x_1, x_1') \right]_{x_1'=x_1} \\ \langle \hat{V}_{\mathrm{ee}} \rangle &= \frac{1}{2} \int \mathrm{d}x_1 \, \mathrm{d}x_2 \, \frac{n_2(x_1, x_2)}{|\vec{r}_1 - \vec{r}_2|} \end{aligned} \tag{2.88}$$

For the present discussion, we adopt $\hat{\Omega}$ to denote a typical energy-like operator composed of one-body, kinetic, and interaction contributions. Its expectation value then follows as:

$$\begin{aligned} \langle \hat{\Omega} \rangle = &\int \mathrm{d}x_1 \, \Omega_1(x_1) \, n(x_1) \\ &- \frac{1}{2} \int \mathrm{d}x_1 \, \left[\nabla^2_{x_1'} \, \rho(x_1, x_1') \right]_{x_1'=x_1} \\ &+ \frac{1}{2} \int \mathrm{d}x_1 \, \mathrm{d}x_2 \, \frac{n_2(x_1, x_2)}{|\vec{r}_1 - \vec{r}_2|} \end{aligned} \tag{2.89}$$

Up to now, we do not have closed-form expressions for the objects appearing in equation (2.89), and evaluating them still requires many-body information beyond practical reach. This motivates the introduction of a compact descriptor: rather than working with the full wave function $\Psi(x_1, \cdots, x_N)$, we seek a simpler variable Q that is nevertheless sufficient to determine the observables of interest.

More precisely, let $\{\hat{\Omega}_i\}_{i=1,2,\cdots}$ denote the operators representing the target physical quantities, and let $\Omega_i[\Psi]$ denote their ground-state expectation values regarded as functionals of the wave function. Our goal is to find a single descriptor Q such that, once Q is known, all these observables can be evaluated without explicit reference to the wave function Ψ:

$$\Psi \mapsto Q, \quad \Omega_i[\Psi] = \Omega_i[Q], \quad (i = 1, 2, \cdots) \tag{2.90}$$

In other words, Q is chosen to retain exactly the information needed for the set $\{\Omega_i\}$, while discarding irrelevant microscopic details.

In particular, we set $Q \coloneqq n(x)$. In the following, we take the particle density $n(x)$ as the basic variable, as it provides a minimal yet powerful descriptor of the ground-state. In this statement, x denotes a combined space-spin coordinate, $x = (\vec{r}, \sigma)$, so that we can represent the one-body (spin-resolved) density as $n(x)$:

Firstly, the particle density $n(x)$ yields exactly all local one-body observables. For an operator of the form $\hat{\Omega}^{(1)} = \sum_{j=1}^{N} \Omega_1(x_j)$, one has:

$$\langle \hat{\Omega}^{(1)} \rangle = \int \mathrm{d}x \, \Omega_1(x) \, n(x) \tag{2.91}$$

Secondly, the particle density $n(x)$ is extremely compact compared to the wave function Ψ. Moreover, in the case of a non-degenerate ground-state, the Hohenberg–Kohn theorem implies a bijective correspondence between the ground-state density and the external potential. As a result, $n(x)$ determines v_{ext} up to an additive constant. This fixes the hamiltonian and thereby fixes the ground-state up to an overall phase. Therefore, all ground-state observables may be regarded as density functionals, and we seek $\Omega_i[n]$ rather than $\Omega_i[\Psi]$.

Therefore, we explore the variational route for n. Define the energy-like functional:

$$E[n; v_{\mathrm{ext}}] = T[n] + E_{\mathrm{int}}[n] + \int \mathrm{d}x \, v_{\mathrm{ext}}(x) n(x) \tag{2.92}$$

and obtain the ground-state density from

$$\frac{\delta}{\delta n} \left(E[n; v_{\mathrm{ext}}] - \mu \, N[n] \right) = 0 \;\Rightarrow\; n_g(x) \tag{2.93}$$

Once n_g is known, all desired observables are evaluated as $\Omega_i[n_g]$; for non-local one-body or two-body quantities, this amounts to adopting controlled approximations for kinetic energy functional $T[n]$ and interaction energy functional $E_{\mathrm{int}}[n]$ such as the Kohn–Sham framework, which we will introduce later.

2.3 An introduction to density functionals

In the previous discussion, we chose the particle density as the basic variable for the ground-state problem. We now introduce the concept of a functional and then specialize to density functionals in quantum mechanics. This idea forms the conceptual basis of density functional theory, and leads to the Hohenberg–Kohn theorem.

2.3.1 What is the functional?

A functional is a mapping that takes a function as its input and returns a number (or sometimes another function).

The reason we introduce density functional theory in condensed matter physics research is that solving the full many-body or many-electron Schrödinger equation scales hopelessly with the number of electrons [14]. DFT reformulates it as a variational problem in terms of the three-dimensional electron density $n(\vec{r})$, making predictive calculations of structures, band structures, magnetism, and reaction energetics feasible.

For example, in condensed matter physics, it often appears as a map from electron density $n(\vec{r})$ to the total energy $E[n]$ given by an external potential:

$$E_{\text{ext}}[n] = \int \mathrm{d}\vec{r}\, n(\vec{r}) v_{\text{ext}}(\vec{r}) \tag{2.94}$$

It also often appears that we use a non-local functional to describe the classical electrostatic term or the Hartree term:

$$E_{\text{H}}[n] = \frac{1}{2} \int \mathrm{d}\vec{r}_i\, \mathrm{d}\vec{r}_j\, \frac{n(\vec{r}_i) n(\vec{r}_j)}{|\vec{r}_i - \vec{r}_j|} \tag{2.95}$$

In this thesis, $E_{\text{H}}[n]$ denotes the Hartree term.

Next, let us consider the expression for kinetic energy in terms of a functional. Within the Thomas–Fermi approximation, the noninteracting kinetic energy is treated locally as that of a uniform (homogeneous) electron gas whose density equals the actual

electron density $n(\vec{r})$ at each point in space. This leads to the Thomas–Fermi kinetic energy functional:

$$T_{\mathrm{TF}}[n] = C \int \mathrm{d}\vec{r}\, n^{5/3}(\vec{r}) \tag{2.96}$$

here, $C = \frac{3}{10}(3\pi^2)^{2/3}$ for a spin-degenerate electron gas in atomic units. This form of kinetic energy functional was introduced independently in the seminal works by L. H. Thomas [15] and E. Fermi [16].

Moreover, to account for inhomogeneity effects beyond the purely local Thomas–Fermi form, one often adds the von Weizsäcker gradient correction to the kinetic energy functional. A detailed discussion can be found in the review by C. J. García-Cervera [17]. In atomic units, it is generally written as:

$$T_{\mathrm{vW}}[n] = \frac{1}{8} \int \mathrm{d}\vec{r}\, \frac{\nabla n(\vec{r}) \cdot \nabla n(\vec{r})}{n(\vec{r})} \tag{2.97}$$

It is neither possible nor necessary to enumerate all examples of functionals in condensed matter physics. In practice, many such functionals depend explicitly on the electron density $n(\vec{r})$ and, in some cases, also on its gradient $\nabla n(\vec{r})$. For this reason, they are commonly referred to as density functionals.

2.3.2 Density functional in quantum mechanics

In quantum mechanics, it is often convenient to shift the basic variable from the many-body wave function to the electron density $n(\vec{r})$. From this point of view, every relevant physical quantity can be expressed as a functional of the density. Therefore, we can realize that a functional takes the density $n(\vec{r})$ as input and returns a quantum-mechanical observable as output.

To describe this process, we first introduce the density operator:

$$\hat{n}(\vec{r}) = \sum_{i=1}^{N} \delta(\vec{r} - \hat{\vec{r}}_i) \tag{2.98}$$

Next, we can define the electron density of ground-state as its expectation value in the ground-state:

$$n(\vec{r}) \coloneqq \langle \Psi_0 | \, \hat{n}(\vec{r}) \, | \Psi_0 \rangle \tag{2.99}$$

here, $|\Psi_0\rangle$ denotes the ground-state many-body wave function, which encodes the full complexity of the current interacting N-electron system. Given an observable represented by an operator $\hat{\Omega}$, its ground-state value is evaluated through the expectation value Ω:

$$\Omega \coloneqq \langle \Psi_0 | \, \hat{\Omega} \, | \Psi_0 \rangle \tag{2.100}$$

and hence every ground-state observable can be regarded, at least formally, as a functional of the many-body wave function $\Omega[\Psi_0] = \Omega$.

However, the kernel idea of density functional theory is here: for the ground-state properties, the wave function dependence can be reduced even further. In our statement, if the density of ground-state $n(\vec{r})$ uniquely specifies the corresponding wave function of ground-state Ψ_0, the expectation value of any operator $\hat{\Omega}$ in the ground-state is uniquely determined by the density $n(\vec{r})$.

Consequently, every ground-state observable can be written as a (unique) density functional:

$$\langle \Psi_0 | \, \hat{\Omega} \, | \Psi_0 \rangle = \Omega[n(\vec{r})] \tag{2.101}$$

Under the usual Hohenberg–Kohn setting (in particular, for a non-degenerate ground-state), the density $n(\vec{r})$ uniquely determines the external potential $v_{\text{ext}}(\vec{r})$ up to an additive constant. Consequently, $n(\vec{r})$ fixes the hamiltonian and hence the ground-state $|\Psi_0\rangle$ (up to an overall phase). This statement is essentially the central result of the Hohenberg–Kohn theorem [8], namely that in the ground state all observables are uniquely determined by the electron density. Accordingly, any ground-state observable may be written as a functional of the density alone:

$$\Omega = \Omega[n(\vec{r})] \tag{2.102}$$

This is not merely a change of notation; it marks the conceptual compression from a function of $3N$ variables, $\Psi_0(\vec{r}_1, \cdots, \vec{r}_N)$, to a function of only three spatial variables, $n(\vec{r})$, which is the cornerstone of the practical formulation of DFT. Historically, the density functional formalism was established in the 1960s through the pioneering work of Hohenberg and Kohn, and was later made computationally practical by the Kohn–Sham scheme [18].

Since the many-body wave function Ψ encodes the full complexity of the system, a natural question is the following: what is the ingredient that actually distinguishes one many-electron system from another?

To answer this, we recall that the quantum dynamics is specified once the hamiltonian is given.

For electronic systems, the kinetic energy operator $\hat{T}$ and the interaction operator $\hat{V}_{\text{int}}$ have fixed forms, and in the present context we identify $\hat{V}_{\text{int}}$ with the electron-electron interaction $\hat{V}_{\text{ee}}$ as in equation (2.58).

Therefore, the system-specific information enters primarily through the external potential $\hat{V}_{\text{ext}}$, which is generated by the nuclei and any applied fields. With this in mind, we review the form of hamiltonian:

$$\hat{H} = \hat{T} + \hat{V}_{\text{int}} + \hat{V}_{\text{ext}} \tag{2.103}$$

In the subsection 2.1.3: Many-body Schrödinger equation, we have discussed in detail the explicit forms and physical meanings of the kinetic, external potential, and interaction terms in the many-body hamiltonian.

Let us return to the reduced expression for the hamiltonian in equation (2.29). We observe that two terms are universal, in the sense that their operator forms do not depend on the specific system.

These are the electronic kinetic energy (2.22), and the electron-electron Coulomb repulsion (2.24).

Here, we say the external potential $\hat{V}_{\text{ext}}$ is not universal, because it depends on the specific nuclear configuration and any applied fields. In contrast, the electronic kinetic energy and the electron-electron Coulomb interaction have fixed operator forms.

Therefore, the external potential is the system-defining ingredient: it distinguishes one many-electron system from another.

In fact, once $\hat{V}_{\text{ext}}$ is given, the hamiltonian is fully specified. Conversely, if the hamiltonian is known, then in principle we may solve its eigenvalue problem to obtain the eigenstates and eigenvalues. With the wave function known, any observable can be evaluated as the expectation value of the corresponding operator.

This viewpoint motivates the search for compact descriptors that retain the essential information of the ground-state. A central example is the electron density, for an arbitrary state $|\Psi\rangle$, the electron density is defined as the expectation value of the density operator $\hat{n}(\vec{r})$:

$$\begin{aligned} n(\vec{r}) &= \langle\Psi|\,\hat{n}(\vec{r})\,|\Psi\rangle \\ &= \langle\Psi|\sum_i \delta(\vec{r}-\hat{\vec{r}}_i)\,|\Psi\rangle \end{aligned} \tag{2.104}$$

Consider a fixed form of interaction, the external potential $v_{\text{ext}}(\vec{r})$ specifies the corresponding hamiltonian and therefore the ground-state structure. Here $v_{\text{ext}}(\vec{r})$ is a scalar potential field in real space, whereas $\hat{V}_{\text{ext}}$ is the associated many-body operator acting on the N-electron wave function.

Given definition of external potential operator (2.58), in coordinate representation, $\hat{V}_{\text{ext}}$ acts by multiplication with $\sum_i v_{\text{ext}}(\hat{\vec{r}}_i)$.

For later convenience, we introduce the particle-density operator in the Schrödinger picture (2.42) and express the same operator in terms of it. Since $v_{\text{ext}}(\vec{r})$ then couples locally to the density operator $\hat{n}(\vec{r})$, we obtain:

$$\hat{V}_{\text{ext}} = \int \mathrm{d}\vec{r}\, v_{\text{ext}}(\vec{r})\,\hat{n}(\vec{r}) \tag{2.105}$$

Taking the expectation value on an arbitrary state $|\Psi\rangle$ yields the corresponding external potential energy. Using the expectation of density operator as in equation (2.104), we have:

$$\langle\hat{V}_{\text{ext}}\rangle = \int \mathrm{d}\vec{r}\, v_{\text{ext}}(\vec{r})\, n(\vec{r}) \tag{2.106}$$

When the ground-state is non-degenerate, $v_{\text{ext}}(\vec{r})$ stands in a one-to-one correspondence (up to an additive constant) with the ground-state wave function Ψ_0.

Consequently, the ground-state expectation value of any observable $\hat{\Omega}$ may be viewed as a unique functional of the external potential:

$$\langle \hat{\Omega} \rangle \coloneqq \langle \Psi_0 | \, \hat{\Omega} \, | \Psi_0 \rangle = \Omega[v_{\text{ext}}] \tag{2.107}$$

The central move of density functional theory is to further replace the dependence on the wave function by a dependence on the electron density $n(\vec{r})$. The key question is whether the map from external potential $v_{\text{ext}}(\vec{r})$ to the ground-state density $n(\vec{r})$ is likewise invertible in an appropriate sense. The first Hohenberg–Kohn theorem answers this in the affirmative for non-degenerate ground-states: the ground-state density $n(\vec{r})$ uniquely determines $v_{\text{ext}}(\vec{r})$ (again up to an additive constant), and hence uniquely fixes the hamiltonian and the corresponding ground-state. In this sense, all ground-state observables can be regarded as functionals of the density:

$$\langle \hat{\Omega} \rangle = \Omega[n(\vec{r})] \tag{2.108}$$

In the presence of ground-state degeneracy, the strict one-to-one correspondence must be understood at the level of the ground-state subspace; certain observables may then depend on the particular choice of degenerate ground-state. This issue is commonly handled through an ensemble formulation (or via a symmetry-breaking limit) to recover a single-valued density-functional description.

2.3.3 The Hohenberg–Kohn theorem

In this subsection, we focus on the Hohenberg–Kohn theorem, which provides the conceptual foundation of density functional theory: for a fixed form of electron-electron interaction and a non-degenerate ground-state, it establishes a one-to-one correspondence between the external potential and the ground-state density (up to an additive constant), and it leads to a variational characterization of the ground-state.

We start by assuming for contradiction that two different external potentials lead to the same ground-state wave function $|\Psi_0\rangle$, while not differing by a mere additive constant:

$$v_{\text{ext}}^{(1)}(\vec{r}) \neq v_{\text{ext}}^{(2)}(\vec{r}) + \text{const.} \tag{2.109}$$

Next, consider the corresponding hamiltonians and their ground-state eigenvalue equations:

$$\begin{aligned} \hat{H}_1 = \hat{T} + \hat{V}_{\text{int}} + \hat{V}_{\text{ext}}^{(1)} &\implies \hat{H}_1 |\Psi_0^{(1)}\rangle = E_1 |\Psi_0^{(1)}\rangle \\ \hat{H}_2 = \hat{T} + \hat{V}_{\text{int}} + \hat{V}_{\text{ext}}^{(2)} &\implies \hat{H}_2 |\Psi_0^{(2)}\rangle = E_2 |\Psi_0^{(2)}\rangle \end{aligned} \tag{2.110}$$

Subtracting the two equations yields:

$$(\hat{H}_1 - \hat{H}_2) |\Psi_0\rangle = (E_1 - E_2) |\Psi_0\rangle \tag{2.111}$$

Review the expression of the external potential operator in equation (2.58). Since $\hat{T}$ and $\hat{V}_{\text{int}}$ are identical in $\hat{H}_1$ and $\hat{H}_2$, the difference comes entirely from the external potential term. Therefore, for an N-electron system, we have:

$$\begin{aligned} \sum_{i=1}^{N} \left[v_{\text{ext}}^{(1)}(\vec{r}_i) - v_{\text{ext}}^{(2)}(\vec{r}_i) \right] |\Psi_0\rangle &= (\hat{V}_{\text{ext}}^{(1)} - \hat{V}_{\text{ext}}^{(2)}) |\Psi_0\rangle \\ &= (E_1 - E_2) |\Psi_0\rangle \end{aligned} \tag{2.112}$$

Whenever the wave function $\Psi_0(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N) \neq 0$, we can divide both sides by Ψ_0 and obtain:

$$\sum_{i=1}^{N} \left[v_{\text{ext}}^{(1)}(\vec{r}_i) - v_{\text{ext}}^{(2)}(\vec{r}_i) \right] = E_1 - E_2 \tag{2.113}$$

The equality above implies that, for any configuration $(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N)$ where the wave function $\Psi_0(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N) \neq 0$, the factor multiplying Ψ_0 must be constant. Dividing both sides by Ψ_0 wherever the ground-state wave function $\Psi_0(\vec{r}_1, \ldots, \vec{r}_N)$ is nonzero, we obtain a constraint on the difference between the two external potentials.

To make this explicit, we define the potential shift function:

$$g(\vec{r}) \coloneqq v_{\text{ext}}^{(1)}(\vec{r}) - v_{\text{ext}}^{(2)}(\vec{r}) \tag{2.114}$$

Then, we can rewrite the previous relation as:

$$g(\vec{r}_1) + g(\vec{r}_2) + \cdots + g(\vec{r}_N) = E_1 - E_2 \tag{2.115}$$

for all configurations on which the wave function $\Psi_0(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N)$ is nonzero.

Now, we pick any set of coordinates $(\vec{r}_2, \cdots, \vec{r}_N)$ such that the wave function does not vanish, and keep them fixed. With these coordinates held fixed, the sum of the last $N-1$ terms, $g(\vec{r}_2) + \cdots + g(\vec{r}_N)$, is just a scalar. The equation then reduces to:

$$g(\vec{r}_1) = (E_1 - E_2) - [g(\vec{r}_2) + \cdots + g(\vec{r}_N)] \tag{2.116}$$

which shows that $g(\vec{r}_1)$ cannot change as $\vec{r}_1$ varies within any region where the wave function remains nonzero. Otherwise the left-hand side would vary with $\vec{r}_1$, while the right-hand side remains fixed.

Since the same argument applies when we vary any other particle coordinate $\vec{r}_j$ with the remaining coordinates fixed, we conclude that $g(\vec{r})$ must be a constant almost everywhere on the set where $\Psi_0(\vec{r}_1, \ldots, \vec{r}_N)$ is nonzero. Thus, we arrive at the following relation:

$$v_{\text{ext}}^{(1)}(\vec{r}) = v_{\text{ext}}^{(2)}(\vec{r}) + \text{const.} \tag{2.117}$$

Hence $(E_1 - E_2)/N$ is a constant. This contradicts the assumption in equation (2.109). Therefore, we arrive at the conclusion that two different external potentials cannot lead to the same non-degenerate ground-state wave function unless they differ only by an additive constant, as stated in equation (2.117).

Equivalently, up to a constant shift in the potential and for a fixed form of interaction, the mapping from the external potential to the non-degenerate ground-state wave function is one-to-one.

Since any ground-state observable follows uniquely from the ground-state wave function via its expectation value, equation (2.107) implies that $|\Psi_0\rangle$ is uniquely determined by v_{ext}, and hence the ground-state observables are unique functionals of v_{ext}. The external potential is defined to an additive constant shift.

A natural question then is: what is the relation between the external potential $v_{\text{ext}}(\vec{r})$ and the density of the current system $n(\vec{r})$?

To state this more precisely, we next define the equivalence class of external potential $v_{\text{ext}}(\vec{r})$ and the ground-state wave function $|\Psi_0\rangle$ as:

$$\begin{aligned} [v_{\text{ext}}] &\coloneqq \{v_{\text{ext}}(\vec{r}) + \text{const.}\} \\ [|\Psi_0\rangle] &\coloneqq \left\{ \mathrm{e}^{\mathrm{i}\theta} \, |\Psi_0\rangle \,\middle|\, \theta \in \mathbb{R} \right\} \end{aligned} \tag{2.118}$$

To address this question, we fix the form of interaction and restrict ourselves to a non-degenerate ground-state. Under these assumptions, there is a one-to-one correspondence between the equivalence class of external potential $[v_{\text{ext}}(\vec{r})]$ and the ground-state wave function $[|\Psi_0\rangle]$, where $[v_{\text{ext}}(\vec{r})]$ is defined up to an additive constant and $[|\Psi_0\rangle]$ up to an overall phase.

Equivalently, $[v_{\text{ext}}(\vec{r})]$ and $[|\Psi_0\rangle]$ are in bijection:

$$[v_{\text{ext}}(\vec{r})] \longleftrightarrow [|\Psi_0\rangle] \tag{2.119}$$

Moreover, the wave function is directly related to the density through the standard marginalization over the remaining $N-1$ particles. As a result, we obtain a unique ground-state density, given by the usual marginalization of the squared modulus of the wave function over all but one space variables:

$$n(\vec{r}) = N \int \mathrm{d}\vec{r}_2 \cdots \mathrm{d}\vec{r}_N \; |\Psi(\vec{r}, \vec{r}_2, \cdots, \vec{r}_N)|^2 \tag{2.120}$$

here, Ψ is normalized and N is the number of particles. This means that the external potential v_{ext} induces a mapping to the ground-state density $n(\vec{r})$, which this subsection further clarifies the character of this mapping.

The strategy to establish this relation is also a reductio ad absurdum argument. Suppose there exist two different external potentials $v_{\text{ext}}^{(1)}(\vec{r})$ and $v_{\text{ext}}^{(2)}(\vec{r})$ such that

$v_{\text{ext}}^{(1)}(\vec{r}) \neq v_{\text{ext}}^{(2)}(\vec{r}) + \text{const}$. Let their corresponding non-degenerate ground-state wave functions be denoted as $\Psi_0^{(1)}$ and $\Psi_0^{(2)}$.

Recall the Schrödinger equation (2.110). The external potentials $v_{\text{ext}}^{(1)}(\vec{r})$ and $v_{\text{ext}}^{(2)}(\vec{r})$ lead to the eigenvalues E_1 and E_2, and the corresponding ground-state wave functions $|\Psi_0^{(1)}\rangle$ and $|\Psi_0^{(2)}\rangle$, respectively. Since the ground-states are assumed non-degenerate $|\Psi_0^{(1)}\rangle \neq |\Psi_0^{(2)}\rangle$. For each external potential, the total energy can be written as the expectation value of the hamiltonian in its ground-state:

$$\begin{aligned} E_1 &= \langle \Psi_0^{(1)} | \, \hat{H}_1 \, | \Psi_0^{(1)} \rangle \\ E_2 &= \langle \Psi_0^{(2)} | \, \hat{H}_2 \, | \Psi_0^{(2)} \rangle \end{aligned} \tag{2.121}$$

We now make the key assumption for contradiction: although the two external potentials, and hence the two ground-state wave functions, are different, they lead to the same ground-state density $n^{(1)}(\vec{r}) = n^{(2)}(\vec{r})$, where $n^{(1)}(\vec{r})$ and $n^{(2)}(\vec{r})$ are obtained from $\Psi_0^{(1)}$ and $\Psi_0^{(2)}$, respectively, via equation (2.120)

Through the Rayleigh–Ritz variational principle: for the ground-state energy E_0 of a given hamiltonian $\hat{H}$, the energy expectation value over any normalized trial wave function Φ provides an upper bound to the ground-state energy:

$$E_0 \leq \langle \Phi | \, \hat{H} \, | \Phi \rangle \tag{2.122}$$

Therefore, using $|\Psi_0^{(2)}\rangle$ as a trial state for $\hat{H}_1$ and using $|\Psi_0^{(1)}\rangle$ as a trial state for $\hat{H}_2$, we obtain:

$$\begin{aligned} E_1 &= \langle \Psi_0^{(1)} | \, \hat{H}_1 \, | \Psi_0^{(1)} \rangle < \langle \Psi_0^{(2)} | \, \hat{H}_1 \, | \Psi_0^{(2)} \rangle \\ E_2 &= \langle \Psi_0^{(2)} | \, \hat{H}_2 \, | \Psi_0^{(2)} \rangle < \langle \Psi_0^{(1)} | \, \hat{H}_2 \, | \Psi_0^{(1)} \rangle \end{aligned} \tag{2.123}$$

here, the strict inequalities follow from the assumption that the ground-state is non-degenerate and that $|\Psi_0^{(1)}\rangle \neq |\Psi_0^{(2)}\rangle$.

Let us rewrite the variational bound for E_1. Recall the eigenvalue equation (2.111) and the definition of the hamiltonian difference in equation (2.112). We can expand the right-hand side as:

$$\begin{aligned} E_1 = \langle \Psi_0^{(1)} | \, \hat{H}_1 \, | \Psi_0^{(1)} \rangle < \langle \Psi_0^{(2)} | \, \hat{H}_1 \, | \Psi_0^{(2)} \rangle &= \langle \Psi_0^{(2)} | \, (\hat{H}_1 - \hat{H}_2 + \hat{H}_2) \, | \Psi_0^{(2)} \rangle \\ &= \langle \Psi_0^{(2)} | \, \hat{H}_2 \, | \Psi_0^{(2)} \rangle + \langle \Psi_0^{(2)} | \, (\hat{H}_1 - \hat{H}_2) \, | \Psi_0^{(2)} \rangle \\ &= E_2 + \langle \Psi_0^{(2)} | \, (\hat{V}_{\text{ext}}^{(1)} - \hat{V}_{\text{ext}}^{(2)}) \, | \Psi_0^{(2)} \rangle \end{aligned} \tag{2.124}$$

The further step is to rewrite the last term in a density form. Let us apply the definition of the electron density function (2.104) as:

$$n^{(2)}(\vec{r}) = \langle \Psi_0^{(2)} | \, \hat{n}(\vec{r}) \, | \Psi_0^{(2)} \rangle \tag{2.125}$$

Then, we apply the definition of the operator for external potential (2.105), and obtain:

$$\begin{aligned} \langle \Psi_0^{(2)} | \, (\hat{V}_{\text{ext}}^{(1)} - \hat{V}_{\text{ext}}^{(2)}) \, | \Psi_0^{(2)} \rangle &= \int \mathrm{d}\vec{r} \, \left[v_{\text{ext}}^{(1)}(\vec{r}) - v_{\text{ext}}^{(2)}(\vec{r}) \right] \, \langle \Psi_0^{(2)} | \, \hat{n}(\vec{r}) \, | \Psi_0^{(2)} \rangle \\ &= \int \mathrm{d}\vec{r} \, \left[v_{\text{ext}}^{(1)}(\vec{r}) - v_{\text{ext}}^{(2)}(\vec{r}) \right] \, n^{(2)}(\vec{r}) \end{aligned} \tag{2.126}$$

As a result, the total energy of the first system satisfies the following inequality:

$$E_1 < E_2 + \langle \Psi_0^{(2)} | \, \left(\hat{V}_{\text{ext}}^{(1)} - \hat{V}_{\text{ext}}^{(2)} \right) \, | \Psi_0^{(2)} \rangle \tag{2.127}$$

Therefore, we arrive at the following inequality for the ground-state energies:

$$E_1 < E_2 + \int \mathrm{d}\vec{r} \, \left[v_{\text{ext}}^{(1)}(\vec{r}) - v_{\text{ext}}^{(2)}(\vec{r}) \right] \, n^{(2)}(\vec{r}) \tag{2.128}$$

Vice versa, upon interchanging the roles of the two systems, we analogously obtain:

$$E_2 < E_1 + \int \mathrm{d}\vec{r} \, \left[v_{\text{ext}}^{(2)}(\vec{r}) - v_{\text{ext}}^{(1)}(\vec{r}) \right] \, n^{(1)}(\vec{r}) \tag{2.129}$$

We are now ready to complete the reductio ad absurdum argument. Revisit our assumption that two distinct external potentials $v_{\text{ext}}^{(1)}(\vec{r}) \neq v_{\text{ext}}^{(2)}(\vec{r})$ yield the same

ground-state density $n(\vec{r}) = n^{(1)}(\vec{r}) = n^{(2)}(\vec{r})$, the Rayleigh–Ritz variational principle leads to the previous ground-state energy inequalities.

Using $n(\vec{r}) = n^{(1)}(\vec{r}) = n^{(2)}(\vec{r})$ and adding equations (2.128) and (2.129), the integral terms cancel identically, which leads to:

$$E_1 + E_2 < E_1 + E_2 \tag{2.130}$$

It is obviously a contradiction. This demonstrates that, for a fixed form of interaction and a non-degenerate ground-state, the ground-state density $n(\vec{r})$ uniquely determines the external potential $v_{\text{ext}}(\vec{r})$ up to an additive constant, and hence uniquely determines the hamiltonian and the ground-state up to an overall phase.

Let us recollect the definition of the equivalence class of the external potential $v_{\text{ext}}(\vec{r})$ and the ground-state wave function $|\Psi_0\rangle$ in equation (2.118), as well as the fact that the equivalence classes $[v_{\text{ext}}]$ and $[|\Psi_0\rangle]$ are in bijection in equation (2.119).

Moreover, the ground-state wave function uniquely determines the ground-state density via the standard marginalization:

$$[|\Psi_0\rangle] \longmapsto n(\vec{r}) \tag{2.131}$$

We emphasize that the map in equation (2.131) is not injective for any arbitrary many-body wave function. In the present setting, however, we restrict ourselves to non-degenerate ground-states. Within this restricted class, the first Hohenberg–Kohn theorem provides a unique inverse implication, namely that the ground-state density uniquely determines the equivalence class of the external potential:

$$n(\vec{r}) \longmapsto [v_{\text{ext}}(\vec{r})] \tag{2.132}$$

Combining equations (2.119), (2.131), and(2.132), we finally arrive at a one-to-one correspondence among the basic ground-state descriptors:

$$[v_{\text{ext}}(\vec{r})] \longleftrightarrow n(\vec{r}) \longleftrightarrow [|\Psi_0\rangle] \tag{2.133}$$

The two previous equations (2.132) and (2.133) summarize the content of the Hohenberg–Kohn theorem [8, 19]. For a fixed form of interaction and a non-degenerate ground-state,

the ground-state density $n(\vec{r})$ is in bijection with the equivalence class of the external potential $[v_{\text{ext}}(\vec{r})]$ up to an additive constant. Consequently, the ground-state density uniquely determines the hamiltonian and the ground-state wave function up to an overall phase, and any ground-state observable is, in principle, a unique functional of $n(\vec{r})$.

Therefore, the ground-state expectation definition (2.107) can be reformulated as:

$$\langle\hat{\Omega}\rangle = \langle\Psi_0|\,\hat{\Omega}\,|\Psi_0\rangle = \Omega[n(\vec{r})] \tag{2.134}$$

here, $\Omega[n(\vec{r})]$ denotes the density functional that yields the ground-state expectation value of $\hat{\Omega}$. This observation forms the central premise of density functional theory and motivates the use of the density, rather than the wave function, as the fundamental variable.

As described by equation (2.134), every ground-state observable is a unique functional of the electron density.

Recall equation (2.63), which expresses the ground-state energy as the expectation value of the hamiltonian. According to decomposition of the hamiltonian in equation (2.103), we write:

$$\begin{aligned} E_0 &= \langle\Psi_0|\,\hat{H}\,|\Psi_0\rangle \\ &= \langle\Psi_0|\,(\hat{T} + \hat{V}_{\text{int}} + \hat{V}_{\text{ext}})\,|\Psi_0\rangle \end{aligned} \tag{2.135}$$

To generalize this variational picture, for a given admissible density $n(\vec{r})$ we consider wave functions Ψ that yield n, and define

$$\begin{aligned} E[n] &= \min_{\Psi\to n} \langle\Psi|\,(\hat{T} + \hat{V}_{\text{int}} + \hat{V}_{\text{ext}})\,|\Psi\rangle \\ &= \langle\Psi[n]|\,(\hat{T} + \hat{V}_{\text{int}} + \hat{V}_{\text{ext}})\,|\Psi[n]\rangle \end{aligned} \tag{2.136}$$

which reduces to E_0 when $n = n_0$.

Here the minimization is taken over all normalized antisymmetric N-electron wave functions $|\Psi\rangle$ compatible with the fixed interaction.

The kinetic energy functional $T[n]$, the interaction energy functional $E_{\text{int}}[n]$, and the external potential energy functional $E_{\text{ext}}[n]$ are evaluated on a minimizing wave function $\Psi[n]$ that yields the density n. We now write their definitions:

$$\begin{aligned} T[n] &:= \langle \Psi[n] | \, \hat{T} \, | \Psi[n] \rangle \\ E_{\text{int}}[n] &:= \langle \Psi[n] | \, \hat{V}_{\text{int}} \, | \Psi[n] \rangle \\ E_{\text{ext}}[n] &:= \langle \Psi[n] | \, \hat{V}_{\text{ext}} \, | \Psi[n] \rangle \end{aligned} \tag{2.137}$$

Using the local form of the external potential operator in equation (2.105), we get the local form of the external potential energy functional:

$$E_{\text{ext}}[n] = \int \mathrm{d}\vec{r} \, v_{\text{ext}}(\vec{r}) \, n(\vec{r}) \tag{2.138}$$

According to the definition of the ground-state density in equation (2.99), as well as the definitions of the kinetic energy and interaction energy functionals in equation (2.137), we obtain the decomposition of exact ground-state energy functional:

$$\begin{aligned} E[n] &= \langle \Psi[n] | \, (\hat{T} + \hat{V}_{\text{int}} + \hat{V}_{\text{ext}}) \, | \Psi[n] \rangle \\ &= \langle \Psi[n] | \, (\hat{T} + \hat{V}_{\text{int}}) \, | \Psi[n] \rangle + \int \mathrm{d}\vec{r} \, v_{\text{ext}}(\vec{r}) \, n(\vec{r}) \\ &= T[n] + E_{\text{int}}[n] + \int \mathrm{d}\vec{r} \, v_{\text{ext}}(\vec{r}) \, n(\vec{r}) \end{aligned} \tag{2.139}$$

Recall the Slater determinant representation (2.34), under this independent-particle form, the expectation value of the kinetic energy operator becomes a sum over orbitals:

$$\langle \Psi | \hat{T} | \Psi \rangle = -\frac{1}{2} \sum_i \langle \psi_i | \nabla_i^2 | \psi_i \rangle \tag{2.140}$$

Substituting equation (2.140), we achieve:

$$E[n] = -\frac{1}{2} \sum_i \langle \psi_i | \, \nabla_i^2 \, | \psi_i \rangle + E_{\text{int}}[n] + \int \mathrm{d}\vec{r} \, v_{\text{ext}}(\vec{r}) \, n(\vec{r}) \tag{2.141}$$

To proceed, we separate the system-specific external potential contribution from the remaining part of the energy. This motivates introducing the universal Hohenberg–Kohn functional $E_{\mathrm{HK}}[n]$, defined by a constrained-search formulation [12, 20]:

$$\begin{aligned} E_{\mathrm{HK}}[n] &= \min_{\Psi \to n} \langle \Psi | \left(\hat{T} + \hat{V}_{\mathrm{int}} \right) | \Psi \rangle \\ &= T[n] + E_{\mathrm{int}}[n] \end{aligned} \tag{2.142}$$

For a fixed form of the electron-electron interaction, $E_{\mathrm{HK}}[n]$ is independent of the external potential $v_{\mathrm{ext}}(\vec{r})$, and the system specific information enters only through v_{ext}.

This allows us to separate out the external potential contribution and write:

$$E[n] = E_{\mathrm{HK}}[n] + \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, n(\vec{r}) \tag{2.143}$$

The Hohenberg–Kohn theorem then allows us to cast the ground-state problem as a constrained search over admissible densities:

$$E_0 = \min_{n(\vec{r})} \left\{ E\left[n(\vec{r})\right] \middle| \int \mathrm{d}\vec{r}\, n(\vec{r}) = N \right\} = E\left[n_0(\vec{r})\right] \tag{2.144}$$

Conceptually, equation (2.144) defines an underlying energy surface over the space of admissible densities; figure 2.1 provides a two-dimensional schematic slice of such a high-dimensional landscape.

Although density functional theory is not usually implemented in this form in practical calculations, this formulation provides a clear variational picture of the underlying idea.

We now turn to the constrained minimization in equation (2.144). To enforce the fixed particle number, we impose the condition:

$$\int \mathrm{d}\vec{r}\, n(\vec{r}) = N \tag{2.145}$$

by introducing a Lagrange multiplier μ and defining the auxiliary functional:

$$\widetilde{E}[n] \coloneqq E[n] - \mu \left(\int \mathrm{d}\vec{r}\, n(\vec{r}) - N \right) \tag{2.146}$$

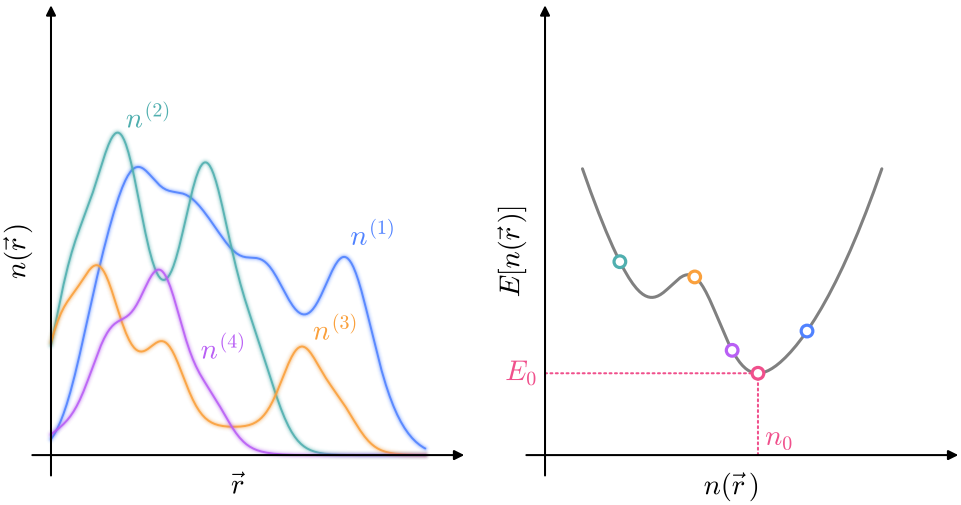

FIGURE 2.1. Schematic illustration of the Hohenberg–Kohn variational picture

At the minimum, the first variation vanishes for all admissible $\delta n(\vec{r})$, which gives the Euler–Lagrange condition:

$$\frac{\delta E[n]}{\delta n(\vec{r})} = \mu \tag{2.147}$$

Combining the Euler–Lagrange condition in equation (2.147) with the standard form of the energy functional in equation (2.143), we obtain:

$$\frac{\delta E_{\mathrm{HK}}[n]}{\delta n(\vec{r})} + v_{\mathrm{ext}}(\vec{r}) = \mu \tag{2.148}$$

evaluated at $n = n_0$.

However, the functional $E_{\mathrm{HK}}[n]$ (and hence its components such as the kinetic and interaction contributions) is not known explicitly as a functional of the density, and must be approximated in practical calculations.

At this point, we return to the standard form of the energy functional in equation (2.143). By rearranging it, we can formally write the universal Hohenberg–Kohn functional as:

$$E_{\mathrm{HK}}[n] = E[n] - \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, n(\vec{r}) \tag{2.149}$$

This relation may suggest that $E_{\mathrm{HK}}[n]$ could be obtained once the total energy functional $E[n]$ is known.

In practice, this is not a constructive route. For a given $v_{\mathrm{ext}}(\vec{r})$, the interacting many-body problem yields only the minimizing density $n_0(\vec{r})$ and the corresponding energy $E[n_0]$. To determine a functional over a broad range of densities, we would in principle need to solve the interacting problem for many different external potentials to sample many ground-state densities. Therefore, the Hohenberg–Kohn theorem should be viewed as an existence and uniqueness statement, rather than an explicit formula for $E_{\mathrm{HK}}[n]$.

These considerations clarify why density functional theory relies on approximations. In what follows, we introduce the Kohn–Sham construction, which reorganizes the problem so that a large part of the functional can be treated exactly, and the remaining unknown part can be approximated systematically.

2.3.4 The Thomas–Fermi approach

Now, we introduce the Thomas–Fermi approach as a first and purely density-based model for the kinetic energy.

The basic idea is to treat the electrons locally as a uniform, spin-degenerate Fermi gas. At each point $\vec{r}$, the local density $n(\vec{r})$ fixes a local Fermi momentum $p_F(\vec{r})$, which allows us to estimate the kinetic energy density from the occupied Fermi sphere in momentum space.

For a spin-degenerate system, the phase-space counting gives:

$$n(\vec{r}) = \frac{2}{h^3}\frac{4\pi}{3}\,p_F^3(\vec{r}) \tag{2.150}$$

and the local Fermi momentum is related to the local single-particle energy through:

$$\mu = \frac{p_F^2(\vec{r})}{2m} + v(\vec{r}) \tag{2.151}$$

Combining these two expressions yields a local relation between $n(\vec{r})$ and $v(\vec{r})$. This relation also leads to an explicit approximation for the kinetic energy functional in terms of $n(\vec{r})$ alone, which we introduce next.

To close the Thomas–Fermi picture, we treat $v(\vec{r})$ as an electrostatic potential generated self-consistently by the electron density. In atomic units, it satisfies the Poisson equation:

$$\nabla^2 v(\vec{r}) = -4\pi\, n(\vec{r}) \tag{2.152}$$

Eliminating $p_F(\vec{r})$ between (2.150) and (2.151), we obtain:

$$n(\vec{r}) = \frac{1}{3\pi^2}\,[2(\mu - v(\vec{r}))]^{3/2} \tag{2.153}$$

Substituting this expression into equation (2.152), We arrive at the Thomas–Fermi equation:

$$\nabla^2 v(\vec{r}) = -\frac{4}{3\pi}\,[2(\mu - v(\vec{r}))]^{3/2} \tag{2.154}$$

This nonlinear equation determines the potential and the density self-consistently within the Thomas–Fermi approximation.

To connect the Thomas–Fermi construction with density functional theory, we rewrite the total energy in a density functional form, denoted by $E_{\mathrm{TF}}[n; v_{\mathrm{ext}}]$. In this sense, the Thomas–Fermi model provides an explicit approximation to the universal functional.

From the Thomas–Fermi approximation, we express the total energy as a density functional composed of a statistical kinetic term $T_{\mathrm{TF}}[n]$, an interaction term $E_{\mathrm{int}}[n]$, and an external contribution $E_{\mathrm{ext}}[n]$ describing the coupling to the external potential $v_{\mathrm{ext}}(\vec{r})$.

Furthermore, there is also a nucleus-nucleus term E_{NN}, but it does not depend on the electron density. Therefore, we do not consider this term at the moment.

Accordingly, we write the Thomas–Fermi total energy functional:

$$E_{\mathrm{TF}}[n; v_{\mathrm{ext}}] = T_{\mathrm{TF}}[n] + E_{\mathrm{int}}[n] + E_{\mathrm{ext}}[n] \tag{2.155}$$

According to the local form of the external potential energy functional (2.138), and the external contribution is given by the coupling between $n(\vec{r})$ and $v_{\mathrm{ext}}(\vec{r})$. For fixed nuclear positions, the external potential $v_{\mathrm{ext}}(\vec{r})$ is a prescribed function of $\vec{r}$ defined in

equation (2.30). Therefore, the external contribution $E_{\text{ext}}[n]$ is linear in the electron density $n(\vec{r})$.

We next approximate the interaction energy functional by its classical Coulomb form associated with the density distribution. Using the electron-electron Coulomb repulsion (2.24), we introduce the Hartree term:

$$\begin{aligned} E_{\text{H}}[n] &\coloneqq E_{\text{int}}[n] - E_{\Delta\text{int}}[n] \\ &= \frac{1}{2} \iint \mathrm{d}\vec{r}\,\mathrm{d}\vec{r}\,' \, \frac{n(\vec{r})n(\vec{r}\,')}{|\vec{r} - \vec{r}\,'|} \end{aligned} \tag{2.156}$$

here, $E_{\Delta\text{int}}[n]$ collects the nonclassical part of the interaction energy beyond the Hartree term. In the Hartree approximation, we set $E_{\Delta\text{int}}[n] = 0$.

We next approximate the interaction energy functional by the classical Coulomb energy associated with the density distribution. Using the electron-electron Coulomb repulsion (2.24), we introduce the Hartree term:

Therefore, we introduce a Hartree approximation at the level of the interaction energy by replacing $E_{\text{int}}[n]$ with its classical counterpart $E_{\text{H}}[n]$:

$$E_{\text{H}}^{\text{int}}[n] \coloneqq T[n] + \int \mathrm{d}^3 r \, v_{\text{ext}}(\vec{r}) n(\vec{r}) + E_{\text{H}}[n] \tag{2.157}$$

here, $T[n]$ is the interacting kinetic energy functional evaluated on the minimizing many-body state $\Psi[n]$ that yields the density n. In other words, this replacement affects only the interaction energy, while the kinetic energy functional remains $T[n]$ at this stage.

We next consider the kinetic energy functional. For a homogeneous spin-unpolarized electron gas with density n, the Fermi wave vector satisfies [21]:

$$k_{\text{F}} = \left(3\pi^2 n\right)^{\frac{1}{3}} \tag{2.158}$$

and the corresponding kinetic energy density reads:

$$t(n) = \frac{3}{10} k_{\text{F}}^2 \, n = \frac{3}{10} \left(3\pi^2\right)^{\frac{2}{3}} n^{\frac{5}{3}} \tag{2.159}$$

Within the local density approximation, we apply this homogeneous result locally by replacing n with $n(\vec{r})$ and integrating over space. Accordingly, this yields the Thomas–Fermi kinetic energy functional [15, 16, 22]:

$$T_{\mathrm{TF}}[n] = C_1 \int \mathrm{d}\vec{r}\, n(\vec{r})^{\frac{5}{3}} \tag{2.160}$$

with the Thomas–Fermi constant:

$$C_1 \coloneqq \frac{3}{10}\left(3\pi^2\right)^{\frac{2}{3}} \tag{2.161}$$

This expression holds for a spin-unpolarized system.

Therefore, we arrive at the full Thomas–Fermi energy functional as a functional of the electron density [17, 23]. The nucleus-nucleus contribution E_{NN} is omitted since it does not depend on $n(\vec{r})$.

$$\begin{aligned} E_{\mathrm{TF}}[n] &= C_1 \int \mathrm{d}\vec{r}\, n(\vec{r})^{\frac{5}{3}} + \frac{1}{2}\iint \mathrm{d}\vec{r}\,\mathrm{d}\vec{r}\,' \frac{n(\vec{r})n(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|} + \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, n(\vec{r}) \\ &= C_1 \int \mathrm{d}\vec{r}\, n(\vec{r})^{\frac{5}{3}} + E_{\mathrm{H}}[n] + \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, n(\vec{r}) \end{aligned} \tag{2.162}$$

This expression gives the total energy in the Thomas–Fermi approximation as a functional of the electron density $n(\vec{r})$ for a given external potential $v_{\mathrm{ext}}(\vec{r})$.

The ground-state of the system is characterized by the density that minimizes the total energy under the constraint of a fixed particle number. This is the variational principle for the ground-state energy in density functional theory.

At finite temperature, the equilibrium density is obtained by minimizing the Helmholtz free energy functional [24]. In the present zero temperature setting, the relevant quantity is the total energy functional $E_{\mathrm{TF}}[n]$.

To determine the ground-state density, we minimize the Thomas–Fermi energy functional under the particle number constraint:

$$\int \mathrm{d}\vec{r}\, n(\vec{r}) = N \tag{2.163}$$

Using the method of Lagrange multipliers, we require the first variation of the constrained functional to vanish.

The Lagrange multiplier λ enforces the particle number constraint and can be identified with the chemical potential:

$$\begin{aligned} 0 &= \delta \left\{ E_{\mathrm{TF}}[n] - \lambda \left(\int \mathrm{d}\vec{r}\, n(\vec{r}) - N \right) \right\} \\ &= \delta \left\{ T_{\mathrm{TF}}[n] + E_{\mathrm{int}}[n] + E_{\mathrm{ext}}[n] - \lambda \left(\int \mathrm{d}\vec{r}\, n(\vec{r}) - N \right) \right\} \end{aligned} \tag{2.164}$$

Here λ enforces the particle number constraint and plays the role of the chemical potential.

We define the Hartree potential as a functional of density:

$$v_{\mathrm{H}}([n];\vec{r}) \coloneqq \int \mathrm{d}\vec{r}\,' \, \frac{n(\vec{r}\,')}{|\vec{r} - \vec{r}\,'|} \tag{2.165}$$

The corresponding first variations of the three contributions can be evaluated explicitly.

For the Thomas–Fermi kinetic term, we obtain:

$$\delta T_{\mathrm{TF}}[n] = \int \mathrm{d}\vec{r}\, \frac{5}{3} C_1\, n(\vec{r})^{\frac{2}{3}}\, \delta n(\vec{r}) \tag{2.166}$$

For the interaction term in the Hartree approximation, the variation introduces the Hartree potential and reads:

$$\delta E_{\mathrm{int}}[n] = \int \mathrm{d}\vec{r}\, v_{\mathrm{H}}([n];\vec{r})\, \delta n(\vec{r}) \tag{2.167}$$

For the external contribution, the variation follows directly from the linear coupling to the external potential:

$$\delta E_{\mathrm{ext}}[n] = \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, \delta n(\vec{r}) \tag{2.168}$$

Substituting equations (2.166)–(2.168) into equation (2.164), we obtain:

$$\int \mathrm{d}\vec{r} \left\{ \frac{5}{3} C_1\, n(\vec{r})^{\frac{2}{3}} + v_{\mathrm{H}}([n];\vec{r}) + v_{\mathrm{ext}}(\vec{r}) - \lambda \right\} \delta n(\vec{r}) = 0 \tag{2.169}$$

Since the density variation $\delta n(\vec{r})$ is arbitrary, the integrand must vanish pointwise, which yields the Thomas–Fermi equation:

$$\frac{5}{3}C_1\, n(\vec{r})^{\frac{2}{3}} + v_{\mathrm{H}}([n];\vec{r}) + v_{\mathrm{ext}}(\vec{r}) = \lambda \tag{2.170}$$

This is convenient to introduce the effective potential:

$$v(\vec{r}) \coloneqq v_{\mathrm{ext}}(\vec{r}) + v_{\mathrm{H}}([n];\vec{r}) \tag{2.171}$$

Equation (2.170) then becomes:

$$\frac{5}{3}C_1\, n(\vec{r})^{\frac{2}{3}} + v(\vec{r}) = \lambda \tag{2.172}$$

Solving for the density, we obtain:

$$n(\vec{r}) = \frac{1}{3\pi^2}\left[2\left(\lambda - v(\vec{r})\right)\right]^{\frac{3}{2}} \tag{2.173}$$

The above expression applies in the region where $\lambda > v(\vec{r})$, and one takes $n(\vec{r}) = 0$ otherwise.

To derive a differential form, we use the Poisson equation. For the nuclear external potential, we have:

$$\nabla^2 v_{\mathrm{ext}}(\vec{r}) = 4\pi \sum_I Z_I\, \delta\left(\vec{r} - \vec{R}_I\right) \tag{2.174}$$

For the Hartree potential, we have:

$$\nabla^2 v_{\mathrm{H}}([n];\vec{r}) = -4\pi n(\vec{r}) \tag{2.175}$$

Combining equations (2.171), (2.174), and (2.175), we find:

$$\nabla^2 v(\vec{r}) = 4\pi \sum_I Z_I\, \delta\left(\vec{r} - \vec{R}_I\right) - 4\pi n(\vec{r}) \tag{2.176}$$

Substituting equation (2.173) into equation (2.176), we arrive at the differential form of the Thomas–Fermi equation:

$$\nabla^2 v(\vec{r}) = 4\pi \sum_I Z_I \, \delta\left(\vec{r} - \vec{R}_I\right) - \frac{4}{3\pi} \left[2\left(\lambda - v(\vec{r})\right)\right]^{\frac{3}{2}} \tag{2.177}$$

This procedure constitutes the Thomas–Fermi method.

2.3.5 Limitations of the Thomas–Fermi approach

For a many-body system, it is instructive to contrast the exact density functional with the Thomas–Fermi approximation. In density functional theory, the total energy can be written as a functional of the electron density [7, 8].

Let us express the exact total energy as a functional of the density:

$$E[n] = T[n] + E_{\mathrm{H}}[n] + E_{\mathrm{xc}}^{\mathrm{int}}[n] + \int \mathrm{d}\vec{r}\, n(\vec{r}) v_{\mathrm{ext}}(\vec{r}) \tag{2.178}$$

Here $T[n]$ denotes the exact kinetic energy functional and $E_{\mathrm{H}}[n]$ is the Hartree energy functional. The exchange-correlation functional $E_{\mathrm{xc}}^{\mathrm{int}}[n]$ collects all nonclassical interaction effects beyond the Hartree term. It includes the exchange contribution originating from the antisymmetry of the many-electron wave function and the correlation contribution accounting for the remaining many-body correlations.

In view of the definition of the universal Hohenberg–Kohn functional in equation (2.142), it is convenient to introduce the exchange-correlation part of the interaction energy functional via:

$$E_{\mathrm{xc}}^{\mathrm{int}}[n] := E_{\mathrm{int}}[n] - E_{\mathrm{H}}[n] \tag{2.179}$$

Equivalently, the universal Hohenberg–Kohn functional can be decomposed as:

$$\begin{aligned} E_{\mathrm{HK}}[n] &= T[n] + E_{\mathrm{int}}[n] \\ &= T[n] + E_{\mathrm{H}}[n] + E_{\mathrm{xc}}^{\mathrm{int}}[n] \end{aligned} \tag{2.180}$$

Accordingly, the exact total energy functional can be written in the following form:

$$E[n] = E_{\mathrm{HK}}[n] + \int \mathrm{d}\vec{r}\, n(\vec{r})\, v_{\mathrm{ext}}(\vec{r}) \tag{2.181}$$

Recall the full Thomas–Fermi energy functional in equation (2.162). Let us compare with the exact decomposition in equation (2.178), the Thomas–Fermi functional neglects the exchange correlation part of the interaction energy functional $E_{\mathrm{xc}}^{\mathrm{int}}[n]$. This omission can lead to sizable errors for total energies even in simple atoms [25, 26].

A first improvement is to approximate $E_{\mathrm{xc}}^{\mathrm{int}}[n]$ by a local exchange functional derived from the homogeneous electron gas. This approximation is commonly associated with the Dirac exchange functional [27]:

$$E_{\mathrm{xc}}^{\mathrm{int}}[n] \approx E_{\mathrm{x}}^{\mathrm{D}}[n] \tag{2.182}$$

Since the Dirac exchange functional is given by:

$$E_{\mathrm{x}}^{\mathrm{D}}[n] \coloneqq -C_2 \int \mathrm{d}\vec{r}\, n(\vec{r})^{\frac{4}{3}} \tag{2.183}$$

here, the corresponding constant reads:

$$C_2 \coloneqq \frac{3}{4}\left(\frac{3}{\pi}\right)^{\frac{1}{3}} \tag{2.184}$$

At this level, correlation effects are not included in $E_{\mathrm{x}}^{\mathrm{D}}[n]$.

Including the Dirac exchange term, we obtain the Thomas–Fermi Dirac functional:

$$E_{\mathrm{TFD}}[n] \coloneqq C_1 \int \mathrm{d}\vec{r}\, n(\vec{r})^{\frac{5}{3}} + E_{\mathrm{H}}[n] - C_2 \int \mathrm{d}\vec{r}\, n(\vec{r})^{\frac{4}{3}} + \int \mathrm{d}\vec{r}\, n(\vec{r})\, v_{\mathrm{ext}}(\vec{r}) \tag{2.185}$$

The Thomas–Fermi Dirac approximation improves total energies by incorporating an explicit exchange contribution. However, the kinetic energy is still treated at the Thomas–Fermi level, which is often insufficient for molecules and solids.

A systematic gradient correction to the kinetic energy can be introduced through the von Weizsäcker form [28]. The corresponding von Weizsäcker kinetic energy functional is defined as:

$$T_{\mathrm{vW}}[n] \coloneqq \frac{1}{2}\int \mathrm{d}\vec{r}\,\left|\nabla\sqrt{n(\vec{r})}\right|^2 \tag{2.186}$$

By adding this gradient correction, we arrive at the Thomas–Fermi Dirac von Weizsäcker model:

$$E_{\mathrm{TFDW}}[n] \coloneqq C_1\int \mathrm{d}\vec{r}\,n(\vec{r})^{\frac{5}{3}} + T_{\mathrm{vW}}[n] + E_{\mathrm{H}}[n] - C_2\int \mathrm{d}\vec{r}\,n(\vec{r})^{\frac{4}{3}} + \int \mathrm{d}\vec{r}\,n(\vec{r})\,v_{\mathrm{ext}}(\vec{r}) \tag{2.187}$$

These extensions provide early orbital free approximations beyond the Thomas–Fermi approach [29]. They also clarify that improving the kinetic energy functional and the exchange correlation functional is essential for a quantitatively reliable description of realistic interacting electron systems.

2.4 From density functional to Kohn–Sham theory

In practical many-body problems, an exact treatment of the hamiltonian is impossible, and approximation is inevitable. This motivates a reformulation in terms of a simpler basic variable.

The electron density is an obvious choice. The Thomas–Fermi approach in the previous subsection provides a first purely density based model, but its limitations show that improved treatments of the kinetic energy and the exchange correlation effects are necessary.

In what follows, we introduce an auxiliary description that retains the density as the central variable and provides a practical starting point for systematic improvements.

2.4.1 Auxiliary system

We introduce an auxiliary system that reproduces the ground-state density $n(\vec{r})$ of the interacting electrons. This maps the many-body problem onto a simpler reference description, with residual interactions captured by the exchange-correlation functional, leading to the Kohn–Sham scheme.

As shown in equation (2.68), the ground-state density can be obtained from the many-body ground-state wave function by integrating out $N-1$ coordinates.

Since one-body observables can be expressed in terms of the density, as summarized in the equation (2.55), the density provides a reduced description that avoids working directly with the full wave function.

This approach motivates introducing an auxiliary description that reproduces the same density.

Firstly, let us start by an interrelated two-body system consisting of two species.

We denote the coordinates of the first species by $x_1, x_2, \cdots, x_N$, and those of the second species by $y_1, y_2, \cdots, y_M$.

Therefore, the full many-body wave function reads $\Psi(x_1, x_2, \cdots, x_N; y_1, y_2, \cdots, y_M)$.

Here, we suppose that there is no interaction between these two species, the wave function factorizes as:

$$\Psi(x_1, x_2, \cdots, x_N; y_1, y_2, \cdots, y_M) = \psi(x_1, x_2, \cdots, x_N)\, \varphi(y_1, y_2, \cdots, y_M) \tag{2.188}$$

In this decoupled limit, the density of the first species is obtained by integrating out the remaining coordinates. To make the cancellation explicit, we use the normalization of φ:

$$\iint \cdots \int \mathrm{d}y_1\, \mathrm{d}y_2 \cdots \mathrm{d}y_M\ |\varphi(y_1, y_2, \cdots, y_M)|^2 = 1 \tag{2.189}$$

We define the particle density associated with ψ as:

$$n_\psi(x) \coloneqq N \iint \cdots \int \mathrm{d}x_2\, \mathrm{d}x_3 \cdots \mathrm{d}x_N\ |\psi(x, x_2, \cdots, x_N)|^2 \tag{2.190}$$

With these definitions, we can express the density of the first species as:

$$n(x) = n_\psi(x) \iint \cdots \int \mathrm{d}y_1\, \mathrm{d}y_2 \cdots \mathrm{d}y_M\ |\varphi(y_1, y_2, \cdots, y_M)|^2 \tag{2.191}$$

Applying the normalization of φ (2.189), therefore, we arrive at:

$$\begin{aligned} n(x) &= n_\psi(x) \\ &= N \iint \cdots \int \mathrm{d}x_2\, \mathrm{d}x_3 \cdots \mathrm{d}x_N\ |\psi(x, x_2, \cdots, x_N)|^2 \end{aligned} \tag{2.192}$$

In general, interactions entangle the two species and the factorization in equation (2.188) is no longer available. The density of one species must then be constructed from the full many-body wave function by integrating out the other degrees of freedom. This illustrates why reduced quantities such as densities are natural objects in many body physics.

Secondly, suppose that we work at fixed nuclear coordinates $\{\vec{R}_I\}$ and obtain the electronic hamiltonian within the Born–Oppenheimer approximation in equation (2.57).

In this situation, the nuclei act as fixed sources of the external potential, and the problem degenerates to an interacting electron system in the prescribed potential

$v_{\text{ext}}(\vec{r})$. The many-body wave function therefore depends only on the electronic coordinates, while the ionic positions enter as parameters.

Therefore, the auxiliary system should be viewed as a simpler description that approximates the target interacting system at the level of selected observables. In our discussion, the density is the central observable, and the auxiliary system is required to reproduce the same density as the original problem.

Moreover, the choice of observables determines how the auxiliary system must be constructed. Suppose that we aim to describe the density, the auxiliary system must have enough flexibility to match the density, but it is not required to reproduce the full many-body wave function.

Accordingly, once an auxiliary system is specified, its predictions should be trusted only for those quantities it is designed to reproduce. For questions outside this design target, the auxiliary system provides no guarantee of accuracy.

Subsequently, we consider an interacting many-electron system and its corresponding auxiliary system.

The difficulty comes from the electron-electron interaction. In the auxiliary system, we drop this interaction and compensate for it by adjusting a one-body potential.

Therefore, the problem can be stated as follows. For a given interacting system with a density $n(\vec{r})$, we seek a noninteracting system with a tuned potential that reproduces exactly the same density. Here we ask only for the density and nothing else. In what follows, we examine whether there is any obstacle to this construction.

Next, let us recall the Born–Oppenheimer hamiltonian in equation (2.57), and write the many-electron Schrödinger equation for the interacting system.

To make the role of $E_{\text{NN}}(\{\vec{R}_I\})$ explicit, we separate the $\hat{H}_{\text{BO}}(\{\vec{R}_I\})$ into an electronic operator $\hat{H}_{\text{el}}(\{\vec{R}_I\})$ and the constant nucleus-nucleus term. Then, we write the eigenvalue equation as:

$$\begin{aligned} \hat{H}_{\text{BO}}(\{\vec{R}_I\})\,|\Psi\rangle &= E_{\text{BO}}(\{\vec{R}_I\})\,|\Psi\rangle \\ &= [\hat{H}_{\text{el}}(\{\vec{R}_I\}) + E_{\text{NN}}(\{\vec{R}_I\})]\,|\Psi\rangle \end{aligned} \tag{2.193}$$

Since $E_{\mathrm{NN}}(\{\vec{R}_I\})$ is a constant with respect to the electronic degrees of freedom, it only shifts the eigenvalue, and equation (2.193) is equivalent to:

$$\hat{H}_{\mathrm{el}}(\{\vec{R}_I\})\,|\Psi\rangle = [E_{\mathrm{BO}}(\{\vec{R}_I\}) - E_{\mathrm{NN}}(\{\vec{R}_I\})]\,|\Psi\rangle \tag{2.194}$$

In what follows, we absorb this constant shift into the eigenvalue and denote the electronic energy by:

$$E \coloneqq E_{\mathrm{BO}}(\{\vec{R}_I\}) - E_{\mathrm{NN}}(\{\vec{R}_I\}) \tag{2.195}$$

with this convention, the Schrödinger equation in the coordinate representation reads:

$$\left(-\frac{1}{2}\sum_i \nabla_i^2 + \sum_i v_{\mathrm{ext}}(\hat{\vec{r}}_i) + \sum_{i<j}\frac{1}{|\vec{r}_i - \vec{r}_j|}\right)\Psi(\vec{r}_1,\vec{r}_2,\cdots,\vec{r}_N) = E\,\Psi(\vec{r}_1,\vec{r}_2,\cdots,\vec{r}_N) \tag{2.196}$$

here, E and Ψ denote the electronic eigenvalue and eigenstate of interest at fixed nuclear coordinates $\{\vec{R}_I\}$, where the constant term $E_{\mathrm{NN}}(\{\vec{R}_I\})$ has been absorbed into E. In particular, the ground-state determines the target density $n(\vec{r})$ through equation (2.68).

Moreover, we introduce a noninteracting auxiliary system by dropping the electron-electron interaction, and compensating for its effect through a tuned one-body potential $v_{\mathrm{aux}}(\vec{r})$:

$$\left(-\frac{1}{2}\sum_i \nabla_i^2 + \sum_i v_{\mathrm{aux}}(\hat{\vec{r}}_i)\right)\Psi_{\mathrm{aux}}(\vec{r}_1,\vec{r}_2,\cdots,\vec{r}_N) = E_{\mathrm{aux}}\Psi_{\mathrm{aux}}(\vec{r}_1,\vec{r}_2,\cdots,\vec{r}_N) \tag{2.197}$$

here, Ψ_{aux} denotes the auxiliary many-body wave function and E_{aux} represents the corresponding eigenvalue.

The density of the auxiliary system is obtained by integrating out $N-1$ coordinates:

$$n_{\mathrm{aux}}(\vec{r}) \coloneqq N \iint\cdots\int \mathrm{d}\vec{r}_2\,\mathrm{d}\vec{r}_3\cdots\mathrm{d}\vec{r}_N\;|\Psi_{\mathrm{aux}}(\vec{r},\vec{r}_2,\cdots,\vec{r}_N)|^2 \tag{2.198}$$

Hence, we define the auxiliary potential $v_{\text{aux}}(\vec{r})$ through the density-matching condition:

$$n_{\text{aux}}(\vec{r}) = n(\vec{r}) \tag{2.199}$$

Consequently, the question becomes whether we can construct a potential $v_{\text{aux}}(\vec{r})$ that gives the same density and, if we can, whether that potential is unique.

Finally, for a noninteracting auxiliary system, the many-body wave function can be constructed from a set of single-particle orbitals. Accordingly, we introduce orbitals $\varphi_l^{\text{aux}}(\vec{r})$ that satisfy the one-body Schrödinger equation:

$$\left[-\frac{\nabla^2}{2} + v_{\text{aux}}(\vec{r})\right] \varphi_l^{\text{aux}}(\vec{r}) = \varepsilon_l^{\text{aux}}\, \varphi_l^{\text{aux}}(\vec{r}) \tag{2.200}$$

The auxiliary density is then expressed in terms of these orbitals as:

$$n_{\text{aux}}(\vec{r}) = \sum_l f_l \, |\varphi_l^{\text{aux}}(\vec{r})|^2 \tag{2.201}$$

here, f_l denotes the occupation number of the orbital φ_l^{aux}.

At this stage, we do not try to reproduce the full many-body wave function of the interacting system; instead, we tune the one-body potential $v_{\text{aux}}(\vec{r})$ in the auxiliary system so that its density matches the target density:

$$n_{\text{aux}}(\vec{r}; v_{\text{aux}}) = n(\vec{r}) \tag{2.202}$$

In other words, we start from a given density $n(\vec{r})$ and ask whether we can find a potential $v_{\text{aux}}(\vec{r})$ that makes a noninteracting system reproduce it.

This is the basic existence question behind the auxiliary construction, and it is often referred to as noninteracting v-representability [30].

2.4.2 The framework of the Kohn–Sham equations

Firstly, we introduce a noninteracting reference system that reproduces the density $n(\vec{r})$, and denote its ground-state wave function by $\Psi_s[n]$.

The wave function $\Psi[n]$ denotes the minimizing interacting many-body wave function that yields the density $n(\vec{r})$, while $\Psi_s[n]$ represents the ground-state of a noninteracting reference system constructed to reproduce the same density.

Accordingly, they share the same one-body density but represent different many-body states:

$$n(\vec{r}) = n_{\Psi}(\vec{r}) = n_{\Psi_s}(\vec{r}) \tag{2.203}$$

In other words, $\Psi_s[n]$ is introduced to match the density, rather than to approximate $\Psi[n]$ at the wave function level.

We define the noninteracting kinetic energy functional as:

$$T_s[n] \coloneqq \langle \Psi_s[n] | \hat{T} | \Psi_s[n] \rangle \tag{2.204}$$

Recall the Slater determinant representation (2.34). For the noninteracting reference state $\Psi_s[n]$, the expectation value of the kinetic energy operator becomes a sum over orbitals:

$$\langle \Psi_s[n] | \hat{T} | \Psi_s[n] \rangle = -\frac{1}{2} \sum_i f_i \, \langle \psi_i | \nabla^2 | \psi_i \rangle \tag{2.205}$$

here, f_i denotes the occupation number of the spin-orbital ψ_i and satisfies:

$$\begin{cases} 0 \leq f_i \leq 1 & \text{Pauli bound} \\ \sum_i f_i = N & \text{particle-number constraint} \end{cases} \tag{2.206}$$

Moreover, we can express the density with f_i:

$$n(\vec{r}) = \sum_i f_i |\psi_i(\vec{r})|^2 \tag{2.207}$$

here, $\psi_i(\vec{r})$ denotes a spin-orbital, and the spin degree of freedom is included implicitly in the index i, and the density $n(\vec{r})$ can be understood as a functional of the set of spin-orbitals $\{\psi_i(\vec{r})\}$.

For comparison, the kinetic energy functional $T[n]$ appearing in the Hohenberg–Kohn functional $E_{\mathrm{HK}}[n]$ is evaluated on the minimizing interacting many-body state $\Psi[n]$ that yields the same density. In general, $T[n]$ and $T_s[n]$ are not equal, and we write their difference as:

$$T[n] = T_s[n] + T_c[n] \tag{2.208}$$

here, $T_c[n]$ measures the kinetic correlation beyond the noninteracting reference. We name it the kinetic-correlation functional.

In equation (2.157), we introduced a Hartree approximation only at the level of the interaction energy by replacing $E_{\mathrm{int}}[n]$ with $E_{\mathrm{H}}[n]$.

To obtain a fully orbital-based reference functional, we further replace the interacting kinetic energy functional $T[n]$ by $T_s[n]$, which amounts to neglecting the kinetic correlation $T_c[n]$ in equation (2.208).

Therefore, the Hartree approximation to the total energy functional reads:

$$E_{\mathrm{H}}^{\mathrm{tot}}[n] = T_s[n] + \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})n(\vec{r}) + E_{\mathrm{H}}[n] \tag{2.209}$$

Secondly, to go beyond the Hartree approximation, we collect the missing contributions into an exchange-correlation functional.

As shown in equation (2.179), the interaction energy functional can be written as a Hartree term plus a remainder $E_{\mathrm{xc}}^{\mathrm{int}}[n]$.

In the Kohn–Sham construction, there is an additional correction coming from the kinetic energy, since $T[n]$ and $T_s[n]$ are not equal, and their difference is measured by $T_c[n]$ in equation (2.208).

Therefore, we define the Kohn–Sham exchange-correlation functional as:

$$E_{\mathrm{xc}}[n] \coloneqq T_c[n] + E_{\mathrm{xc}}^{\mathrm{int}}[n] \tag{2.210}$$

According to the definition of universal Hohenberg–Kohn functional (2.142), and the exchange-correlation part of the interaction energy functional (2.179), we have:

$$\begin{aligned} E_{\mathrm{xc}}[n] &= (T[n] - T_s[n]) + (E_{\mathrm{int}} - E_{\mathrm{H}}[n]) \\ &= (T[n] - T_s[n]) + (E_{\mathrm{HK}} - T[n] - E_{\mathrm{H}}[n]) \\ &= E_{\mathrm{HK}}[n] - T_s[n] - E_{\mathrm{H}}[n] \end{aligned} \tag{2.211}$$

Let us perform some manipulations on exact ground-state energy functional (2.139):

$$\begin{aligned} E[n] &= T[n] + E_{\mathrm{int}}[n] + \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, n(\vec{r}) \\ &= E_{\mathrm{HK}}[n] + \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, n(\vec{r}) + T_s[n] - T_s[n] + E_{\mathrm{H}}[n] - E_{\mathrm{H}}[n] \\ &= T_s[n] + \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, n(\vec{r}) + E_{\mathrm{H}}[n] + (E_{\mathrm{HK}}[n] - T_s[n] - E_{\mathrm{H}}[n]) \end{aligned} \tag{2.212}$$

Recall the Kohn–Sham exchange-correlation functional (2.211), and the Hartree approximation to the total energy functional (2.209), we obtain:

$$\begin{aligned} E[n] &= T_s[n] + \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, n(\vec{r}) + E_{\mathrm{H}}[n] + E_{\mathrm{xc}}[n] \\ &= T_s[n] + \int \mathrm{d}\vec{r}\, v_{\mathrm{ext}}(\vec{r})\, n(\vec{r}) + \frac{1}{2} \iint \mathrm{d}\vec{r}\, \mathrm{d}\vec{r}\,' \, \frac{n(\vec{r})n(\vec{r}\,')}{|\vec{r} - \vec{r}\,'|} + E_{\mathrm{xc}}[n] \\ &= E_{\mathrm{H}}^{\mathrm{tot}}[n] + E_{\mathrm{xc}}[n] \end{aligned} \tag{2.213}$$

Subsequently, we minimize the energy functional $E[n]$ with respect to the spin-orbital wave functions $\psi_i(\vec{r})$. This variational procedure determines the stationary condition of the functional, while enforcing the orthonormality constraints:

Accordingly, we introduce a set of Lagrange multipliers $\{\lambda_{ij}\}$ and consider the constrained variation. Varying with respect to $\psi_i^*(\vec{r})$ generates a one-body eigenvalue problem.

Moreover, the functional derivative of the Hartree term produces the Hartree potential $v_{\mathrm{H}}([n]; \vec{r})$ defined in equation (2.165), while the exchange-correlation contribution defines the exchange-correlation potential as its functional derivative:

$$v_{\mathrm{xc}}([n]; \vec{r}) \coloneqq \frac{\delta E_{\mathrm{xc}}[n]}{\delta n(\vec{r})} \tag{2.214}$$

It is therefore convenient to introduce the effective Kohn–Sham potential:

$$v_{\text{eff}}([n];\vec{r}) \coloneqq v_{\text{ext}}(\vec{r}) + v_{\text{H}}([n];\vec{r}) + v_{\text{xc}}([n];\vec{r}) \tag{2.215}$$

here, the corresponding one-body operator is $\hat{v}_{\text{eff}} \coloneqq v_{\text{eff}}([n];\hat{\vec{r}})$.

Next, in a basis where the Lagrange-multiplier matrix is diagonal, the stationarity condition can be expressed as the Kohn–Sham equations [18]:

$$\left[-\frac{1}{2}\nabla^2 + v_{\text{eff}}([n];\vec{r})\right]\psi_i(\vec{r}) = \lambda_i\,\psi_i(\vec{r}) \tag{2.216}$$

here, the orbitals are determined by the effective potential $v_{\text{eff}}([n];\vec{r})$, therefore, at self-consistency, we may write:

$$\psi_i(\vec{r}) = \psi_i([n];\vec{r}) \tag{2.217}$$

Since the Kohn–Sham density expression (2.207) gives $n(\vec{r})$ once the orbitals are known. Meanwhile, $v_{\text{eff}}([n];\vec{r})$ depends on $n(\vec{r})$ through equation (2.215). Therefore, equations (2.215) and (2.216) are solved self-consistently.

In this sense, density functional theory is built on the Hohenberg–Kohn theorem. It states that the ground-state density uniquely determines the external potential modulo an additive constant [8, 31]. Meanwhile, the density and the external potential are in bijective correspondence:

$$n(\vec{r}) \longleftrightarrow [v_{\text{ext}}(\vec{r})] \tag{2.218}$$

In the Kohn–Sham auxiliary system, the electrons are noninteracting. Therefore, the only potential entering the one-body hamiltonian is the effective potential $v_{\text{eff}}([n];\vec{r})$. It plays the role of the external potential for the auxiliary system.

Consequently, we may also view the Kohn–Sham potential as being uniquely linked to the density:

$$n(\vec{r}) \longleftrightarrow [v_{\text{eff}}([n];\vec{r})] \tag{2.219}$$

here, $[v(\vec{r})]$ denotes the equivalence class of potentials that differ only by an additive constant.

This shows that once the density $n(\vec{r})$ is given, the effective potential $v_{\text{eff}}([n];\vec{r})$ is fixed up to an additive constant.

Therefore, in this noninteracting Kohn–Sham auxiliary system, we process the corresponding total ground-state energy as a functional of the density, and denote it by $E_s[n]$.

For the auxiliary noninteracting system, we achieve the Kohn–Sham auxiliary energy functional:

$$E_s[n] = T_s[n] + \int \mathrm{d}\vec{r}\, n(\vec{r}) v_{\text{eff}}([n];\vec{r}) \tag{2.220}$$

Moreover, we recall the exact energy decomposition in equation (2.139), together with the Hohenberg–Kohn correspondence in equation (2.133). They tell us that the ground-state energy can be viewed as a functional of the density.

Hence, we introduce the key idea of Kohn–Sham equations. For a given interacting system, we assume the existence of an auxiliary system of noninteracting electrons. This auxiliary system is constructed to reproduce the same density:

$$n_s(\vec{r}) = n(\vec{r}) \tag{2.221}$$

here, $n_s(\vec{r})$ denotes the ground-state density of the auxiliary noninteracting system, it is obtained from a set of one-electron orbitals. The corresponding one-body potential is denoted by the Kohn–Sham potential $v_{\text{eff}}([n];\vec{r})$. It is chosen such that equation (2.221) holds.

Consequently, the interacting many-electron problem is mapped to a noninteracting orbital problem with the same density.

2.4.3 From variational derivation to the Kohn–Sham equations

To determine the orbitals in the Kohn–Sham scheme, we start from the density built from orbitals in equation (2.207). We are going to apply the variational principle and minimize the total energy with respect to the orbitals.

Firstly, we introduce the one-body kinetic operator in the single-particle Hilbert space

$$\hat{K} \coloneqq -\frac{1}{2}\nabla^2 \tag{2.222}$$

Accordingly, we represent the noninteracting kinetic energy as an orbital functional:

$$T_s[\{\psi_i(\vec{r})\}] = \sum_i f_i \langle\psi_i| \hat{K} |\psi_i\rangle \tag{2.223}$$

Next, we define the one-body external potential operator:

$$\hat{v}_{\text{ext}} \coloneqq v_{\text{ext}}(\hat{\vec{r}}) \tag{2.224}$$

In coordinate representation, the corresponding orbital expectation value reads:

$$\langle\psi_i| \hat{v}_{\text{ext}} |\psi_i\rangle = \int \mathrm{d}\vec{r}\, \psi_i^*(\vec{r}) v_{\text{ext}}(\vec{r}) \psi_i(\vec{r}) \tag{2.225}$$

Afterwards, using the Kohn–Sham density expression in equation (2.207), we express the external potential energy in the orbital form:

$$\int \mathrm{d}\vec{r}\, n(\vec{r}) v_{\text{ext}}(\vec{r}) = \sum_i f_i \langle\psi_i| \hat{v}_{\text{ext}} |\psi_i\rangle \tag{2.226}$$

Therefore, we view the total energy as an orbital functional through the density $n(\vec{r})$:

$$E[\{\psi_i(\vec{r})\}] \coloneqq T_s[\{\psi_i(\vec{r})\}] + \int \mathrm{d}\vec{r}\, n(\vec{r}) v_{\text{ext}}(\vec{r}) + E_{\text{H}}[n] + E_{\text{xc}}[n] \tag{2.227}$$

here, we reiterate that the density $n(\vec{r})$ is understood as a functional of the set of spin-orbitals $\{\psi_i(\vec{r})\}$ through equation (2.207).

Secondly, we impose the orthonormality constraints:

$$\langle\psi_i|\psi_j\rangle = \delta_{ij} \tag{2.228}$$

Accordingly, we introduce a matrix of Lagrange multipliers λ_{ij} and define this constrained functional:

$$\mathcal{F}[\{\psi_i(\vec{r})\}] \coloneqq E[\{\psi_i(\vec{r})\}] - \sum_{i,j} \lambda_{ij} \left(\langle\psi_i|\psi_j\rangle - \delta_{ij}\right) \tag{2.229}$$

Thirdly, we require stationarity with respect to the orbitals:

$$\frac{\delta\mathcal{F}[\{\psi_i(\vec{r})\}]}{\delta\langle\psi_i|} = 0 \tag{2.230}$$

here, we vary with respect to $\langle\psi_i|$, treating $|\psi_i\rangle$ and $\langle\psi_i|$ as independent variables.

Moreover, we use the density dependence on the orbitals. In coordinate representation, equation (2.207) shows:

$$\frac{\delta n(\vec{r})}{\delta\psi_i^*(\vec{r})} = f_i\,\psi_i(\vec{r}) \tag{2.231}$$

Next, we collect the potential terms including the Hartree potential $v_{\mathrm{H}}([n];\vec{r})$ in equation (2.165), and the exchange-correlation potential $v_{\mathrm{xc}}([n];\vec{r})$ in equation (2.214). Together with the external potential, they form the Kohn–Sham effective potential $v_{\mathrm{eff}}([n];\vec{r})$ in equation (2.215).

Accordingly, we combine the kinetic term and the effective potential into a single one-body operator. This allows us to write the Kohn–Sham equations in a compact form. Therefore, we define the Kohn–Sham one-body operator as:

$$\hat{h}_{\mathrm{KS}} \coloneqq \hat{K} + \hat{v}_{\mathrm{eff}} \tag{2.232}$$

Finally, from the stationarity condition in equation (2.230), we obtain a set of one-particle equations:

$$\hat{h}_{\mathrm{KS}}\,|\psi_i\rangle = \sum_j \lambda_{ij}\,|\psi_j\rangle \tag{2.233}$$

Accordingly, we take λ_{ij} to be Hermitian and diagonalize it in a chosen basis. Equation (2.233) then reduces to the standard eigenvalue form:

$$\hat{h}_{\mathrm{KS}} |\psi_i\rangle = \lambda_i |\psi_i\rangle \tag{2.234}$$

In coordinate representation, using equation (2.234), we get the Kohn–Sham equations (2.216). For ease of understanding and calculation, we expand the Kohn–Sham equation (2.216) as:

$$\begin{aligned} \lambda_i \, \psi_i(\vec{r}) &= \left[-\frac{1}{2}\nabla^2 + v_{\mathrm{ext}}(\vec{r}) + v_{\mathrm{H}}([n];\vec{r}) + v_{\mathrm{xc}}([n];\vec{r}) \right] \psi_i(\vec{r}) \\ &= \left[-\frac{1}{2}\nabla^2 + v_{\mathrm{ext}}(\vec{r}) + \int \mathrm{d}\vec{r}\,' \, \frac{n(\vec{r}\,')}{|\vec{r} - \vec{r}\,'|} + \frac{\delta E_{\mathrm{xc}}[n]}{\delta n(\vec{r})} \right] \psi_i(\vec{r}) \end{aligned} \tag{2.235}$$

Consequently, we have obtained the one-particle equations for the auxiliary noninteracting system, which means each electron moves in the effective potential $v_{\mathrm{eff}}([n];\vec{r})$.

Together with equation (2.207), we can solve the Kohn–Sham equations (2.235), and achieve the exact ground-state density of the interacting system.

2.4.4 Variational construction of the Kohn–Sham scheme

The previous subsection derived the Kohn–Sham equations by varying the orbitals directly. Here we complement that derivation by showing, at the density-functional level, why the auxiliary noninteracting construction is variationally consistent.

We start from the interacting ground-state density $n_0(\vec{r})$, which makes the exact energy functional stationary.

However, we introduce an auxiliary noninteracting energy functional and require that it is minimized by $n_0(\vec{r})$. This gives an auxiliary independent-particle system with the same ground-state density. Then an effective potential exists, and the Kohn–Sham equations follow.

Let us start from the Kohn–Sham auxiliary energy functional in equation (2.220). We search its minimum among admissible densities, and impose the corresponding positivity and normalization conditions:

$$\begin{cases} n(\vec{r}) \geq 0 \\ \int \mathrm{d}\vec{r}\, n(\vec{r}) = N \end{cases} \tag{2.236}$$

Accordingly, we impose the normalization constraint by a Lagrange multiplier λ and require stationarity:

$$\frac{\delta}{\delta n(\vec{r})} \left[E_s[n] - \lambda \int \mathrm{d}\vec{r}\,' \, n(\vec{r}\,') \right] = 0 \tag{2.237}$$

Therefore, at the minimizing density $n_0(\vec{r})$, we obtain the Euler–Lagrange condition for the auxiliary noninteracting energy functional $E_s[n]$:

$$\left. \frac{\delta T_s[n]}{\delta n(\vec{r})} \right|_{n_0} + v_{\text{eff}}([n_0]; \vec{r}) = \lambda \tag{2.238}$$

here, λ is treated as a constant.

Therefore, we equivalently express the Euler–Lagrange condition as an explicit expression for the effective potential $v_{\text{eff}}([n_0]; \vec{r})$:

$$v_{\text{eff}}([n_0]; \vec{r}) = \lambda - \left. \frac{\delta T_s[n]}{\delta n(\vec{r})} \right|_{n_0} = - \left. \frac{\delta T_s[n]}{\delta n(\vec{r})} \right|_{n_0} + \text{const.} \tag{2.239}$$

Consequently, if the functional derivative of the noninteracting kinetic energy functional $T_s[n]$ is well behaved around the ground-state density $n_0(\vec{r})$, the Kohn–Sham effective potential $v_{\text{eff}}(\vec{r})$ can be constructed to reproduce $n_0(\vec{r})$. In this sense, the auxiliary noninteracting system idea is reasonable.

In practice, the Kohn–Sham equations give the exact electron density in principle [32]. The only missing piece is the exchange-correlation energy functional $E_{\text{xc}}[n]$, or equivalently the exchange-correlation potential $v_{\text{xc}}([n]; \vec{r})$. We will introduce approximations for it later.

2.4.5 Remarks on the Kohn–Sham solutions

In practical applications, we have to solve the Kohn–Sham equations (2.235). Once the Kohn–Sham one-body operator is specified, we diagonalize it and obtain the eigenvalues λ_i and eigenvectors $|\psi_i\rangle$. We then construct the electron density through equation (2.207).

However, the density $n(\vec{r})$ is both an output and an input. To build the Kohn–Sham operator, we need the Hartree potential $v_{\mathrm{H}}([n];\vec{r})$, and the exchange-correlation potential $v_{\mathrm{xc}}([n];\vec{r})$. Both of them depend on the density. Therefore, the Kohn–Sham equations are solved self-consistently.

Consequently, we utilize an iterative self-consistent field scheme. Let us start from a trial density. We construct the potentials and diagonalize the Kohn–Sham operator. This generates updated orbitals and a new density. We repeat these steps until the density is converged. This idea goes back to Hartree and his self-consistent treatment [33].

Next, we comment on the Kohn–Sham eigenvalues λ_i. They have the dimension of energy, but belong to the auxiliary noninteracting system. Therefore, in general, they are not electron removal or addition energies, which are defined by total-energy differences.

There is an important exception for finite systems. For the exact functional, the highest occupied Kohn–Sham eigenvalue equals the negative ionization energy [34, 35]. For extended systems, similar interpretations have been discussed [36], but no general proof is available in the same simple form.

In fact, we work with approximate functionals. The resulting eigenvalues are widely used in materials science, for instance as a useful reference for band structure analysis. We will return to this point later.

Afterwards, we look into the Kohn–Sham potential. It consists of the external potential, the Hartree potential, and the exchange-correlation potential:

$$v_{\mathrm{KS}}([n];\vec{r}) = v_{\mathrm{ext}}(\vec{r}) + v_{\mathrm{H}}([n];\vec{r}) + v_{\mathrm{xc}}([n];\vec{r}) \tag{2.240}$$

here, the external potential $v_{\text{ext}}(\vec{r})$ is fixed by the ionic structure. The Hartree potential $v_{\text{H}}([n];\vec{r})$ is a known functional of the density. The main difficulty lies in the exchange-correlation potential $v_{\text{xc}}([n];\vec{r})$.

Moreover, the exchange-correlation potential $v_{\text{xc}}([n];\vec{r})$ is a static and local potential. In coordinate representation, it acts as a multiplicative operator. Nevertheless, its functional dependence on the density can be highly nontrivial. Therefore, practical density functional theory relies on approximations for the exchange-correlation energy functional $E_{\text{xc}}[n]$, or equivalently for $v_{\text{xc}}([n];\vec{r})$. We will discuss these approximations later.

2.4.6 Structure and construction of the Kohn–Sham potential

After the general framework of the Kohn–Sham construction has been established through equations (2.207), (2.235), and (2.240), it allows us to examine several simple exactly solvable model systems. These examples demonstrate the exact form that the Kohn–Sham potential can take. They also elucidate why its exchange-correlation component can remain highly nontrivial even though it is still a local multiplicative potential.

Firstly, let us consider a one-dimensional harmonic well containing two interacting electrons. In atomic units, the interacting problem is defined by the Schrödinger equation:

$$\hat{H}\Psi(x_1, x_2) = E\Psi(x_1, x_2) \tag{2.241}$$

with the many-body hamiltonian:

$$\hat{H} = \sum_{i=1}^{2}\left[-\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}x_i^2} + v_{\text{ext}}(x_i)\right] + v_{\text{ee}}(x_1, x_2) \tag{2.242}$$

here, $v_{\text{ee}}(x_1, x_2)$ denotes the two-body interaction, $v_{\text{ee}}(x_1, x_2) = |x_1 - x_2|^{-1}$. For a harmonic confinement, the external potential takes the form:

$$v_{\text{ext}}(x) = \frac{1}{2}\Omega^2 x^2 \tag{2.243}$$

here, Ω refers to the angular frequency of harmonic confinement.

By solving the Schrödinger equation (2.241), we get the exact density:

$$n(x) = 2 \int \mathrm{d}x_2 \, |\Psi(x, x_2)|^2 \tag{2.244}$$

Secondly, we inquire which local Kohn–Sham potential reproduces the exact density given in equation (2.244).

In one dimension, equations (2.235) and (2.207) may be written schematically as:

$$\left[-\frac{1}{2} \frac{\mathrm{d}^2}{\mathrm{d}x^2} + v_{\mathrm{KS}}([n]; x) \right] \psi_i(x) = \lambda_i \psi_i(x) \tag{2.245}$$

We denote the corresponding density function as:

$$n_{\mathrm{KS}}(x) = \sum_i f_i |\psi_i(x)|^2 \tag{2.246}$$

The Kohn–Sham auxiliary system is designed to reproduce exactly the same ground-state electron density as the true interacting system:

$$n_{\mathrm{KS}}(x) = n(x) \tag{2.247}$$

If one starts from the naive choice:

$$v_{\mathrm{KS}}([n]; x) = v_{\mathrm{ext}}(x) \tag{2.248}$$

Therefore, the density from equation (2.246) is concentrated near the center of the well. This concentration is generally excessive. Therefore, the missing interaction effects are encoded directly in the local Kohn–Sham potential. In the exact solution, this potential develops a central bump [37]. To isolate the interaction-induced contribution, we define the Hartree-exchange-correlation potential:

$$v_{\mathrm{Hxc}}([n]; x) \coloneqq v_{\mathrm{KS}}([n]; x) - v_{\mathrm{ext}}(x) \tag{2.249}$$

Accordingly, we can process this central bump as a feature of the Hartree-exchange–correlation potential $v_{\mathrm{Hxc}}([n]; x)$. It works to suppress charge accumulation near the

origin and thereby enforce equation (2.247). Therefore, the exact Kohn–Sham potential $v_{\mathrm{KS}}([n];x)$ is not merely a shifted harmonic well.

Next, an asymmetric double-well potential provides an even more instructive example. In this interacting system, the electron-electron repulsion may force the two electrons to occupy different wells. By contrast, in our auxiliary noninteracting system, both electrons would tend to occupy the energetically more favorable well.

Therefore, the exact Kohn–Sham potential $v_{\mathrm{KS}}([n];x)$ must develop an additional structure that redistributes the charge correctly. Schematically, it demonstrates as:

$$v_{\mathrm{KS}}([n];x) \approx \begin{cases} v_L(x) & x \text{ in the left well} \\ v_R(x) + C & x \text{ in the right well} \end{cases} \tag{2.250}$$

here, the approximately constant shift C denotes the offset between the two spatial regions. This step raises one well relative to the other and thereby enforces the correct density distribution [38].

Subsequently, from the Kohn–Sham potential decomposition in equation (2.240), the external potential has already been fixed, and the Hartree potential $v_{\mathrm{H}}([n];x)$ has already been defined in equation (2.165). Therefore, the remaining freedom lies mainly in the exchange-correlation potential $v_{\mathrm{xc}}([n];x)$ defined in equation (2.214). Hence, the exact exchange-correlation potential $v_{\mathrm{xc}}([n];x)$ may contain bumps, barriers, and steps [39]. In particular, a pronounced step may appear in a low-density region. This makes it difficult to construct accurate approximations to the exchange-correlation potential $v_{\mathrm{xc}}([n];x)$.

Consequently, we turn to the inverse point of view. If an accurate density is available from an exact method, we seek a local potential that reproduces it. This defines the inverse Kohn–Sham problem [40, 41].

Let us denote the target density by $n_{\mathrm{tar}}(x)$. We require:

$$n_{\mathrm{KS}}(x) = n_{\mathrm{tar}}(x) \tag{2.251}$$

and we take the zeroth-step Kohn–Sham potential as:

$$v_{\mathrm{KS}}^{(0)}(x) = v_{\mathrm{ext}}(x) \tag{2.252}$$

We then solve the Kohn–Sham equation, equation (2.245), and obtain the density at the kth iteration:

$$n^{(k)}(x) = \sum_i f_i |\psi_i^{(k)}(x)|^2 \tag{2.253}$$

here, f_i denotes the occupation number of the ith Kohn–Sham orbital $\psi_i^{(k)}(x)$. We update the Kohn–Sham potential according to the density mismatch. A simple schematic iteration is:

$$v_{\mathrm{KS}}^{(k+1)}(x) = v_{\mathrm{KS}}^{(k)}(x) + \mu \left[\left(n^{(k)}(x) \right)^p - \left(n_{\mathrm{tar}}(x) \right)^p \right] \tag{2.254}$$

here, μ denotes a mixing parameter and p denotes an additional stability parameter.

Accordingly, the Kohn–Sham potential is increased where the density is overestimated and decreased where it is underestimated. At convergence, we arrive at:

$$\begin{cases} n^{(k)}(x) \to n_{\mathrm{tar}}(x) \\ v_{\mathrm{KS}}^{(k)}(x) \to v_{\mathrm{KS}}([n_{\mathrm{tar}}]; x) \end{cases} \qquad \text{as } k \to \infty \tag{2.255}$$

2.4.7 The Hartree equations

Before turning to practical approximations to E_{xc}, it is useful to consider the simpler self-consistent scheme that remains when exchange and correlation are omitted. If we retain the external potential $v_{\mathrm{ext}}(\vec{r})$ and the classical electrostatic interaction generated by the density $n(\vec{r})$, then we arrive at the Hartree equations:

$$\left[-\frac{1}{2}\nabla^2 + v_{\mathrm{ext}}(\vec{r}) + v_{\mathrm{H}}([n]; \vec{r}) \right] \psi_i(\vec{r}) = \lambda_i \psi_i(\vec{r}) \tag{2.256}$$

here, λ_i denotes the Hartree eigenvalue associated with the one-particle orbital $\psi_i(\vec{r})$. The Hartree potential $v_{\mathrm{H}}([n]; \vec{r})$, introduced in equation (2.165), is the classical electrostatic potential generated by the electron density $n(\vec{r})$. Since the density is still constructed from the occupied one-particle orbitals through equation (2.207), the Hartree equations (2.256) must be solved self-consistently.

Next, the Hartree equations already demonstrate the basic one-particle structure of a self-consistent field theory. We start from a trial density, construct the Hartree potential

$v_{\mathrm{H}}([n];\vec{r})$, solve equation (2.256), update the density through equation (2.207), and repeat these steps until convergence is reached. In this sense, the Hartree equations already anticipate the later structure of the Kohn–Sham equations (2.235).

However, the Hartree equations remain incomplete. The Hartree potential $v_{\mathrm{H}}([n];\vec{r})$ contains only the classical electrostatic contribution. It does not incorporate exchange effects or correlation effects. Moreover, since $v_{\mathrm{H}}([n];\vec{r})$ is generated by the total density $n(\vec{r})$, it also contains an unphysical self-interaction contribution [42]. Therefore, the Hartree equations do not yet provide an accurate description of interacting electrons.

Consequently, one must go beyond the Hartree approximation. In the Kohn–Sham framework, this missing physics is collected into the exchange-correlation potential $v_{\mathrm{xc}}([n];\vec{r})$ appearing in equation (2.240). Equivalently, it is described by the exchange-correlation energy functional $E_{\mathrm{xc}}[n]$. As we have already emphasized in the previous subsection, the main difficulty does not lie in the external potential $v_{\mathrm{ext}}(\vec{r})$ or the Hartree potential $v_{\mathrm{H}}([n];\vec{r})$, but in the construction of the exchange-correlation functional itself. Therefore, the next step is to introduce practical approximations for $E_{\mathrm{xc}}[n]$ and $v_{\mathrm{xc}}([n];\vec{r})$.

2.5 Exchange-correlation functional

As we have already discussed in equation (2.214), the exchange-correlation potential $v_{\mathrm{xc}}([n];\vec{r})$ is obtained from the exchange-correlation energy functional $E_{\mathrm{xc}}[n]$. Moreover, from equation (2.240), we have already seen that the external potential $v_{\mathrm{ext}}(\vec{r})$ is fixed by the ionic structure, and the Hartree potential $v_{\mathrm{H}}([n];\vec{r})$ is a known functional of the density. Therefore, in practical density functional theory, the main difficulty lies in the construction of the exchange-correlation functional itself.

Thereafter, we introduce several widely used approximations for the exchange-correlation energy functional $E_{\mathrm{xc}}[n]$ [43]. They differ in how much information about the density is retained. The simplest approximation depends only on the local value of the density. More accurate semilocal approximations also include the density gradient. Beyond them, hybrid functionals incorporate a fraction of exact exchange.

2.5.1 Local density and generalized gradient approximations

Firstly, we consider the local density approximation. Its basic idea is to approximate the exchange-correlation energy locally by that of a homogeneous electron gas with the same density. Accordingly, the exchange-correlation functional is written as:

$$E_{\mathrm{xc}}^{\mathrm{LDA}}[n] = \int \mathrm{d}\vec{r}\, n(\vec{r})\, \varepsilon_{\mathrm{xc}}^{\mathrm{HEG}}(n(\vec{r})) \tag{2.257}$$

here, $\varepsilon_{\mathrm{xc}}^{\mathrm{HEG}}(n)$ refers to the exchange-correlation energy per electron of the homogeneous electron gas at density n [27, 44].

The local density approximation is simple and computationally efficient [7, 45]. For many bulk solids, it already gives a reasonable description. However, it neglects the inhomogeneity of the density. Therefore, it may become insufficient in systems where the density varies strongly in space, for instance in molecules, surfaces, and low-dimensional systems [43, 46].

Next, to improve upon the local density approximation, we include the gradient of the density. This leads to the generalized gradient approximation. In this case, the exchange-correlation functional depends on both the density and its gradient:

$$E_{\mathrm{xc}}^{\mathrm{GGA}}[n] = \int \mathrm{d}\vec{r}\, f(n(\vec{r}), \nabla n(\vec{r})) \tag{2.258}$$

here, $f(n(\vec{r}), \nabla n(\vec{r}))$ refers to a semilocal exchange-correlation energy density.

In practical treatments of the exchange part, the reduced density gradient takes the form:

$$\begin{cases} s(\vec{r}) \coloneqq \dfrac{|\nabla n(\vec{r})|}{2k_{\mathrm{F}}(\vec{r})n(\vec{r})} \\ k_{\mathrm{F}}(\vec{r}) \coloneqq \left(3\pi^2 n(\vec{r})\right)^{1/3} \end{cases} \tag{2.259}$$

here, $k_{\mathrm{F}}(\vec{r})$ refers to the local Fermi wave vector corresponding to the density $n(\vec{r})$, and $s(\vec{r})$ measures the local inhomogeneity of the density.

Accordingly, the exchange functional in a generalized gradient approximation combines the local-density exchange energy per electron with the exchange enhancement factor:

$$E_x^{\mathrm{GGA}}[n] = \int \mathrm{d}\vec{r}\, n(\vec{r})\, \varepsilon_x^{\mathrm{LDA}}(n(\vec{r}))\, F_x(s(\vec{r})) \tag{2.260}$$

here, $\varepsilon_x^{\mathrm{LDA}}(n)$ denotes the local-density exchange energy per electron, and $F_x(s)$ denotes the exchange enhancement factor. The factor $F_x(s)$ incorporates density-gradient effects into the exchange energy.

Subsequently, different generalized gradient approximations choose different exchange enhancement factors $F_x(s)$ and different correlation functionals. Prominent examples include the Perdew–Wang functional and the Perdew–Burke–Ernzerhof functional. These generalized gradient approximations satisfy several known exact constraints on the exchange-correlation energy [47, 48]. Among them, the Perdew–Burke–Ernzerhof functional remains one of the most widely used generalized gradient approximations in first-principles calculations of condensed-matter systems.

Compared with the local density approximation, generalized gradient approximations usually improve equilibrium geometries, atomization energies, and surface properties. However, they still tend to overestimate lattice constants slightly and underestimate band gaps. Meanwhile, their balance between computational efficiency and accuracy keeps generalized gradient approximations in wide use in practical calculations.

2.5.2 Hybrid functionals and the Heyd–Scuseria–Ernzerhof functional

In addition to local and semilocal approximations, hybrid functionals incorporate a fraction of exact exchange into the exchange-correlation functional. The exact Hartree–Fock exchange term partially reduces the self-interaction error and often yields more accurate band gaps, reaction barriers, and localized electronic states.

However, in periodic solids, the direct use of long-range Hartree–Fock exchange is computationally expensive. Therefore, screened hybrid functionals are introduced. Here, a widely used example is the Heyd–Scuseria–Ernzerhof functional [49].

In the Heyd–Scuseria–Ernzerhof construction, the Coulomb interaction is separated into short-range and long-range parts as follows:

$$\frac{1}{|\vec{r}-\vec{r}\,'|} = \frac{\operatorname{erfc}(\omega|\vec{r}-\vec{r}\,'|)}{|\vec{r}-\vec{r}\,'|} + \frac{\operatorname{erf}(\omega|\vec{r}-\vec{r}\,'|)}{|\vec{r}-\vec{r}\,'|} \tag{2.261}$$

here, ω refers to the screening parameter that controls the range separation.

With this range separation, the Heyd–Scuseria–Ernzerhof scheme constructs the exchange-correlation functional as:

$$E_{\mathrm{xc}}^{\mathrm{HSE}}[n] = aE_x^{\mathrm{HF,SR}}[n;\omega] + (1-a)E_x^{\mathrm{PBE,SR}}[n;\omega] + E_x^{\mathrm{PBE,LR}}[n;\omega] + E_c^{\mathrm{PBE}}[n] \tag{2.262}$$

here, $E_x^{\mathrm{HF,SR}}[n;\omega]$ refers to the short-range Hartree–Fock exchange energy,

$E_x^{\mathrm{PBE,SR}}[n;\omega]$ and $E_x^{\mathrm{PBE,LR}}[n;\omega]$ refer to the short-range and long-range parts of the PBE exchange functional, and $E_c^{\mathrm{PBE}}[n]$ refers to the PBE correlation functional. Moreover, a refers to the mixing parameter that determines the fraction of exact exchange.

In the standard HSE06 functional, one usually takes $a = 0.25$. The screening parameter is typically chosen as $\omega = 0.11\,\mathrm{bohr}^{-1}$ [50]. Therefore, only the short-range part of the exchange interaction contains exact Hartree–Fock exchange, while the long-range part is still described by a semilocal approximation.

By restricting exact exchange to the short-range part, the Heyd–Scuseria–Ernzerhof functional lowers the computational cost significantly relative to global hybrid functionals. Meanwhile, its screened exact exchange often yields more accurate band

gaps, more reasonable defect-level positions, and a more reliable description of localized states in semiconductors and insulators [50, 51]. These advantages make the Heyd–Scuseria–Ernzerhof functional one of the most widely used hybrid functionals in first-principles calculations of condensed-matter systems.

Consequently, the local density approximation, the generalized gradient approximation, and hybrid functionals such as the Heyd–Scuseria–Ernzerhof functional form a practical hierarchy of exchange-correlation approximations in density functional theory. The local density approximation gives the simplest local description of exchange and correlation. The generalized gradient approximation includes the density gradient and captures density inhomogeneity. Hybrid functionals incorporate a fraction of exact exchange. Therefore, for the density functional theory framework used in the present work, we arrive at the essential exchange-correlation approximations required in practical first-principles calculations.

Chapter 3

Fundamentals II: Linear Optical Response

We continue our discussion of the fundamentals in this chapter, where we introduce the linear optical response of matter. The interaction between materials and light occurs through the response of electrons to the electric-field component of incident radiation. This response induces polarization and produces electronic excitations, including interband transitions and plasmonic charge excitations. Collectively, these processes determine the dielectric function, plasmonic behaviour, absorption, reflection, refraction, and energy loss. We therefore use linear response theory to connect the electromagnetic field to the dielectric response of the medium.

The discussion follows the route used in the optical analysis of this thesis. We first establish the electrodynamic variables, then formulate the dielectric response, and finally derive the optical observables. In section 3.1, we begin with Maxwell equations in matter and introduce the polarization field $\vec{P}(\vec{r},t)$ and electric displacement field $\vec{D}(\vec{r},t)$. The constitutive relation closes the field equations and leads to the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$. Section 3.2 then describes the complex dielectric function $\varepsilon(\omega) = \varepsilon_1(\omega)+i\varepsilon_2(\omega)$ and its directional form in anisotropic media. Causality constrains the real and imaginary parts of this response, and section 3.3 expresses this constraint through the Kramers–Kronig relations. Finally, section 3.4 derives the absorption coefficient $\alpha(\omega)$, refractive index $n(\omega)$, extinction coefficient $k(\omega)$, reflectivity $R(\omega)$, and energy-loss spectrum $L(\omega)$ from the complex dielectric function.

3.1 Toward electromagnetic response in matter

The previous chapter focused on the ground-state electronic problem. There, the central question was how electrons arrange themselves under a given external potential. However, optical properties do not follow from the ground-state alone. When an electromagnetic field is applied, it drives the electrons away from their static equilibrium and triggers a series of responses.

Therefore, before introducing the dielectric function, we need a language that describes both the applied field and the feedback of matter to that field. That language is electrodynamics. The present section does not aim to review a complete lecture of electrodynamics; instead, we only extract the ingredients required for the later discussion of the dielectric function, the Kramers–Kronig relations, and the optical properties [52, 53]. We begin with electromagnetic fields in matter, and then gradually move toward a frequency-dependent response function that can be connected to the electronic structure.

In the present section, we mainly use SI units for the macroscopic electrodynamic equations. This choice keeps the structure of the Maxwell equations explicit.

3.1.1 Maxwell equations in matter

Let us start by the universal structure of the problem. When an electromagnetic field enters a medium, it drives charge redistribution and induces currents inside the medium, while these induced sources further modify the field itself. Therefore, the optical response of matter should be described within the framework of electrodynamics [52, 53].

At this stage, we do not explore how a specific medium responds; instead, we first identify the general field equations that govern electromagnetic dynamics in matter. They provide the universal backbone of the problem, while the constitutive relations later encode the medium-specific information.

Firstly, we write the macroscopic Maxwell equations for the electric field $\vec{E}(\vec{r},t)$ and magnetic field $\vec{B}(\vec{r},t)$ as:

$$\begin{cases} \nabla\cdot\vec{E}(\vec{r},t)=\dfrac{\rho(\vec{r},t)}{\varepsilon_0} & \text{Gauss's law for electric field} \\ \nabla\cdot\vec{B}(\vec{r},t)=0 & \text{Gauss's law for magnetic field} \\ \nabla\times\vec{E}(\vec{r},t)=-\dfrac{\partial}{\partial t}\vec{B}(\vec{r},t) & \text{Faraday's law} \\ \nabla\times\vec{B}(\vec{r},t)=\mu_0\vec{J}(\vec{r},t)+\mu_0\varepsilon_0\dfrac{\partial}{\partial t}\vec{E}(\vec{r},t) & \text{Ampère–Maxwell law} \end{cases} \tag{3.1}$$

here, $\rho(\vec{r},t)$ and $\vec{J}(\vec{r},t)$ denote the total charge density and total current density, respectively.

These equations already imply a local conservation law. More specifically, we take the divergence of the Ampère–Maxwell law, and use Gauss's law for the electric field to obtain the continuity equation for charge at the macroscopic level [52]:

$$\frac{\partial}{\partial t}\rho(\vec{r},t)+\nabla\cdot\vec{J}(\vec{r},t)=0 \tag{3.2}$$

here, the conservation principle encountered in the previous chapter reappears in electrodynamic language.

Secondly, for media, it proves useful to decompose the sources into free and bound contributions:

$$\begin{aligned} \rho(\vec{r},t)&=\rho_{\text{free}}(\vec{r},t)+\rho_{\text{bound}}(\vec{r},t) \\ \vec{J}(\vec{r},t)&=\vec{J}_{\text{free}}(\vec{r},t)+\vec{J}_{\text{bound}}(\vec{r},t) \end{aligned} \tag{3.3}$$

here, the free part describes externally supplied or mobile macroscopic sources, while the bound part describes the response generated internally by the medium.

For discussing the bound electric response, we introduce the polarization field $\vec{P}(\vec{r},t)$ as:

$$\rho_{\text{bound}}(\vec{r},t)\coloneqq-\nabla\cdot\vec{P}(\vec{r},t) \tag{3.4}$$

Accordingly, we define the electric displacement field as:

$$\vec{D}(\vec{r},t) \coloneqq \varepsilon_0 \vec{E}(\vec{r},t) + \vec{P}(\vec{r},t) \tag{3.5}$$

For the nonmagnetic media considered in the present work, the electric response is the central object. Therefore, it is sufficient here to keep the Maxwell equations in the form:

$$\begin{cases} \nabla \cdot \vec{D}(\vec{r},t) = \rho_{\text{free}}(\vec{r},t) & \text{Gauss's law for electric displacement} \\ \nabla \cdot \vec{B}(\vec{r},t) = 0 & \text{Gauss's law for magnetic field} \\ \nabla \times \vec{E}(\vec{r},t) = -\dfrac{\partial}{\partial t}\vec{B}(\vec{r},t) & \text{Faraday's law} \\ \dfrac{1}{\mu_0}\nabla \times \vec{B}(\vec{r},t) = \vec{J}_{\text{free}}(\vec{r},t) + \dfrac{\partial}{\partial t}\vec{D}(\vec{r},t) & \text{Ampère–Maxwell law in matter} \end{cases} \tag{3.6}$$

Consequently, we can now characterize the structure of the problem. The Maxwell equations themselves are universal, while the medium dependence enters through the induced polarization $\vec{P}(\vec{r},t)$, and therefore through the electric displacement field $\vec{D}(\vec{r},t)$.

In many optical problems, one considers the response to an external electromagnetic perturbation in the absence of free charges and free currents inside the medium. Under this condition, equation (3.6) shows that the central issue is no longer the field equations themselves, but how the medium develops polarization under the applied electric field. Once the relation between $\vec{P}(\vec{r},t)$ and $\vec{E}(\vec{r},t)$ is specified, the electromagnetic response becomes well defined. This is precisely the point where macroscopic electrodynamics meets microscopic electronic structure, and it also provides the starting point for introducing the dielectric function [52, 54, 55].

3.1.2 Polarization, electric displacement, and constitutive relations

We have established the universal field equations in the previous subsection. However, those equations do not explore how matter responds to the applied field. To close the problem, we still need a relation that connects the induced polarization to the electric field. This relation is the constitutive relation, and it is here that the medium-specific information enters [52, 54].

Here, the central object is the polarization field $\vec{P}(\vec{r},t)$. It describes how the medium rearranges its internal charges under the applied electric field. Once $\vec{P}(\vec{r},t)$ is known, equation (3.5) immediately gives the electric displacement field $\vec{D}(\vec{r},t)$. Therefore, instead of treating $\vec{D}(\vec{r},t)$ as an independent quantity, we may regard it as a convenient way to encode the electric response of matter.

Firstly, we start from the most general linear-response form. For any time t, the polarization may depend not only on the electric field at the same point, but also on fields acting earlier and in a surrounding spatial region. Therefore, before introducing the simplified form used in the present work, we write the constitutive relation in its general nonlocal form in space and time [56]:

$$P_\alpha(\vec{r},t) = \varepsilon_0 \sum_\beta \int \mathrm{d}^3 r' \int_{-\infty}^{t} \mathrm{d}t'\, \chi_{\alpha\beta}(\vec{r},t;\vec{r}',t')\, E_\beta(\vec{r}',t') \tag{3.7}$$

here, $\chi_{\alpha\beta}(\vec{r},t;\vec{r}',t')$ describes the contribution of the electric field acting at $(\vec{r}',t')$ to the polarization that develops later at $(\vec{r},t)$. In this sense, the medium does not have to respond only at one point and one instant; instead, it may spread the response over space and retain memory in time. At the same time, the upper limit t in equation (3.7) keeps the causal order explicit: the perturbation acts first, and the response follows.

For the present work, we focus on the linear response in the long-wavelength regime. Under this approximation, spatial nonlocality drops out, and the constitutive relation retains only the memory of the earlier perturbation. Therefore, it reduces to the time-domain constitutive relation:

$$P_\alpha(\vec{r},t) = \varepsilon_0 \sum_\beta \int_{-\infty}^{t} \mathrm{d}t'\, \chi_{\alpha\beta}(t-t')\, E_\beta(\vec{r},t') \tag{3.8}$$

here, $\chi_{\alpha\beta}(t-t')$ denotes the time-dependent susceptibility tensor. This expression demonstrates that the polarization does not follow the electric field instantaneously; instead, the medium retains a memory of the earlier perturbation.

Optical experiments usually probe the response of a medium under an oscillating electromagnetic field. Therefore, once we enter the linear-response regime, the frequency

domain provides the natural language for discussing the optical response [57]. Next, we transform equation (3.8) to the frequency domain, and obtain:

$$P_\alpha(\vec{r},\omega) = \varepsilon_0 \sum_\beta \chi_{\alpha\beta}(\omega)\, E_\beta(\vec{r},\omega) \tag{3.9}$$

here, $\chi_{\alpha\beta}(\omega)$ denotes the frequency-dependent electric susceptibility tensor. This equation already shows that the susceptibility determines the linear electric response of the medium.

Substituting equation (3.9) into equation (3.5), we then express the electric displacement field in terms of the electric susceptibility tensor:

$$\begin{aligned} D_\alpha(\vec{r},\omega) &= \varepsilon_0 E_\alpha(\vec{r},\omega) + P_\alpha(\vec{r},\omega) \\ &= \varepsilon_0 \sum_\beta \left[\delta_{\alpha\beta} + \chi_{\alpha\beta}(\omega)\right] E_\beta(\vec{r},\omega) \end{aligned} \tag{3.10}$$

It now shows that the combination $\delta_{\alpha\beta} + \chi_{\alpha\beta}(\omega)$ naturally enters the linear electric response.

This observation directly motivates the definition of the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$:

$$\varepsilon_{\alpha\beta}(\omega) \coloneqq \delta_{\alpha\beta} + \chi_{\alpha\beta}(\omega) \tag{3.11}$$

Then, the frequency-domain constitutive relation takes the compact form:

$$D_\alpha(\vec{r},\omega) = \varepsilon_0 \sum_\beta \varepsilon_{\alpha\beta}(\omega) E_\beta(\vec{r},\omega) \tag{3.12}$$

The dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ thus provides a compact description of the linear electric response. Starting from the polarization field $\vec{P}(\vec{r},t)$, we introduced the electric susceptibility tensor $\chi_{\alpha\beta}(\omega)$, and then arrived at the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$. Therefore, this tensor emerges naturally from the constitutive relation that closes Maxwell electrodynamics in matter.

Besides its directional dependence, the dielectric response is also dynamical. In linear response theory, causality constrains this dynamical character, which means that the polarization at a given time can only depend on the electric field applied at earlier times. This causal structure will become important in the later discussion, where

we introduce the complex dielectric function and then derive the Kramers–Kronig relations from the analytic structure of the response.

3.1.3 Anisotropy and the dielectric tensor

The previous subsection presented the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ as a compact description of the linear electric response. We now examine how symmetry constrains its form [58]. When the medium responds identically along all directions, the dielectric tensor reduces to a scalar form. If the response changes with direction, the full tensor form must be retained.

Firstly, for an isotropic medium, the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ reduces to a scalar dielectric function $\varepsilon(\omega)$ [59]:

$$\varepsilon_{\alpha\beta}(\omega) = \varepsilon(\omega)\delta_{\alpha\beta} \tag{3.13}$$

here, the scalar dielectric function $\varepsilon(\omega)$ already determines the full linear electric response, since the isotropic medium does not distinguish one Cartesian direction from another. The tensor structure, therefore, no longer carries additional directional information.

Secondly, crystalline media usually distinguish directions. In that case, anisotropy survives, and we must retain the full tensor form. When we choose the Cartesian axes along the principal directions of the crystal, the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ takes the principal-axis form [60]:

$$[\varepsilon_{\alpha\beta}(\omega)] = \begin{pmatrix} \varepsilon_{xx}(\omega) & 0 & 0 \\ 0 & \varepsilon_{yy}(\omega) & 0 \\ 0 & 0 & \varepsilon_{zz}(\omega) \end{pmatrix} \tag{3.14}$$

here, each diagonal component describes the dielectric response along one principal axis, and anisotropy enters through the difference among these components.

Here, this directional dependence plays a central role [61]. We later compare the optical response within the structural plane and perpendicular to that plane. Therefore, it is

convenient to introduce the in-plane dielectric function $\varepsilon_{/\!/}(\omega)$ and the out-of-plane dielectric function $\varepsilon_{\perp}(\omega)$ [60, 62, 63]:

$$[\varepsilon_{\alpha\beta}(\omega)] = \begin{pmatrix} \varepsilon_{/\!/}(\omega) & 0 & 0 \\ 0 & \varepsilon_{/\!/}(\omega) & 0 \\ 0 & 0 & \varepsilon_{\perp}(\omega) \end{pmatrix} \tag{3.15}$$

here, the in-plane dielectric function $\varepsilon_{/\!/}(\omega)$ describes the response to an electric field within the structural plane, while the out-of-plane dielectric function $\varepsilon_{\perp}(\omega)$ describes the response to an electric field normal to that plane. This notation directly prepares the comparison between the in-plane and out-of-plane dielectric spectra and optical properties [61, 62].

This anisotropy also propagates to other optical properties derived from the dielectric tensor [61, 64]. Once the dielectric response changes with direction, the refractive index, the extinction coefficient, the reflectivity, and the energy-loss spectrum also inherit that directional dependence. Therefore, the tensor structure does not only refine the notation. It also determines how we organize and interpret the later optical results.

At this stage, the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ has been introduced as the quantity that encodes the directional linear electric response of the medium. When anisotropy is present, its tensor form makes this directional dependence explicit.

Once the principal directions of the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ are chosen as the working axes, the electric field $\vec{E}(\omega)$ can be decomposed as:

$$\vec{E}(\omega) = \sum_{\lambda} E_{\lambda}(\omega)\, \hat{e}_{\lambda} \tag{3.16}$$

where $\hat{e}_{\lambda}$ denotes the unit vector along the principal direction λ. In the same basis, the induced polarization $\vec{P}(\omega)$ takes the form:

$$\vec{P}(\omega) = \varepsilon_0 \sum_{\lambda} \chi_{\lambda}(\omega)\, E_{\lambda}(\omega)\, \hat{e}_{\lambda} \tag{3.17}$$

Accordingly, we denote the dielectric response along each principal direction by $\varepsilon_\lambda(\omega)$ and express:

$$\varepsilon_\lambda(\omega) = 1 + \chi_\lambda(\omega) \tag{3.18}$$

This direction-resolved form provides the natural bridge from the tensor description to the dielectric function discussed in the next section.

3.2 From electronic response to the dielectric function

The previous section established the electrodynamic framework of optical response. There, we introduced the polarization field $\vec{P}(\vec{r},t)$, the electric susceptibility tensor $\chi_{\alpha\beta}(\omega)$, and the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$. These quantities complete Maxwell electrodynamics in matter. We now recover the response function that electronic structure calculations actually provide, which is the frequency-dependent dielectric function [65, 66, 67].

In this section, we start by expressing the polarization response in the language of electric susceptibility and the complex dielectric function. We then identify the microscopic electronic processes that generate its imaginary part. Next, we formulate this response for anisotropic systems, since the present work compares the in-plane and out-of-plane optical channels explicitly. Therefore, we advance from the general constitutive picture to a response function that can be calculated, interpreted, and subsequently converted into optical observables.

3.2.1 From polarization response to the complex dielectric function

The previous section established the constitutive chain from the polarization field $\vec{P}(\vec{r},t)$ to the electric susceptibility tensor $\chi_{\alpha\beta}(\omega)$ and the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ through equations (3.9) and (3.11). We now turn to the dielectric function, which serves as the central quantity in optical response. In physical terms, the time-dependent electric field of incident light drives this response by perturbing electrons away from equilibrium and inducing the aforementioned polarization.

The electric susceptibility tensor $\chi_{\alpha\beta}(\omega)$ measures how strongly the polarization component P_α responds to the field component E_β, while the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ collects this response into the form most directly used in optical analysis [66].

Under a monochromatic electric field, part of the polarization stays in phase with the field, while another part lags behind it. The first part stores energy in the medium, and the second part transfers energy from the field into the medium. We therefore write the dielectric tensor in complex form as:

$$\varepsilon_{\alpha\beta}(\omega) = \varepsilon_{1,\alpha\beta}(\omega) + \mathrm{i}\,\varepsilon_{2,\alpha\beta}(\omega) \tag{3.19}$$

here, $\varepsilon_{1,\alpha\beta}(\omega)$ denotes the real part of the dielectric tensor, and $\varepsilon_{2,\alpha\beta}(\omega)$ denotes the imaginary part of the dielectric tensor.

The real part denotes the dispersive part of the response. It describes how the medium polarizes with the field and therefore how it stores electromagnetic energy.

The imaginary part denotes the absorptive part of the response. It describes how the field drives electronic excitations and therefore how it transfers energy into the medium [67].

For the later discussion on optical observables, when the field polarization is fixed or working along a principal direction of the dielectric tensor, we suppress the tensor indices and write the dielectric response in scalar form.

Therefore, along a principal direction, the complex dielectric response can be expressed in scalar form as:

$$\varepsilon(\omega) = \varepsilon_1(\omega) + \mathrm{i}\,\varepsilon_2(\omega) \tag{3.20}$$

here, $\varepsilon_1(\omega)$ denotes the real part of the dielectric function, and $\varepsilon_2(\omega)$ denotes the imaginary part of the dielectric function.

Under an external electric field, this scalar form represents the dielectric response along a principal direction. We can therefore derive the absorption coefficient $\alpha(\omega)$, the refractive index $n(\omega)$, the extinction coefficient $k(\omega)$, the reflectivity $R(\omega)$, and the energy-loss spectrum $L(\omega)$. This complex dielectric function therefore provides a unified framework for the linear optical properties [65, 68, 69].

For the xene systems considered in the present work, we accordingly resolve the dielectric response for the in-plane and out-of-plane channels for separate analysis. In anisotropic xenes, the response along the structural plane usually differs substantially from that along the perpendicular direction [61, 70, 71, 72].

At this stage, we arrive at the complex dielectric function as the central response quantity for discussing optical performance. When we enter the frequency domain, the optical response no longer appears as a static redistribution of charge; instead, the field drives the electrons continuously, and the medium responds with a frequency-dependent polarization. Within this complex quantity, microscopic electronic processes generate the imaginary part $\varepsilon_2(\omega)$, while the Kramers–Kronig relation determines the

real part $\varepsilon_1(\omega)$ [73, 74, 75]. We now turn to the microscopic origin of the imaginary part $\varepsilon_2(\omega)$.

3.2.2 Microscopic origin of the imaginary part

For the complex dielectric function, the previous subsection discussed that the imaginary part $\varepsilon_2(\omega)$ characterizes the absorptive response of the medium. We now connect this quantity to the microscopic electronic transitions in the crystal.

When a monochromatic electric field acts on a crystal, it drives electrons from occupied states to empty states, and therefore induces interband transitions. In this process, the external field transfers the energy $\hbar\omega$ to the electronic system, and the medium absorbs this energy through a dissipative response [76]. To describe this dissipative response quantitatively, we now introduce the optical conductivity tensor $\sigma_{\alpha\beta}(\omega)$ within linear response theory [77].

Firstly, we express the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ through the optical conductivity tensor $\sigma_{\alpha\beta}(\omega)$ as [65, 77]:

$$\varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} + \frac{\mathrm{i}}{\varepsilon_0\omega}\,\sigma_{\alpha\beta}(\omega) \tag{3.21}$$

Once we decompose this optical conductivity tensor into its real part $\sigma_{1,\alpha\beta}(\omega)$ and imaginary part $\sigma_{2,\alpha\beta}(\omega)$, the previous equation immediately gives:

$$\varepsilon_{2,\alpha\beta}(\omega) = \frac{\sigma_{1,\alpha\beta}(\omega)}{\varepsilon_0\omega} \tag{3.22}$$

This relation demonstrates that the imaginary part of the dielectric response directly describes the energy-absorbing part of the induced current, which leads to dissipation under the external field.

Next, we describe this microscopic process in a periodic solid as an interband transition from an occupied valence state $|\psi_{v\vec{k}}\rangle$ to an empty conduction state $|\psi_{c\vec{k}}\rangle$. Here, $\vec{k}$ denotes the crystal momentum in the first Brillouin zone. In the optical regime, the photon momentum is negligible on the scale of the Brillouin zone, so the transition is effectively vertical in $\vec{k}$-space [78]. Accordingly, the transition strength is determined by the velocity operator $\hat{v}_\alpha$ rather than by the band energies alone.

For the present work, we adopt the independent-particle approximation (IPA). The imaginary part of the dielectric tensor $\varepsilon_{2,\alpha\beta}(\omega)$ takes the standard interband form [79]:

$$\varepsilon_{2,\alpha\beta}(\omega) \propto \sum_{v,c} \int_{\mathrm{BZ}} \frac{\mathrm{d}^3k}{(2\pi)^3} \langle \psi_{c\vec{k}} | \hat{v}_\alpha | \psi_{v\vec{k}} \rangle \langle \psi_{v\vec{k}} | \hat{v}_\beta | \psi_{c\vec{k}} \rangle \delta \left(E_{c\vec{k}} - E_{v\vec{k}} - \hbar\omega \right) \tag{3.23}$$

here, the valence-band index v and conduction-band index c label the occupied and empty bands, respectively. The band energies $E_{v\vec{k}}$ and $E_{c\vec{k}}$ describe the initial and final states. The velocity operator $\hat{v}_\alpha$ determines the transition strength along the α direction. Meanwhile, the Dirac delta function $\delta \left(E_{c\vec{k}} - E_{v\vec{k}} - \hbar\omega \right)$ enforces the energy-conservation condition:

$$E_{c\vec{k}} - E_{v\vec{k}} = \hbar\omega \tag{3.24}$$

Equation (3.23) describes the microscopic origin of the absorptive response. For a transition to contribute to the absorptive response, the initial state must be occupied, the final state must be empty, the transition energy $E_{c\vec{k}} - E_{v\vec{k}}$ must match the photon energy $\hbar\omega$, and the velocity matrix elements $\langle \psi_{c\vec{k}} | \hat{v}_\alpha | \psi_{v\vec{k}} \rangle$ and $\langle \psi_{v\vec{k}} | \hat{v}_\beta | \psi_{c\vec{k}} \rangle$ must remain finite. The corresponding matrix elements therefore serve as the criterion for optically allowed transitions. Therefore, when many optically allowed transitions share the same excitation energy and show strong transition strength, a peak appears in $\varepsilon_{2,\alpha\beta}(\omega)$.

This point clarifies what the calculated spectra describe. The imaginary part $\varepsilon_{2,\alpha\beta}(\omega)$ reflects the band gap, the band dispersion, the occupation pattern, and the direction-resolved transition matrix elements. At this stage, $\varepsilon_{2,\alpha\beta}(\omega)$ provides a direct bridge between the electronic structure and the optical absorption channel.

In the present work, we further adopt the long-wavelength limit. Accordingly, we evaluate the dielectric response within the independent-particle approximation built from the ground-state electronic structure. Under this approximation, the calculated $\varepsilon_{2,\alpha\beta}(\omega)$ describes the single-particle interband absorption, and it identifies the main transition channels in the optical spectra. Accordingly, it captures the principal onset energies, the major absorption peaks, and the directional contrast relevant to the systems studied here.

The central point is that the tensor indices α and β enter directly through the velocity matrix elements. Therefore, the microscopic transition strength already resolves the

field polarization and the crystal directions. Once the crystal responds differently along different directions, the imaginary part $\varepsilon_{2,\alpha\beta}(\omega)$ already exhibits this anisotropy at the dielectric-response level. We thus arrive at a microscopic description of $\varepsilon_{2,\alpha\beta}(\omega)$, which collects the energy-conserving interband transitions and weights them by their directional dependence. This result naturally leads us to the dielectric response in anisotropic media [71].

3.2.3 Dielectric response in anisotropic media

The previous subsection established the microscopic expression for the imaginary part $\varepsilon_{2,\alpha\beta}(\omega)$. We now describe the dielectric response of anisotropic media in a direction-resolved form [61].

For an anisotropic medium, once we determine the principal directions, equation (3.14) describes the principal-axis form of the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$. For each principal component with $\lambda = x, y, z$, we write the complex dielectric function in a direction-resolved form [60]:

$$\varepsilon_{\lambda\lambda}(\omega) = \varepsilon_{1,\lambda\lambda}(\omega) + \mathrm{i}\,\varepsilon_{2,\lambda\lambda}(\omega) \tag{3.25}$$

here, the real part $\varepsilon_{1,\lambda\lambda}(\omega)$ describes the directional dispersive response, whereas the imaginary part $\varepsilon_{2,\lambda\lambda}(\omega)$ describes the directional absorptive response. As different principal components produce different optical responses along different directions, optical anisotropy appears [60].

In what follows, we denote the dielectric response within the structural plane by the in-plane dielectric function $\varepsilon_{/\!/}(\omega)$, whereas we denote the response normal to that plane by the out-of-plane dielectric function $\varepsilon_{\perp}(\omega)$.

Once we decompose the dielectric response into directional channels, the derived optical properties inherit the same anisotropy [80].

A representative example of the direction-resolved dielectric response is shown in the following figure. The real and imaginary parts exhibit distinct in-plane and out-of-plane features over the full photon-energy range.

At this stage, we have established the direction-resolved complex dielectric response for anisotropic media. For each principal direction, the real and imaginary parts are not independent. In the next section, we introduce causality as the fundamental

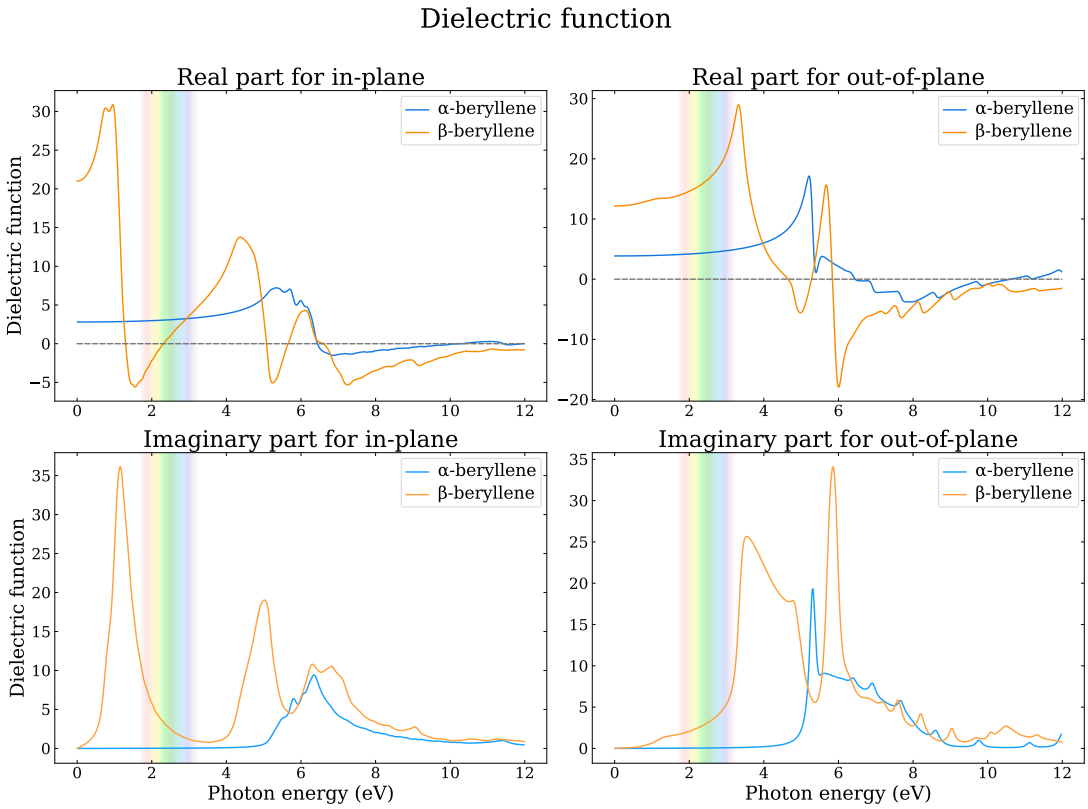

FIGURE 3.1. Representative real and imaginary parts of the in-plane and out-of-plane dielectric function as a function of photon energy for α-Beryllene and β-Beryllene. The figure illustrates the direction-resolved complex dielectric response discussed in the text.

constraint on the linear response, and then derive the Kramers–Kronig relations for each directional component.

3.3 Causality and the Kramers–Kronig relations

The previous section established a direction-resolved complex dielectric response for anisotropic media. We now identify the constraint that connects the real and imaginary parts of each directional component.

Here, the major physical constraint is causality [74]. Specifically, causality means that the polarization response cannot occur before the external electric field is applied. Equivalently, in the time domain, the response kernel vanishes when the response time precedes the driving time. This temporal order determines the analytic structure of the response function in frequency space. As a result, for each complex component, the real and imaginary parts cannot vary independently; instead, they are related through the Kramers–Kronig relations [73, 74, 81].

Therefore, for the dielectric response, once we calculate the absorptive part, we can determine the dispersive part and construct the complete dielectric response. This connection completes the dielectric response and opens the derivation of the optical properties.

3.3.1 Causality and analytic structure of response functions

Firstly, to express causality explicitly, we start from the time-domain linear relation between the polarization field $P_\alpha(\vec{r},t)$ and the electric field $E_\beta(\vec{r}\,',t')$ [82]:

$$P_\alpha(\vec{r},t) = \varepsilon_0 \sum_\beta \int_{-\infty}^{\infty} \mathrm{d}t' \int \mathrm{d}^3 r'\, \chi_{\alpha\beta}(\vec{r},\vec{r}\,',t-t') E_\beta(\vec{r}\,',t') \tag{3.26}$$

here, $\chi_{\alpha\beta}(\vec{r},\vec{r}\,',t-t')$ denotes the time-domain electric susceptibility tensor. This equation characterizes the polarization response at position $\vec{r}$ and time t to an electric field applied at position $\vec{r}\,'$ and time t'.

Therefore, causality requires that the medium responds only to earlier external driving fields [83]:

$$\chi_{\alpha\beta}(\vec{r},\vec{r}\,',t-t') = 0 \qquad \text{for } t < t' \tag{3.27}$$

This condition defines a retarded response kernel [52]. It excludes any polarization response before the external electric field acts.

In the present work, we consider the optical long-wavelength limit, where the spatial nonlocality of the dielectric response becomes negligible [84]. Afterwards, the linear response then reduces to a local frequency-dependent tensor relation [85]:

$$P_\alpha(\omega) = \varepsilon_0 \sum_\beta \chi_{\alpha\beta}(\omega)\, E_\beta(\omega) \tag{3.28}$$

here, $\chi_{\alpha\beta}(\omega)$ denotes the frequency-domain electric susceptibility tensor. It characterizes the response amplitude for each Cartesian component at angular frequency ω. Meanwhile, it also describes the phase delay of the response at the same frequency [86].

We then express the dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ via the electric susceptibility tensor $\chi_{\alpha\beta}(\omega)$ [87]:

$$\varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} + \chi_{\alpha\beta}(\omega) \tag{3.29}$$

Subsequently, the causal condition in equation (3.27) gives the frequency-domain susceptibility in the form:

$$\chi_{\alpha\beta}(\omega) = \int_0^\infty \chi_{\alpha\beta}(t)\, \mathrm{e}^{\mathrm{i}\omega t} \mathrm{d}t \tag{3.30}$$

This representation reveals the analytic structure directly. Since the time integral starts from $t = 0$, the response function remains analytic in the upper half of the complex frequency plane [88].

For a real time-domain response kernel, the susceptibility $\chi_{\alpha\beta}(\omega)$ and dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$ further satisfy the following condition:

$$\begin{cases} \chi_{\alpha\beta}(-\omega) = \chi^*_{\alpha\beta}(\omega) \\ \varepsilon_{\alpha\beta}(-\omega) = \varepsilon^*_{\alpha\beta}(\omega) \end{cases} \tag{3.31}$$

Therefore, in the principal-axis representation, for each complex dielectric component $\varepsilon_{\lambda\lambda}(\omega)$, the real part $\varepsilon_{1,\lambda\lambda}(\omega)$ is an even function of frequency, whereas the imaginary part $\varepsilon_{2,\lambda\lambda}(\omega)$ is an odd function:

$$\begin{cases} \varepsilon_{1,\lambda\lambda}(-\omega) = \varepsilon_{1,\lambda\lambda}(\omega) \\ \varepsilon_{2,\lambda\lambda}(-\omega) = -\varepsilon_{2,\lambda\lambda}(\omega) \end{cases} \tag{3.32}$$

At this stage, the previous discussion has established the analytic structure of the dielectric response in frequency space [87, 88, 89]. We next derive the Kramers–Kronig relations between its real and imaginary parts.

3.3.2 Kramers–Kronig relations for the dielectric function

The previous subsection established the analytic structure of the dielectric response in frequency space. We now derive the Kramers–Kronig relations for the dielectric function [74]. Firstly, the causal structure in equation (3.30) and the reality condition in equation (3.31) establish a Hilbert-transform relation between the real and imaginary parts of the dielectric response [83, 86, 87]. We therefore represent the Kramers–Kronig relations for the dielectric tensor in the general frequency-domain form as:

$$\begin{cases} \varepsilon_{1,\alpha\beta}(\omega) - \delta_{\alpha\beta} = \dfrac{1}{\pi}\mathcal{P}\displaystyle\int_{-\infty}^{\infty} \mathrm{d}\omega' \, \dfrac{\varepsilon_{2,\alpha\beta}(\omega')}{\omega' - \omega} \\ \varepsilon_{2,\alpha\beta}(\omega) = -\dfrac{1}{\pi}\mathcal{P}\displaystyle\int_{-\infty}^{\infty} \mathrm{d}\omega' \, \dfrac{\varepsilon_{1,\alpha\beta}(\omega') - \delta_{\alpha\beta}}{\omega' - \omega} \end{cases} \tag{3.33}$$

here, $\mathcal{P}$ denotes the Cauchy principal value. It removes the singular contribution at $\omega' = \omega$. The integral therefore remains well defined [83, 86].

This expression identifies the physical roles of the two parts of the dielectric response. The imaginary part $\varepsilon_{2,\alpha\beta}(\omega)$ characterizes the absorptive channel, whereas the real part $\varepsilon_{1,\alpha\beta}(\omega)$ characterizes the dispersive channel. Consequently, causality unifies these two channels into a single complex response function.

Next, the symmetry relations in equation (3.32) reduce the full-frequency integrals to positive-frequency integrals. In the principal-axis representation, each complex dielectric component $\varepsilon_{\lambda\lambda}(\omega)$ satisfies the Kramers–Kronig relations in the form [90]:

$$\begin{cases} \varepsilon_{1,\lambda\lambda}(\omega) = 1 + \dfrac{2}{\pi}\mathcal{P}\displaystyle\int_{0}^{\infty} \mathrm{d}\omega' \, \dfrac{\omega' \varepsilon_{2,\lambda\lambda}(\omega')}{\omega'^2 - \omega^2} \\ \varepsilon_{2,\lambda\lambda}(\omega) = -\dfrac{2\omega}{\pi}\mathcal{P}\displaystyle\int_{0}^{\infty} \mathrm{d}\omega' \, \dfrac{\varepsilon_{1,\lambda\lambda}(\omega') - 1}{\omega'^2 - \omega^2} \end{cases} \tag{3.34}$$

These relations provide the positive-frequency form of the dielectric response in the present work. Once we obtain the absorptive part $\varepsilon_{2,\lambda\lambda}(\omega)$ from the electronic transitions, equation (3.34) determines the dispersive part $\varepsilon_{1,\lambda\lambda}(\omega)$. This relation also preserves the causal connection between the two parts of the dielectric response.

Consequently, the complex dielectric function describes the absorptive and dispersive responses of the medium within a unified framework. The imaginary part characterizes the energy absorbed by the medium at a given frequency, whereas the real part describes the associated dispersion generated by the same response process. Within this framework, equation (3.34) directly determines the static limit. In the next subsection, we derive the corresponding static expression and clarify its physical meaning.

3.3.3 Static limit and physical interpretation

The previous subsection established the positive-frequency form of the Kramers–Kronig relations for the dielectric response. We now describe the static limit and clarify its physical interpretation [91, 92].

Firstly, the zero-frequency limit of the dispersive part determines the static dielectric response. In the principal-axis representation, we set $\omega \to 0$ in positive-frequency Kramers–Kronig relations (3.34). Then we obtain the static-limit form of the Kramers–Kronig relations:

$$\varepsilon_{1,\lambda\lambda}(0) = 1 + \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega' \, \frac{\varepsilon_{2,\lambda\lambda}(\omega')}{\omega'} \tag{3.35}$$

here, $\varepsilon_{1,\lambda\lambda}(0)$ characterizes the static dielectric response along the principal direction λ. This expression shows that the static limit is not an independent quantity; instead, the full directional absorption spectrum $\varepsilon_{2,\lambda\lambda}(\omega)$ determines it through causality [93].

Next, the factor $1/\omega'$ gives higher weight to low-energy spectral contributions. The low-energy electronic transitions dominate the static dielectric response, whereas the higher-energy transitions contribute secondarily. Therefore, the static limit represents an accumulated response over the full absorption spectrum, with the dominant contribution controlled by its low-energy part.

Meanwhile, anisotropic media generally exhibit different spectral-weight distributions along different principal directions [94, 95].

The Kramers–Kronig relations (3.35) describe a direct directional dependence in the static limit. The different directional absorption spectra produce different directional static dielectric responses [94]. This static limit shows that directional absorption produces directional dielectric response.

Consequently, the static limit reveals the physical interpretation of the Kramers–Kronig relations. The imaginary part characterizes the spectral weight generated by electronic excitations, whereas the real part determines the corresponding zero-frequency dielectric response. Therefore, the complex dielectric function provides the complete information required for the subsequent optical properties [81, 92].

At this stage, we arrive at the static physical interpretation of the Kramers–Kronig relations for anisotropic media. We next use the same dielectric response to derive the absorption coefficient, refractive index, extinction coefficient, reflectivity, and energy-loss spectrum.

3.4 From the dielectric function to optical properties

The previous section established the complex dielectric function. It describes the optical response of the present systems. We now apply this dielectric response to determine the corresponding optical properties.

We start from the absorption coefficient $\alpha(\omega)$, which characterizes the attenuation of incident light in the medium. We then turn to the refractive index $n(\omega)$ and extinction coefficient $k(\omega)$, which describe phase propagation and amplitude decay. Next, we determine the reflectivity $R(\omega)$, which represents the reflected portion of the electromagnetic wave at the interface. Finally, we examine the energy-loss spectrum $L(\omega)$, which identifies the dominant resonant energy-loss channels in the medium.

In what follows, we present the general relations that connect these quantities to the complex dielectric function $\varepsilon(\omega)$ [59, 78, 82, 96]. Recent first-principles studies have also analyzed the dielectric response together with the derived optical properties in low-dimensional materials [69, 71, 72, 97, 98]. Afterwards, we analyze the in-plane and out-of-plane optical response of the present systems.

3.4.1 Absorption coefficient

We first consider the absorption coefficient $\alpha_\lambda(\omega)$, which characterizes the optical attenuation along the principal direction λ [99].

For a monochromatic wave propagating through the medium, the intensity follows:

$$I_\lambda(z,\omega) = I_{\lambda,0}(\omega)\, e^{-\alpha_\lambda(\omega) z} \tag{3.36}$$

here, we use z to represent the propagation distance in the medium. $I_{\lambda,0}(\omega)$ and $I_\lambda(z,\omega)$ denote the incident and transmitted intensities, respectively. Therefore, a larger absorption coefficient corresponds to stronger attenuation and shorter optical penetration depth.

We next introduce the complex refractive index along the principal direction λ:

$$N_\lambda(\omega) = n_\lambda(\omega) + \mathrm{i}\, k_\lambda(\omega) \tag{3.37}$$

here, $n_\lambda(\omega)$ denotes the refractive index, whereas $k_\lambda(\omega)$ denotes the extinction coefficient. We then express the absorption coefficient $\alpha_\lambda(\omega)$ as:

$$\alpha_\lambda(\omega) = \frac{2\,\omega}{c} k_\lambda(\omega) \tag{3.38}$$

here, c is the speed of light in vacuum. This expression connects the absorption coefficient to the imaginary part of the complex refractive index. That imaginary part drives the amplitude decay in the medium [66, 78, 82, 96].

We now combine the defining relation between the complex refractive index $N_\lambda(\omega)$ and the complex dielectric function $\varepsilon_{\lambda\lambda}(\omega)$ as [59]:

$$N_\lambda^2(\omega) = \varepsilon_{\lambda\lambda}(\omega) \tag{3.39}$$

Using this relation together with equation (3.25), we then derive the absorption coefficient $\alpha_\lambda(\omega)$ from the complex dielectric function $\varepsilon_{\lambda\lambda}(\omega)$ [59]:

$$\alpha_\lambda(\omega) = \frac{\sqrt{2}\,\omega}{c} \left[\sqrt{\varepsilon_{1,\lambda\lambda}^2(\omega) + \varepsilon_{2,\lambda\lambda}^2(\omega)} - \varepsilon_{1,\lambda\lambda}(\omega)\right]^{1/2} \tag{3.40}$$

This equation expresses the absorption coefficient as a response quantity derived from the dielectric function. Here, the imaginary part $\varepsilon_{2,\lambda\lambda}(\omega)$ provides the spectral weight generated by electronic transitions, whereas the real part $\varepsilon_{1,\lambda\lambda}(\omega)$ modifies the magnitude and energy dependence of the resulting attenuation.

The following figure presents the calculated absorption coefficient $\alpha_\lambda(\omega)$ spectra of the present systems for the in-plane and out-of-plane directions [69, 71, 72, 94, 97, 100, 101]:

The absorption coefficient spectra demonstrate differences between the two optical directions and between the two materials. In the in-plane channel, β-Beryllene shows a stronger absorption response than α-Beryllene over a broad photon-energy range, whereas α-Beryllene develops its main absorption only beyond the low-energy range and then forms a broader spectral feature. In the out-of-plane channel, both materials develop stronger peak structures, while β-Beryllene still shows the stronger overall absorption. These trends indicate that the optical attenuation depends on both the propagation direction and the crystal structure of the medium.

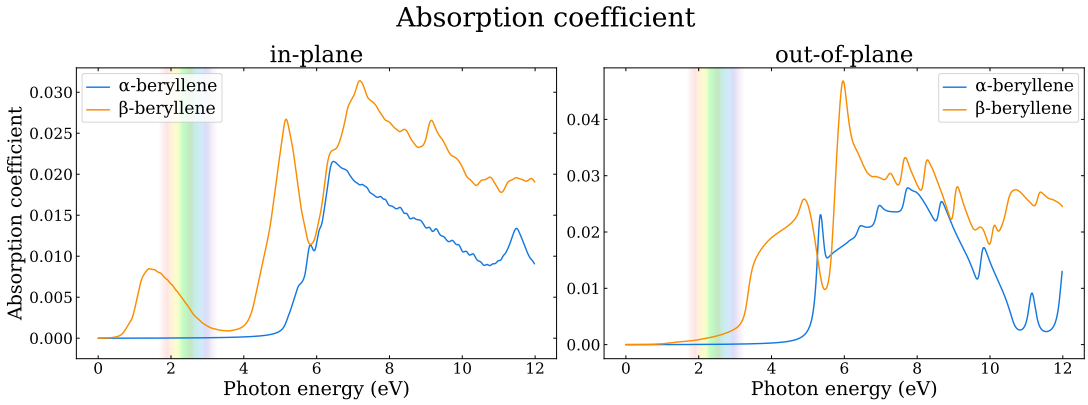

FIGURE 3.2. The absorption coefficients for the in-plane and out-of-plane directions as functions of photon energy for α-Beryllene and β-Beryllene.

Once the electronic transitions generate different spectral weights along different directions, the corresponding optical attenuation also acquires directional dependence [94].

The low-energy increase of $\alpha_\lambda(\omega)$ identifies the beginning of the optical transitions. As the photon energy increases, broader peaks emerge in the absorption spectrum. These peaks indicate energy ranges where interband transitions contribute more strongly to the absorption process [102].

Therefore, the absorption coefficient $\alpha_\lambda(\omega)$ reflects the directional excitation spectrum of the medium and characterizes the rate of optical attenuation [69, 97, 102, 103]. We next apply the same dielectric response to determine the refractive index $n_\lambda(\omega)$ and extinction coefficient $k_\lambda(\omega)$, which describe the propagation of the attenuated wave.

3.4.2 Refractive index and extinction coefficient

The previous subsection described the absorption coefficient as the optical quantity that characterizes energy dissipation in the medium. We now continue this description of light propagation in the medium by introducing the refractive index $n_\lambda(\omega)$ and the extinction coefficient $k_\lambda(\omega)$. The refractive index determines the phase evolution, whereas the extinction coefficient describes the attenuation of the wave amplitude. Starting from the dielectric response, we derive explicit expressions for $n_\lambda(\omega)$ and $k_\lambda(\omega)$ [59, 78, 82, 96].

For a monochromatic wave propagating along the principal direction λ, we first represent the electric field as [104]:

$$\begin{aligned} E_\lambda(z,t) &= E_{\lambda,0}\,\exp\left[\mathrm{i}\frac{\omega}{c}N_\lambda(\omega)z - \mathrm{i}\,\omega t\right] \\ &= E_{\lambda,0}\,\exp\left[\mathrm{i}\frac{\omega}{c}n_\lambda(\omega)z\right]\exp\left[-\frac{\omega}{c}k_\lambda(\omega)z\right]\exp\left(-\mathrm{i}\,\omega t\right) \end{aligned} \tag{3.41}$$

here, $N_\lambda(\omega)$ denotes the complex refractive index. Its real part $n_\lambda(\omega)$ denotes the refractive index, and its imaginary part $k_\lambda(\omega)$ denotes the extinction coefficient.

This expression separates phase evolution from amplitude attenuation [59, 82, 96]. In equation (3.41), the refractive index $n_\lambda(\omega)$ determines the phase evolution of the electromagnetic wave, whereas the extinction coefficient $k_\lambda(\omega)$ describes the exponential attenuation of its amplitude in the medium.

We then write the general relation between the complex refractive index $N_\lambda(\omega)$, dielectric function $\varepsilon_{\lambda\lambda}(\omega)$, and magnetic permeability $\mu_{\lambda\lambda}(\omega)$ as [59, 82, 96, 104]:

$$N_\lambda^2(\omega) = \varepsilon_{\lambda\lambda}(\omega)\,\mu_{\lambda\lambda}(\omega) \tag{3.42}$$

here, $\mu_{\lambda\lambda}(\omega)$ denotes the magnetic permeability. For the non-magnetic systems considered in the present work, we take $\mu_{\lambda\lambda}(\omega) \approx 1$. Therefore, the relation reduces to [59, 82, 96]:

$$N_\lambda^2(\omega) = \varepsilon_{\lambda\lambda}(\omega) \tag{3.43}$$

We substitute equation (3.25) and the definition of the complex refractive index $N_\lambda(\omega)$ in equation (3.37) into equation (3.43), and then separate the real and imaginary parts. We therefore derive the coupled relations [78, 82, 96]:

$$\begin{aligned} n_\lambda^2(\omega) - k_\lambda^2(\omega) &= \varepsilon_{1,\lambda\lambda}(\omega) \\ 2\,n_\lambda(\omega)\,k_\lambda(\omega) &= \varepsilon_{2,\lambda\lambda}(\omega) \end{aligned} \tag{3.44}$$

These two equations connect the refractive index $n_\lambda(\omega)$ and extinction coefficient $k_\lambda(\omega)$ to the real part $\varepsilon_{1,\lambda\lambda}(\omega)$ and imaginary part $\varepsilon_{2,\lambda\lambda}(\omega)$ of the dielectric function $\varepsilon_{\lambda\lambda}(\omega)$.

By solving these coupled relations, we express the refractive index $n_\lambda(\omega)$ as [58, 72, 101]:

$$n_\lambda(\omega) = \left[\frac{1}{2}\left(\sqrt{\varepsilon_{1,\lambda\lambda}^2(\omega) + \varepsilon_{2,\lambda\lambda}^2(\omega)} + \varepsilon_{1,\lambda\lambda}(\omega)\right)\right]^{1/2} \tag{3.45}$$

Similarly, we express the extinction coefficient $k_\lambda(\omega)$ as:

$$k_\lambda(\omega) = \left[\frac{1}{2}\left(\sqrt{\varepsilon_{1,\lambda\lambda}^2(\omega) + \varepsilon_{2,\lambda\lambda}^2(\omega)} - \varepsilon_{1,\lambda\lambda}(\omega)\right)\right]^{1/2} \tag{3.46}$$

With these expressions, we present the calculated refractive index $n_\lambda(\omega)$ spectra of α-Beryllene and β-Beryllene [69, 71, 72, 94, 97, 100, 101]:

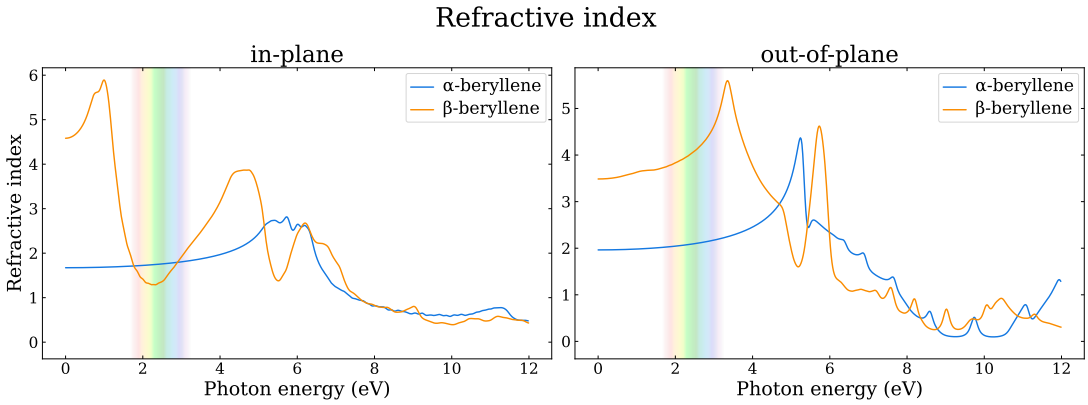

FIGURE 3.3. The refractive indices for the in-plane and out-of-plane directions as functions of photon energy for α-Beryllene and β-Beryllene.

The refractive index spectra demonstrate distinct directional and structural contrasts. In the in-plane channel, β-Beryllene develops two pronounced enhancements in the infrared and near-ultraviolet regions, with a minimum in the visible range, whereas α-Beryllene varies more smoothly over the same interval. In the out-of-plane channel, β-Beryllene remains stronger than α-Beryllene over the infrared to near-ultraviolet range. These trends indicate that the phase response depends on both the optical direction and the crystal structure of the medium [69, 71, 72, 94, 97, 101].

We next present the extinction coefficient $k_\lambda(\omega)$ spectra, which follow the absorptive part of the dielectric response more directly [78, 82, 96]:

The extinction coefficient spectra demonstrate a similar directional contrast. In the in-plane channel, β-Beryllene exhibits a strong infrared peak, followed by a pronounced

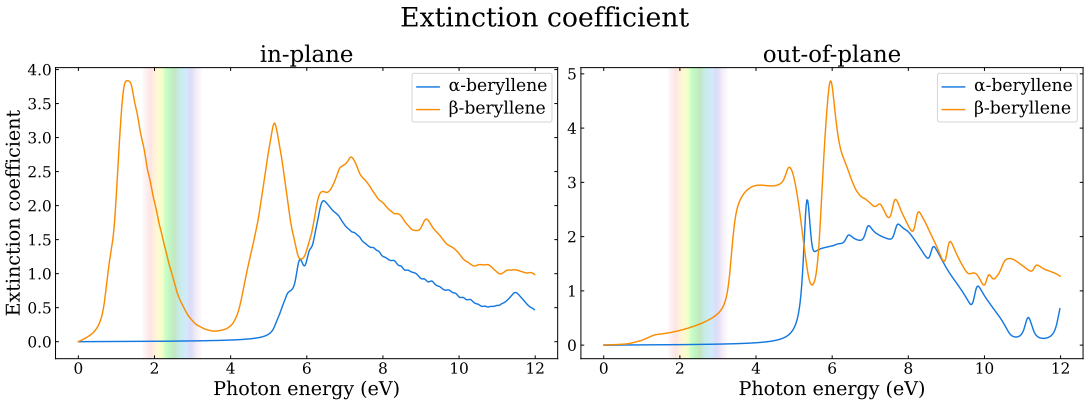

FIGURE 3.4. The extinction coefficients for the in-plane and out-of-plane directions as functions of photon energy for α-Beryllene and β-Beryllene.

trough from the visible to near-ultraviolet range and another strong feature in the ultraviolet region. By contrast, α-Beryllene remains weak at low photon energies and develops a broader ultraviolet response before gradually decreasing at higher energies. In the out-of-plane channel, β-Beryllene is stronger than α-Beryllene from the infrared to the near-ultraviolet range and reaches its largest response in the ultraviolet region. Both phases then show oscillatory features and an overall decrease toward higher photon energies. This behavior shows that amplitude attenuation also depends on both the optical direction and the crystal structure of the medium [69, 71, 72, 94, 97, 100, 101].

Accordingly, the refractive index $n_\lambda(\omega)$ and extinction coefficient $k_\lambda(\omega)$ connect the dielectric function to the interface response through the Fresnel equations [59, 82, 96]. We next use them to determine the reflectivity, which transfers the bulk optical response to the interface.

3.4.3 Reflectivity

The previous subsection determined the refractive index $n_\lambda(\omega)$ and extinction coefficient $k_\lambda(\omega)$, which characterize the phase propagation and amplitude attenuation in the medium. We now turn from propagation inside the medium to reflection at its surface.

The reflectivity $R_\lambda(\omega)$ characterizes the fraction of incident electromagnetic energy reflected at the interface. We then describe it through the Fresnel coefficient at normal incidence [59, 78, 82, 96].

When an electromagnetic wave reaches the interface between two media, the mismatch of optical response generates a reflected wave and a transmitted wave. For a plane wave incident from an ambient medium with complex refractive index $N_{0,\lambda}(\omega)$ onto the present system with complex refractive index $N_\lambda(\omega)$, the amplitude reflection coefficient at normal incidence takes the general form [59, 96]:

$$r_\lambda(\omega) = \frac{N_\lambda(\omega) - N_{0,\lambda}(\omega)}{N_\lambda(\omega) + N_{0,\lambda}(\omega)} \tag{3.47}$$

here, $r_\lambda(\omega)$ denotes the amplitude reflection coefficient, $N_{0,\lambda}(\omega)$ denotes the complex refractive index of the incident medium, and $N_\lambda(\omega)$ denotes the complex refractive index of the present medium.

For the xene systems considered in the present work, we investigate reflection from vacuum to a non-magnetic medium. The incident medium therefore satisfies $N_{0,\lambda}(\omega) \approx 1$. With the complex refractive index $N_\lambda(\omega)$ in equation (3.37), we then write the amplitude reflection coefficient as:

$$r_\lambda(\omega) = \frac{n_\lambda(\omega) - 1 + \mathrm{i}\, k_\lambda(\omega)}{n_\lambda(\omega) + 1 + \mathrm{i}\, k_\lambda(\omega)} \tag{3.48}$$

here, the refractive index $n_\lambda(\omega)$ determines the phase mismatch across the interface, whereas the extinction coefficient $k_\lambda(\omega)$ represents the dissipative part of the optical response. These two quantities together determine the reflected amplitude at the surface.

We next express the reflectivity $R_\lambda(\omega)$ as the reflected power ratio. It equals the squared modulus of the amplitude reflection coefficient [59, 82, 96]:

$$R_\lambda(\omega) = |r_\lambda(\omega)|^2 \tag{3.49}$$

After substituting equation (3.48), this equation produces the reflectivity through the refractive index $n_\lambda(\omega)$ and the extinction coefficient $k_\lambda(\omega)$:

$$R_\lambda(\omega) = \frac{[n_\lambda(\omega) - 1]^2 + k_\lambda^2(\omega)}{[n_\lambda(\omega) + 1]^2 + k_\lambda^2(\omega)} \tag{3.50}$$

This equation links the surface reflection to the refractive index $n_\lambda(\omega)$ and the extinction coefficient $k_\lambda(\omega)$. A larger mismatch between $n_\lambda(\omega)$ and the vacuum value enhances reflection, whereas $k_\lambda(\omega)$ modifies the reflected intensity through optical dissipation. Therefore, the reflectivity $R_\lambda(\omega)$ characterizes the reflected optical response at a specific photon energy.

The following figure presents the calculated reflectivity $R_\lambda(\omega)$ spectra of α-Beryllene and β-Beryllene for the in-plane and out-of-plane directions [69, 71, 72, 94, 97, 100, 101]:

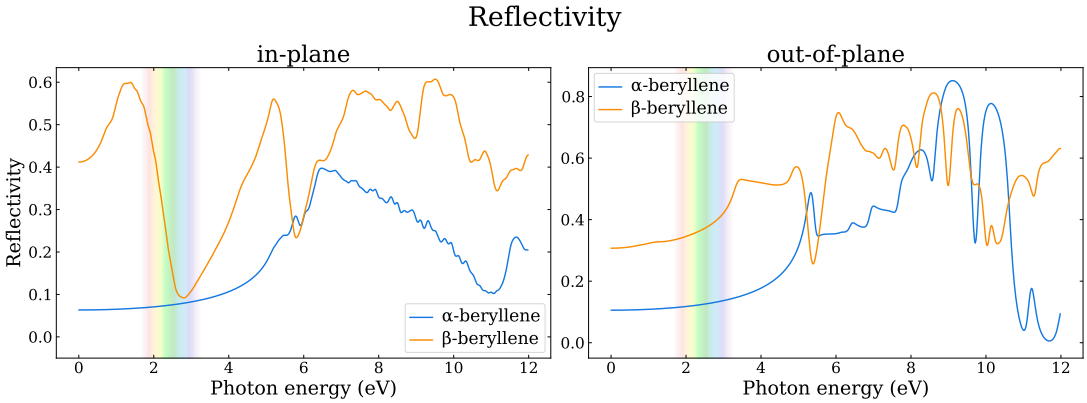

FIGURE 3.5. The reflectivities for the in-plane and out-of-plane directions as functions of photon energy for α-Beryllene and β-Beryllene.

The reflectivity spectra follow the directional contrast already demonstrated in the extinction coefficient. In the in-plane channel, β-Beryllene exhibits strong low-energy reflection, followed by a pronounced minimum across the visible range and renewed strong reflection from the near-ultraviolet into the ultraviolet region, whereas α-Beryllene varies more smoothly with a broad intermediate-energy feature. In the out-of-plane channel, β-Beryllene remains stronger over the low-energy range and then develops a series of rapid oscillations toward higher photon energies, while α-Beryllene shows pronounced peaks in the ultraviolet region. These trends reflect the combined role of the refractive index $n_\lambda(\omega)$ and the extinction coefficient $k_\lambda(\omega)$ in determining the surface reflectivity [69, 71, 72, 94, 97, 98, 100].

Accordingly, the complex refractive index $N_\lambda(\omega)$ directly determines the reflectivity $R_\lambda(\omega)$, which characterizes the surface reflection of the medium. We next turn to the energy-loss spectrum $L_\lambda(\omega)$ and identify the photon-energy range where the dielectric screening gives way to the strongest resonant dissipation.

3.4.4 Energy-loss spectrum

The previous subsection introduced the reflectivity $R_\lambda(\omega)$ as the optical response of the surface, characterized by the refractive index $n_\lambda(\omega)$ and the extinction coefficient $k_\lambda(\omega)$. We now move from surface reflection to energy dissipation inside the medium by examining the energy-loss spectrum $L_\lambda(\omega)$.

The energy-loss spectrum $L_\lambda(\omega)$ characterizes the energy transferred from an external probe to the medium at a given photon energy. It therefore describes the strength of longitudinal energy dissipation and identifies the energy range where the loss response peaks. The energy-loss spectrum $L_\lambda(\omega)$ is directly related to the loss channels measured in electron energy-loss spectroscopy.

We first express the general form of the energy-loss spectrum $L_\lambda(\omega)$ along the principal direction λ as [65, 78, 82, 96]:

$$L_\lambda(\omega) = \mathrm{Im}\left[-\frac{1}{\varepsilon_{\lambda\lambda}(\omega)}\right] \tag{3.51}$$

This expression indicates that the energy-loss spectrum $L_\lambda(\omega)$ is governed by the inverse dielectric function $1/\varepsilon_{\lambda\lambda}(\omega)$. For a medium, it therefore follows not only the imaginary part $\varepsilon_{2,\lambda\lambda}(\omega)$, but also the screening effect determined by the full complex dielectric function $\varepsilon_{\lambda\lambda}(\omega)$.

When the dielectric screening weakens, the energy-loss spectrum $L_\lambda(\omega)$ intensifies, near the point where the real part $\varepsilon_{1,\lambda\lambda}(\omega)$ of the dielectric function approaches zero and the imaginary part $\varepsilon_{2,\lambda\lambda}(\omega)$ remains finite [65, 82, 89].

Substituting equation (3.25) into equation (3.51), we then obtain the explicit expression adopted in the present work:

$$L_\lambda(\omega) = \frac{\varepsilon_{2,\lambda\lambda}(\omega)}{\varepsilon_{1,\lambda\lambda}^2(\omega) + \varepsilon_{2,\lambda\lambda}^2(\omega)} \tag{3.52}$$

This expression separates the dissipative channel from the dielectric screening. The imaginary part $\varepsilon_{2,\lambda\lambda}(\omega)$ provides the dissipative channel, whereas the denominator contains the total dielectric screening. Therefore, the energy-loss spectrum $L_\lambda(\omega)$ intensifies when the medium supports strong dissipation but provides limited dielectric

screening against the external perturbation. Meanwhile, it identifies the dominant dissipation channels more directly than the absorption coefficient $\alpha_\lambda(\omega)$ or the reflectivity $R_\lambda(\omega)$ [65, 82, 89, 102].

The following figure presents the calculated energy-loss $L_\lambda(\omega)$ spectra of α-Beryllene and β-Beryllene for the in-plane and out-of-plane directions [69, 71, 72, 94, 97, 100, 101]:

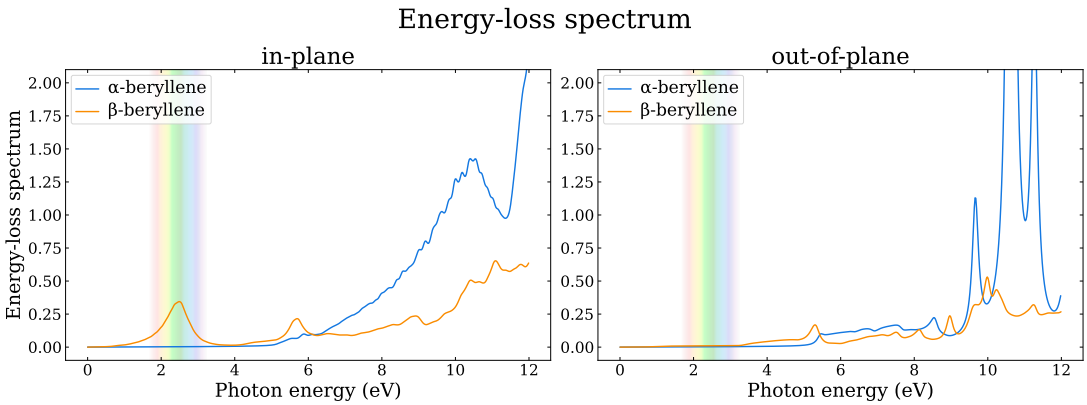

FIGURE 3.6. The energy-loss spectra for the in-plane and out-of-plane directions as functions of photon energy for α-Beryllene and β-Beryllene.

The energy-loss spectra demonstrate clear directional and structural contrasts. In the in-plane channel, β-Beryllene develops pronounced peaks in the visible and deep-ultraviolet regions, with the visible-range peak being stronger, whereas α-Beryllene remains weak at low photon energy and then rises gradually on entering the deep-ultraviolet region. In the out-of-plane channel, both materials remain nearly zero in the visible range and then increase gradually toward the near-ultraviolet region, while β-Beryllene rises slightly faster and develops a more evident peak in the deep-ultraviolet range.

These results demonstrate that the energy-loss spectrum $L_\lambda(\omega)$ varies with both the optical direction and the crystal structure of the medium. As $L_\lambda(\omega)$ follows the inverse dielectric function, its peaks identify the energy range where dielectric screening changes most critically. Therefore, the energy-loss spectrum $L_\lambda(\omega)$ characterizes the energy transferred from an external probe to the medium and more directly identifies the dominant dissipation channels than absorption or reflectivity alone [69, 71, 72, 89, 94, 97, 100].

Consequently, we arrive at the final optical quantity derived from the dielectric function $\varepsilon_{\lambda\lambda}(\omega)$ and complete the chain from the absorption coefficient $\alpha_\lambda(\omega)$, to the refractive index $n_\lambda(\omega)$ and extinction coefficient $k_\lambda(\omega)$, to the reflectivity $R_\lambda(\omega)$, and finally to the energy-loss spectrum $L_\lambda(\omega)$.

Chapter 4

Electronic and Optical Properties of 2D Heterostructure Bilayers of Graphene, Borophene and 2D Boron Carbides from First-Principles

4.1 Abstract

In this chapter, we investigate the atomic, electronic, and optical properties of two-dimensional heterostructure bilayers, namely Graphene-BC_3, Graphene-Borophene, and Graphene-B_4C_3, as well as their constituent monolayers, on the basis of first-principles calculations using the HSE06 hybrid functional. The calculations show that while borophene is metallic, both monolayer BC_3 and B_4C_3 are indirect semiconductors, with band gaps of 1.822 eV and 2.381 eV as obtained using HSE06. The Graphene-BC_3 and Graphene-B_4C_3 bilayer heterojunction systems maintain the Dirac point of graphene at the K point and are semimetals, while Graphene-Borophene is metallic. All bilayer heterostructure systems exhibit absorption in the visible region, where the resonance frequency and resonance absorption peak intensity vary between structures. In addition, all heterojunctions support plasmons within the range 16.5 eV to 18.5 eV, while Graphene-B_4C_3 and Graphene-Borophene exhibit a π-type plasmon within the region 4 eV to 6 eV, with the latter possessing an additional plasmon at the lower energy of 1.5 eV to 3 eV. The dielectric tensor for Graphene-B_4C_3 exhibits complex off-diagonal elements due to the lower P3 space-group symmetry, indicating anisotropic dielectric properties and possible optically active (chiral) effects. These results show that the two-dimensional heterostructures have desirable optical properties and broaden their potential applications.

4.2 Introduction

Graphene has excellent electrical conductivity, desirable mechanical and thermal properties, and high light transmittance in the visible-light and infrared regions. It also has an extremely high quantum efficiency for light-matter interactions and is strongly optically nonlinear. These properties have therefore enabled applications in e.g. electronics, energy storage, biomedical devices, and other fields [105]. Relative to conventional plasmonic materials, graphene possesses highly confined plasmons with much longer lifetimes. In addition, graphene plasmons are active in an extended wavelength range, namely, the mid-infrared and terahertz regime. This wavelength range overlaps with the characteristic signals of most organic and biomolecules, which has broadened graphene's applications towards plasmonic biological and chemical sensors [106] in addition to plasmonic waveguides and infrared photodetectors. Moreover, forming graphene-hybrid materials can tune the properties of graphene, finding applications in biosensors, chemical sensors, optical sensors, and other types of sensors [107].

In other areas, however, the use of graphene remains limited by its zero band gap. One method used to expand the application of graphene is to form heterostructures. Stacking different two-dimensional (2D) materials together can form double-layer or even multi-layer artificial materials that are maintained by van der Waals interactions [108]. Such materials are known as van der Waals heterojunctions. Such stacking can generate a wide range of physical properties, making van der Waals heterojunctions even more important than the individual 2D materials themselves. There are vast numbers of 2D materials, and by forming heterostructures thereof, a diverse range of electrical, photonic, plasmonic, mechanical, and chemical properties can be theoretically predicted through *ab initio* calculations and machine learning [109, 110], and experimentally engineered [99, 102].

Recently, the prospect of developing plasmonic devices employing 2D semiconductors has attracted considerable attention [111, 112]. Plasmonic modes in each class of van der Waals semiconductors have their own peculiarities, along with potential technological capabilities. Graphene-like BC_3 has also been synthesized experimentally [113, 114, 115] and is found to be semiconducting with an indirect band gap. Calculations have shown that nanostructures of graphene-like BC_3 exhibit desirable absorption in the visible region and, by changing the size of the nanostructure, the resonance peak position of the absorption spectrum can be effectively regulated [116]. Recent studies have also predicted a stable B_4C_3 monolayer, which is even more stable than

the previously synthesized BC_3 monolayer [117]. Boron carbides are known for their exceptional hardness, thermal stability, and chemical inertness. The emergence of 2D boron carbides adds a new dimension to the study of these materials, offering novel properties and opportunities for technological innovation [118, 119].

Borophene is another 2D material of high current interest. This monolayer allotrope of boron forms several structures and is characterized by its unique atomic structure and exceptional electronic properties, which have opened new avenues for innovation in the areas of electronics, energy storage, catalysis, plasmonics, superconductivity, and sensors [112]. One particularly intriguing aspect of borophene's versatility lies in its ability to form heterojunctions—interfaces between different borophene structures or between borophene and other materials. Borophene-graphene heterostructures have been realized [120], where they have shown good stability and the ability to function as humidity sensors, exhibiting a sensitivity about 700 times higher than that of pristine graphene, and 27 times higher than that of borophene. Recently, both lateral and vertical integration of borophene with graphene has also been achieved [121], as has the formation of bilayer (α-phase) borophene on the Ag(111) surface [122].

The synergistic combination of graphene's exceptional electrical conductivity, mechanical strength, and flexibility with boron carbide's high hardness, thermal stability, and chemical resistance, and with borophene's attractive electronic properties, holds significant potential for a wide range of applications spanning from electronics and optoelectronics to energy storage and sensing. As research in this field continues to advance, the development of scalable synthesis methods, the elucidation of structure-property relationships, and the exploration of novel applications will be key focal points. To date, little is known about the physical properties of graphene-based bilayer heterojunctions with borophene and boron-carbide materials. This motivates the present chapter to gain an understanding of these heterojunctions through the study of their electronic and optical properties on the basis of first-principles calculations. Such understanding is crucial for their future technological application, where the unique properties of these materials can be harnessed.

This chapter is organised as follows: In Section 4.3, the calculation method is described, followed by the Results and Discussion in Section 4.4. Herein, we describe the physical properties of the monolayers and heterostructures, including band structure, density of states, electron density difference distributions, Schottky barriers and the optical properties. This is followed by the Conclusion in Section 4.5.

4.3 Calculation methods

We perform first-principles calculations using density functional theory (DFT) as implemented in the Vienna Ab initio Simulation Package (VASP) [123, 124, 125]. Projector augmented wave (PAW) pseudopotentials account for the electron-ion interactions [126]. For the structural optimization calculations, we use the generalized gradient approximation (GGA) of Perdew-Burke-Ernzerhof (PBE) as the exchange-correlation functional. The Becke-Johnson damped D3(BJ) dispersion correction accounts for long-range van der Waals (vdW) interactions. The structural calculations use a Γ-centered $\vec{k}$-point sampling mesh of $11 \times 11 \times 1$, except for the smaller graphene unit cell shown in figure 4.1, for which a denser mesh of $23 \times 23 \times 1$ is used. Here, the plane-wave energy cutoff is set to 450 eV. The electronic convergence criterion is set to 1×10^{-6} eV, and the systems are allowed to relax until all forces on each atom are below 0.01 eV Å^{-1}. For smearing at the Fermi level, we use the Gaussian smearing method implemented in VASP.

For the calculation of electronic properties, the PBE exchange-correlation functional is known to underestimate band gap energies. Therefore, we also use the screened hybrid exchange-correlation functional of Heyd, Scuseria, and Ernzerhof (HSE06), which gives more accurate band gap energies [49, 50, 127].

All density of states (DoS) calculations use a Γ-centered $\vec{k}$-point sampling mesh of $33 \times 33 \times 1$, with smearing at the Fermi level calculated using the Gaussian method and a broadening value of 0.1 eV. The HSE06 exchange-correlation functional is also used for the calculation of optical properties, in particular, the frequency-dependent dielectric function with Γ-centered $\vec{k}$-point sampling meshes of $17 \times 17 \times 1$ and $65 \times 65 \times 1$. The complex frequency-dependent dielectric function, $\varepsilon(\omega)$, is given by equation 3.20; here, $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$ are the real and imaginary parts, respectively, and ω denotes the photon energy in eV.

We then use the dielectric function to calculate the absorption coefficient, refractive index, extinction coefficient, reflectivity, and energy-loss spectrum, following the relations introduced in section 3.4.

The interlayer binding energy E_b follows as:

$$E_\mathrm{b} = E_\mathrm{H} - E_\mathrm{BC/B} - E_\mathrm{G} \tag{4.1}$$

here, E_{H} denotes the total energy of the heterostructure, $E_{\mathrm{BC/B}}$ corresponds to the total energy of the isolated boron-carbide or borophene monolayer, and E_{G} denotes the total energy of the isolated graphene monolayer.

Afterwards, the charge density difference $\Delta\rho(\vec{r})$ takes the form:

$$\Delta\rho(\vec{r}) = \rho_{\mathrm{AB}}(\vec{r}) - \rho_{\mathrm{A}}(\vec{r}) - \rho_{\mathrm{B}}(\vec{r}) \tag{4.2}$$

here, $\rho_{\mathrm{AB}}(\vec{r},)$ denotes the total charge density of the heterostructure. The terms $\rho_{\mathrm{A}}(\vec{r},)$ and $\rho_{\mathrm{B}}(\vec{r},)$ represent the charge densities of the isolated monolayers forming the heterostructure.

4.4 Results and discussion

4.4.1 Monolayer properties

We first examine the structural and electronic properties of the three non-graphene monolayers, namely BC_3, B_4C_3, and borophene. Graphene is also investigated because it always forms one layer of the heterostructure systems. The optimized structures of BC_3, B_4C_3, borophene, and graphene are shown in figure 4.1. Borophene monolayers have been predicted to possess various low-energy structures, among which the β_{12}, χ_3, and α' structures have been synthesised experimentally [128]. For this study, we choose to investigate the α' structure of borophene, not only because it has been synthesised experimentally, but also because GGA-level *ab initio* calculations have shown it to have the most favourable cohesive energy [129].

The structural properties and band gap energies of the monolayer systems are shown in table 4.1. For graphene, the calculated lattice constant is $a = 2.467$ Å, while those of borophene, BC_3, and B_4C_3 are 5.051 Å, 5.169 Å, and 4.690 Å, respectively. In agreement with previous calculations, graphene is a semimetal and borophene is metallic, whereas BC_3 and B_4C_3 are indirect semiconductors [130, 131]. The calculated indirect band gaps of BC_3 and B_4C_3 are 1.822 eV (0.634 eV) and 2.381 eV (1.647 eV), respectively, as obtained using HSE06 (PBE).

Figure 4.16 shows the optimization of the lattice constants for the monolayers, while figure 4.2 presents the band structures of the four monolayer systems.

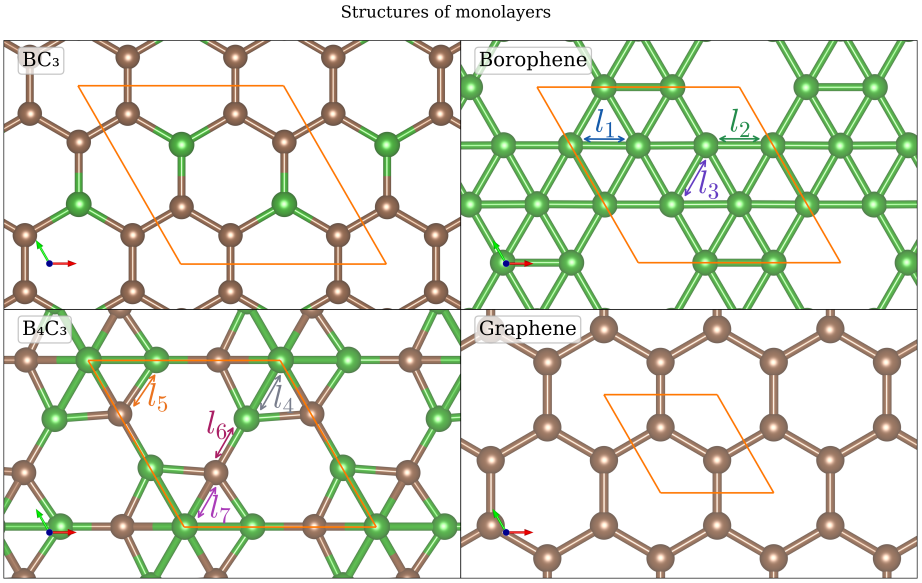

FIGURE 4.1. Optimized atomic structures of BC_3, borophene, B_4C_3, and graphene. Green and brown spheres denote boron and carbon atoms, respectively. Borophene has three unique bonds indicated by l_1, l_2 and l_3, while B_4C_3 has four unique bonds indicated by l_4, l_5, l_6 and l_7. Orange lines highlight the unit cells.

system	a ($a = b$) (Å)	B-B (Å)	C-C (Å)	B-C (Å)	E_{g-PBE} (eV)	$E_{g-HSE06}$ (eV)	Nature
BC_3	5.169	-	1.421	1.563	0.634 (0.62–0.66) [130, 132, 133]	1.822 (1.83)[133]	indirect
Borophene	5.051	1.689 (l_1) 1.672 (l_2) 1.707 (l_3)	-	-	-	-	metallic
B_4C_3	4.690	1.689 (l_4)	-	1.592 (l_5) 1.517 (l_6) 1.548 (l_7)	1.647	2.381 (2.39)[134]	indirect
Graphene	2.467	-	1.424	-	-	-	semimetal

TABLE 4.1. Optimized lattice constants, interatomic distances, and band gaps of monolayer BC_3, borophene, B_4C_3, and graphene. Here, a is the lattice parameter with $a = b$ for all systems, and B-B, C-C, and B-C are the boron-boron, carbon-carbon, and boron-carbon bond lengths, respectively. Borophene and B_4C_3 have B-B and B-C bonds of different lengths, which are identified by labels $l_1, \cdots, l_7$ in figure 4.1. The band gap E_g is calculated using the PBE and HSE06 exchange-correlation functionals; values in brackets are from previous theoretical studies. "Nature" indicates the electronic character of the system.

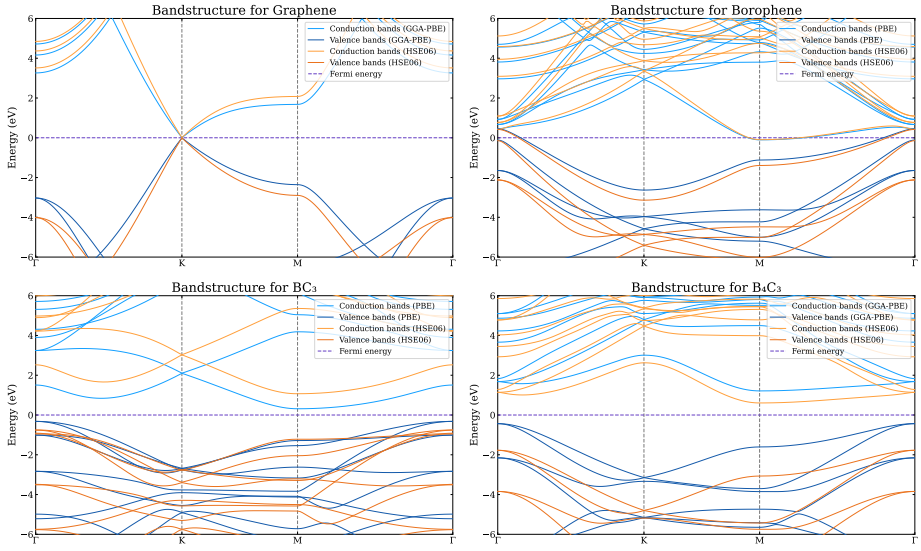

FIGURE 4.2. Band structures of graphene, borophene, BC_3, and B_4C_3 calculated using PBE (blue) and HSE06 (orange).

4.4.2 Heterostructure

4.4.2.1 Structural properties

We next determine the structures of the bilayer heterojunctions, which consist of monolayer graphene stacked with monolayer BC_3, borophene, and B_4C_3, namely Graphene-BC_3, Graphene-Borophene, and Graphene-B_4C_3, respectively. To create the heterostructures, we use a $2 \times 2 \times 1$ cell. For each system, we consider several lateral stacking coordinations of these layers. Specifically, we consider "Hollow", ‘"Bridge", and "Top" coordinations for the Graphene-BC_3 bilayer system, and "Hollow 1", "Hollow 2", "Bridge", and "Top" coordinations for the Graphene-Borophene and Graphene-B_4C_3 bilayer systems. We calculate the relative energies, optimized lattice constants, and interlayer distances for all structures. Figures 4.18–4.20 give details of this procedure. Due to the weak van der Waals interaction between the layers, the energy differences among the lateral positions are very small, as shown in table 4.3.

The results show that the lowest-energy lateral position for the Graphene-BC_3 system is the ‘Hollow" stacking configuration. As shown in figure 4.3, each atom in BC_3 projects to the center of a hexagon in graphene. Both the Graphene-Borophene and Graphene-B_4C_3 bilayer systems favor the ‘Top" stacking configuration. In this arrangement, each atom-free hexagon in the bottom layer projects to the center of a hexagon in graphene. However, the converse does not hold, because not every hexagon

center in graphene aligns with an atom-free hexagon in the bottom layer, as illustrated in figure 4.3.

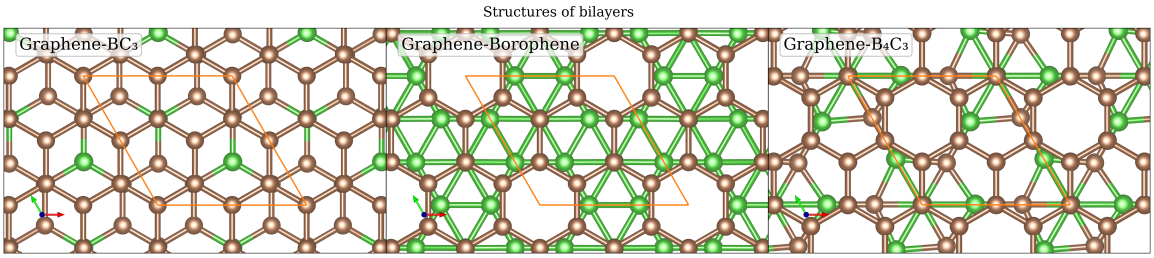

FIGURE 4.3. Top views of the optimized atomic structures of Graphene-BC_3, Graphene-Borophene, and Graphene-B_4C_3. Green and brown spheres denote boron and carbon atoms, respectively. The orange parallelogram indicates the unit cell.

system	coordination	a ($a = b$) (Å)	S_G	$S_{BC/B}$	d_{IL} (Å)	Nature
Graphene-BC_3	Hollow	5.044	2.23 %	−2.42 %	3.375	semimetal
Graphene-Borophene	Top	4.979	0.91 %	−1.43 %	3.496	metallic
Graphene-B_4C_3	Top	4.846	−1.78 %	3.33 %	3.514	semimetal

TABLE 4.2. Optimized structural properties of each bilayer heterojunction system, including the lowest-energy lateral stacking coordination relative to graphene, the optimized lattice constant a, the strains relative to the respective monolayers, S_G and $S_{BC/B}$, the interlayer distance d_{IL}, and the electronic nature. Negative and positive percentage strains denote compressive and tensile strains, respectively.

Table 4.2 shows that Graphene-Borophene is metallic, whereas Graphene-BC_3 and Graphene-B_4C_3 are semimetals. Among all heterostructure systems, B_4C_3 shows the largest tensile strain of 3.33 %. For Graphene-BC_3, graphene is under tensile strain and BC_3 is under compressive strain. For Graphene-Borophene, graphene has a slight tensile strain, while borophene experiences compressive strain. For Graphene-B_4C_3, graphene is under compressive strain, while B_4C_3 is under a larger tensile strain. In general, tensile strain tends to narrow the band gap and increase conductivity [135]. Conversely, compressive strain tends to widen the band gap and decrease conductivity [136].

4.4.2.2 Charge density difference

The charge density difference distributions of the heterostructure systems are shown in figure 4.4. We calculate these distributions using equation 4.2. Specifically, we subtract the charge densities of the isolated monolayers from the charge density of the corresponding heterostructure.

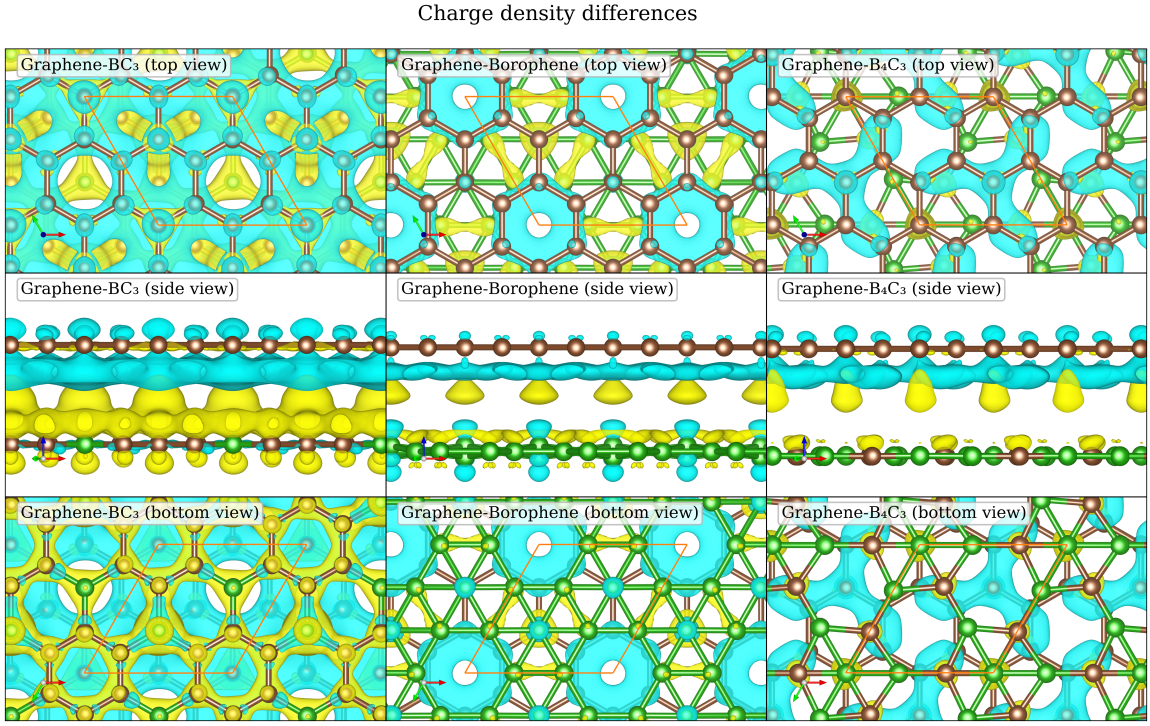

Figure 4.4. Charge density difference ($\Delta\rho(\vec{r})$, calculated using equation 4.2) between the monolayers and the heterostructures. Regions of charge accumulation are shown in yellow, and regions of charge depletion are shown in blue. The isosurface level is 1.5×10^{-4} a_0^{-3} and the top layer is always graphene.

The side views in the middle panels of figure 4.4 show charge depletion from the graphene layer and charge accumulation in the corresponding non-graphene monolayer for all three systems. Among these systems, Graphene-BC_3 exhibits the largest charge redistribution. This larger charge redistribution is consistent with the smaller interlayer distance associated with the hollow-site coordination. In contrast, Graphene-Borophene and Graphene-B_4C_3 adopt top-site coordination.

We also observe that atoms in the lower layers that directly coordinate with a carbon atom in the graphene layer have the largest charge accumulation regions. In contrast, the carbon atoms in the graphene layer with bridge or hollow coordination to the atoms of the lower layer show the largest regions of charge depletion. These coordination-dependent charge redistributions produce the accumulation and depletion patterns shown in the top and bottom views of figure 4.4. Therefore, the relative coordination of the two monolayers directly influences the interfacial electronic structure. The

top and bottom views of the Graphene-B_4C_3 heterostructure reveal a clear threefold rotational symmetry in the upper and lower plots of the rightmost panel.

In the next sections, we investigate the band structures and densities of states to further characterise the electronic structures of these heterostructure systems.

4.4.2.3 Band structure and density of states

For the bilayer systems, the band structures and total density of states (DoS) are shown in figures 4.5–4.7 as calculated using the HSE06 and PBE functionals. The corresponding atom-projected (or partial) PDoS are presented in figures 4.21–4.23.

The Graphene-BC_3 heterojunction exhibits a sizeable density of states just above the Fermi level, as shown in figure 4.22, which is related to the relatively flat band near the M point and along the K–Γ path in the Brillouin zone. The Graphene-Borophene system exhibits metallic behaviour due to the small occupation of states at the K point; these states are mainly carbon-related, as shown by the PDoS in figure 4.21. The Graphene-B_4C_3 bilayer exhibits essentially semimetallic behaviour, with the B_4C_3-derived states located away from the Fermi level, mainly above approximately 1.2 eV and below approximately −0.8 eV, as shown by the PDoS in figure 4.23.

Across these three bilayers, HSE06 mainly shifts the conduction bands upward and the valence bands downward relative to PBE. At the K point, the graphene bands exhibit very small band-gap openings of 20 meV and 50 meV for Graphene-B_4C_3 and Graphene-BC_3, respectively.

Figures 4.5–4.7 show the band structures and total DoS of the three bilayer systems calculated using the PBE and HSE06 functionals. In each figure, the PBE and HSE06 results are shown in blue and orange, respectively, and the Fermi level is indicated by the purple dashed line.

4.4.2.4 Schottky barrier height

Bilayer heterostructures composed of graphene and boron-carbide monolayers can function as Schottky diodes. A high Schottky barrier blocks the reverse current from the metal to the semiconductor and enables the rectification effect of a diode. Based on the Schottky–Mott model shown in figure 4.8, the Schottky barrier at a metal–semiconductor interface [137] is determined by the relative positions of the Fermi level and the semiconductor band edges.

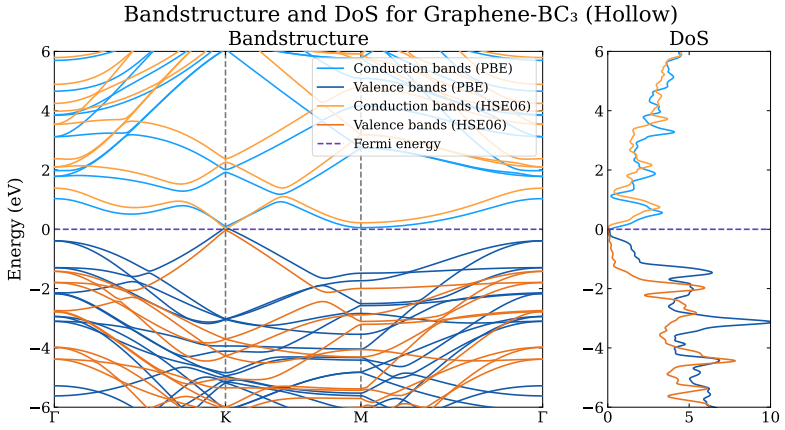

FIGURE 4.5. Band structure and total DoS of Graphene-BC_3.

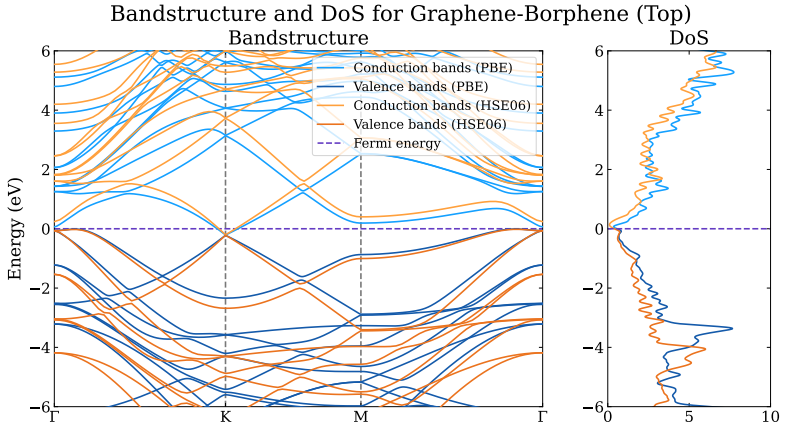

FIGURE 4.6. Band structure and total DoS of Graphene-Borophene.

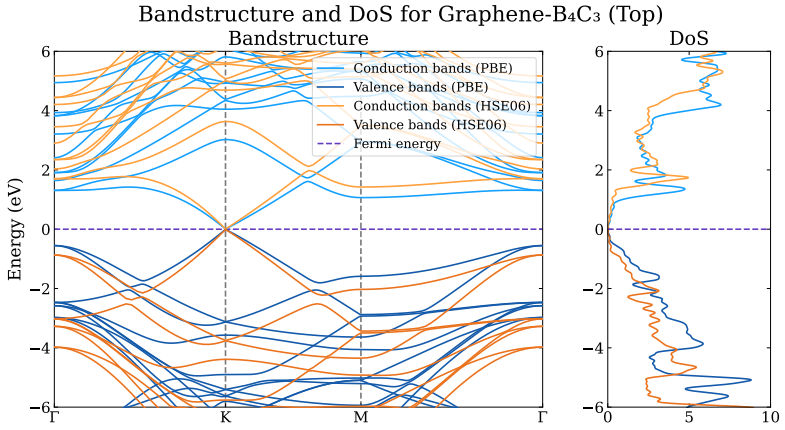

FIGURE 4.7. Band structure and total DoS of Graphene-B_4C_3.

For n-type materials, the Schottky barrier Φ_n is defined as the energy difference between the conduction band minimum E_C and the Fermi level E_F:

$$\Phi_n = E_C - E_F \tag{4.3}$$

Similarly, for p-type materials, the Schottky barrier Φ_p is defined as the energy difference between the Fermi level E_F and the valence band maximum E_V:

$$\Phi_p = E_F - E_V \tag{4.4}$$

The sum of these two Schottky barriers is approximately equal to the semiconductor band gap E_g:

$$\Phi_n + \Phi_p \approx E_g \tag{4.5}$$

For the Graphene-B_4C_3 system illustrated in figure 4.8, we find $\Phi_n = 1.42\,\text{eV}$ (1.07 eV) and $\Phi_p = 0.87\,\text{eV}$ (0.55 eV) using HSE06 (PBE). Since $\Phi_p < \Phi_n$, this heterojunction forms a p-type Schottky barrier. For the Graphene-BC_3 heterostructure bilayer, we find $\Phi_n = 0.19\,\text{eV}$ (0.003 eV) and $\Phi_p = 1.42\,\text{eV}$ (0.39 eV) using HSE06 (PBE). This result indicates that the Graphene-BC_3 heterojunction forms an n-type Schottky barrier. For device applications, a smaller Schottky barrier height, or even an Ohmic contact, is preferred to reduce contact resistance.

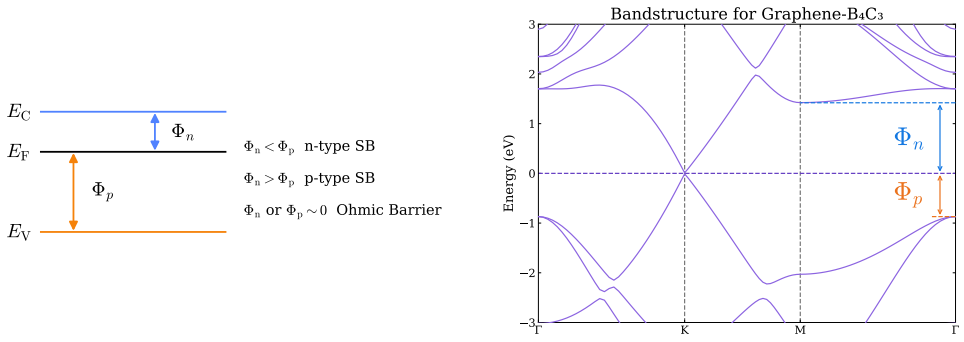

FIGURE 4.8. The left panel shows a schematic of the Schottky–Mott model, including the valence and conduction band energies, the Fermi energy, and the n-type and p-type Schottky barriers labelled E_V, E_C, E_F, Φ_n, and Φ_p, respectively. The right panel shows an example of the HSE06-calculated Graphene-B_4C_3 heterojunction and the Schottky barrier height determined from the band structure.

4.4.3 Dielectric function

Having determined the atomic and electronic properties of the bilayer systems, we calculate the real part ε_1 and imaginary part ε_2 of the dielectric function, from which the other optical properties can be determined, as given by equations 3.40, 3.45, 3.46, 3.50, and 3.51. The real part of the frequency-dependent dielectric function $\varepsilon_1(\omega)$ characterizes the strength of the dynamical screening from charge excitations, whereas the imaginary part $\varepsilon_2(\omega)$ describes light absorption from neutral and plasmonic charge excitations. A negative value of $\varepsilon_1(\omega)$ indicates metallic character in the optical response over the corresponding photon-energy range.

It is first important to establish the appropriate $\vec{k}$-point set to yield converged results. For graphene, we test a number of $\vec{k}$-point meshes and find that a $129 \times 129 \times 1$ mesh gives converged results, as shown in figure 4.24, in good agreement with previous results [138]. We therefore use this $\vec{k}$-point sampling density for the other single monolayers, whose unit cells are 2×2 relative to graphene, namely a $65 \times 65 \times 1$ $\vec{k}$-point mesh. The dielectric functions for all the monolayers are shown in figure 4.25, where the results are consistent with previous calculations, namely graphene [138], BC_3 [116, 139], B_4C_3 [134], and borophene [140, 141].

The anisotropic in-plane and out-of-plane components of the dielectric function show that the optical response depends strongly on the polarization direction of the incoming light. For the in-plane direction, the well-known π plasmon of graphene emerges around 4 eV, and the $\pi - \sigma$ plasmon emerges around 14 eV. The other monolayer systems also have a π-like plasmon, which emerges for B_4C_3 around 3 eV, for BC_3 around 2 eV, and for borophene around 1 eV. It can furthermore be seen from figure 4.25 that, for the real part of the dielectric function in the out-of-plane direction for borophene, a plasmon emerges at 9.8 eV, as evidenced by the negative value that becomes positive around 10.5 eV.

As previously reported, the latter three monolayers exhibit visible-light absorption, which is also indicated by the in-plane imaginary part of the dielectric function shown in figure 4.25. The static dielectric constants, namely the real part of the dielectric function at zero energy, for BC_3 are 4.0 for in-plane polarization and 1.4 for out-of-plane polarization as obtained using PBE. These values can be compared with previously reported PBE values of 4.86 for in-plane polarization and 1.59 for out-of-plane polarization [139].

For B_4C_3, the in-plane static dielectric constant is 3.9, and the out-of-plane value is calculated to be 1.5. We note that these values reflect the combined monolayer-vacuum system. Figure 4.26 shows the corresponding optical properties for the monolayer systems.

Before presenting the results for the three heterostructure bilayer systems, we first perform a convergence check for the number of unoccupied bands included in the calculation of the dielectric function. Figure 4.27 shows that 128 bands yield well-converged results for the representative Graphene-B_4C_3 system, and we therefore use this value for all three bilayer calculations. Further tests of the energy convergence criterion are reported in figure 4.28.

The results for the dielectric function of the Graphene-Borophene heterojunction are shown in figure 4.9, as obtained using both the PBE and HSE06 functionals. The corresponding full dielectric tensor components are provided in figure 4.29. Compared with the PBE results, HSE06 yields a blue shift of the features to higher energies. The real part exhibits characteristics of both graphene and borophene, with plasmons emerging around the same energies as those of the respective monolayers shown in figure 4.25, namely around 5 eV (4 eV) and 2 eV (1 eV) for HSE06 (PBE), respectively. There is, however, a significant difference in the out-of-plane dielectric function compared with the monolayer systems. The borophene-related plasmon that emerges at around 9.8 eV in the monolayer shifts to higher energies and now appears at around 17 eV (14 eV) for HSE06 (PBE), with a plasmon resonance frequency at about 18 eV (15 eV) for HSE06 (PBE), as determined by the energy at which the real part of the dielectric function changes from negative to positive. In addition, a new feature with two peaks occurs around 2 eV to 5 eV (1 eV to 4 eV) for HSE06 (PBE) in both the real and imaginary parts of the dielectric function. This feature in the imaginary part is attributed to interband transitions involving borophene- and graphene-derived states around the K point and along the direction towards the M and Γ points. A similar feature appears for the other bilayer heterojunctions, although it shows only one main peak. For in-plane polarization, the first main peak of ε_2 occurs in the visible range and is related to $\pi - \pi^*$ transitions. The main peaks along the out-of-plane direction are broad and occur in the energy range 9 eV to 18.5 eV (8 eV to 17.5 eV) for HSE06 (PBE), being related to $\pi - \sigma^*$ and $\pi^* - \sigma$ transitions.

The results for the dielectric function of the Graphene-BC_3 heterostructure are shown in figure 4.10, as obtained using both the PBE and HSE06 functionals. The corresponding full dielectric tensor components are provided in figure 4.30. Interestingly, and in

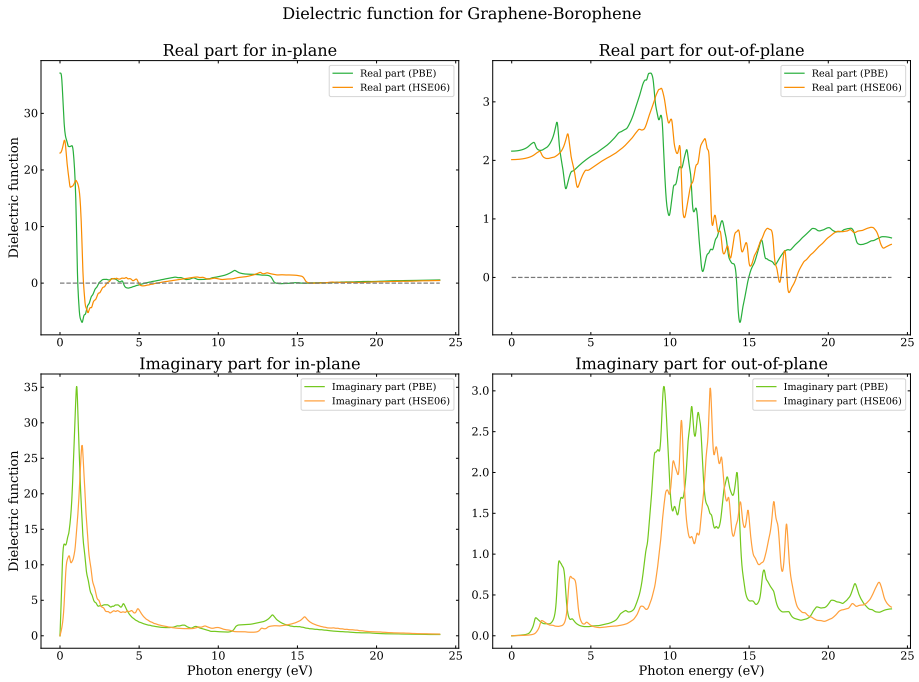

Figure 4.9. In-plane and out-of-plane dielectric-function components of the Graphene-Borophene heterostructure.

contrast to Graphene-Borophene, the dielectric tensor for this system contains complex non-zero off-diagonal components. These components satisfy the antisymmetric relations $\varepsilon_{xy} = -\varepsilon_{yx}$, $\varepsilon_{xz} = -\varepsilon_{zx}$, and $\varepsilon_{yz} = -\varepsilon_{zy}$. The dielectric tensor describes the response of a material to an external electric field, or equivalently, how the polarization of the material changes under the applied electric field. For an incident photon, the tensor components describe the response of the material to the photon electric-field component along different directions. For example, the elements ε_{xy} and ε_{yx} describe the coupling between the x and y directions, namely the response to an electric field applied along one direction resulting in polarization along the other direction, and vice versa. However, these off-diagonal components are mainly non-zero only in the low-energy region and remain small in magnitude. Therefore, the following discussion focuses on the in-plane and out-of-plane diagonal components, which describe the dominant optical response for the polarization directions considered here.

In contrast to the Graphene-Borophene system, the real part of the dielectric function is positive over the whole energy range considered. There are, however, energy ranges where it approaches zero, namely in the region 2.5 eV to 5 eV and from 13 eV (15 eV) onwards for PBE (HSE06). When ε_1 is zero but not negative, this suggests that the

system is close to resonance but has not fully entered the regime where strong optical reflection or plasmonic modes are dominant, since such modes typically require $\varepsilon_1 < 0$. Another difference from the Graphene-Borophene system is the sizeable in-plane value of both ε_1 and ε_2 in the region 10 eV to 15 eV. Similar to the Graphene-Borophene system, however, the out-of-plane real part of the dielectric function becomes negative at around 16 eV to 17 eV (14 eV to 15 eV) for HSE06 (PBE), indicating the presence of a plasmon. However, monolayer BC_3 does not exhibit this feature, unlike borophene, as shown in figure 4.25. It is therefore presumably related to the $\pi - \sigma$-type plasmon present for graphene at approximately 15 eV, but involves transitions between states associated with both monolayers. For in-plane polarization, the first main peaks of ε_2 occur in the visible range and are related to $\pi - \pi^*$ transitions. The main peaks along the out-of-plane direction are broad and occur in the energy range 11 eV to 17 eV (9 eV to 16 eV) for HSE06 (PBE), being related to $\pi - \sigma^*$ and $\pi^* - \sigma$ transitions.

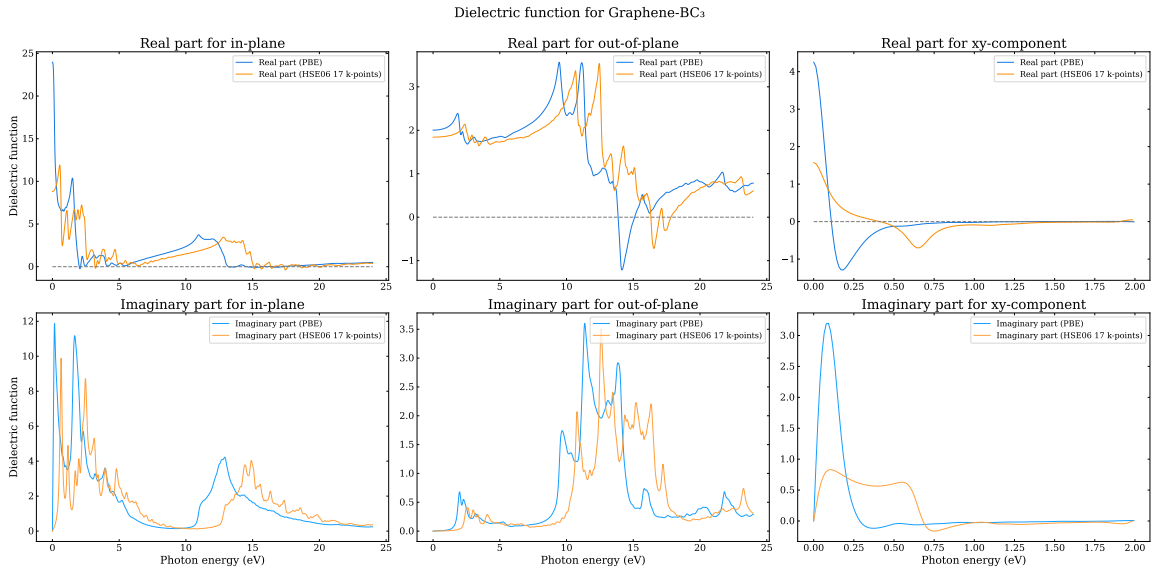

FIGURE 4.10. In-plane, out-of-plane, and off-diagonal ε_{xy} components of the dielectric function for the Graphene-BC_3 heterostructure.

The results for the dielectric function of the Graphene-B_4C_3 heterostructure are shown in figure 4.11, as obtained using both the PBE and HSE06 functionals. The corresponding full dielectric tensor components are provided in figure 4.31. The off-diagonal ε_{xy} component is also included in figure 4.11. The real part of the in-plane dielectric function is negative in two regions: 5 eV to 6 eV for HSE06 (4 eV to 5 eV for PBE) and 17 eV to 18 eV for HSE06 (15 eV to 16 eV for PBE). These negative regions indicate the presence of plasmons in the ultraviolet region. The out-of-plane

component also displays negative values at about 17 eV to 18 eV for HSE06 (14.5 eV to 15.5 eV for PBE).

Similar to Graphene-BC_3, the first main peaks of ε_2 for in-plane polarization are also in the visible range. The broad main peaks along the out-of-plane direction occur in the energy range 10 eV to 18 eV (9 eV to 17 eV) for HSE06 (PBE).

The dielectric tensor for this system also contains complex non-zero off-diagonal components. These components satisfy the antisymmetric relations $\varepsilon_{xy} = -\varepsilon_{yx}$, $\varepsilon_{xz} = -\varepsilon_{zx}$, and $\varepsilon_{yz} = -\varepsilon_{zy}$, as shown in figure 4.31. In contrast to Graphene-BC_3, however, these components remain non-zero over a considerable energy range, namely from 0 eV to 5 eV and from 10 eV to 15 eV. This behaviour is consistent with the lower symmetry of Graphene-B_4C_3. The system has a P3 space group, with threefold rotational symmetry along the c axis, corresponding to the z direction. It also lacks mirror-symmetry planes perpendicular to this axis or within the a, b plane, corresponding to the x, y plane. This lower symmetry allows the presence of off-diagonal elements in the dielectric tensor. Therefore, the dielectric response of this material can exhibit anisotropic coupling between different crystallographic directions. The presence of complex off-diagonal elements in the dielectric tensor of a material with the P3 space group indicates anisotropic and possibly chiral optical properties, which could lead to effects such as optical activity. It also indicates that Graphene-B_4C_3 dissipates energy differently depending on the direction of the electric field.

4.4.4 Derived optical properties

To further examine the optical properties of the heterostructure bilayers, we calculate the absorption coefficient, energy-loss spectrum, reflectivity, refractive index, and extinction coefficient.

Figure 4.12 shows the HSE06 absorption coefficients and energy-loss spectra of the three systems for in-plane and out-of-plane polarizations. The corresponding component-resolved PBE and HSE06 results are provided in figures 4.32–4.36.

The absorption threshold corresponds to the first peak of $\varepsilon_2(\omega)$. For all systems, the absorption coefficient starts from a very small value and increases with photon energy, consistent with the lowest direct transition energies in the band structure. All three systems absorb in the visible region, 1.7 eV to 3.3 eV. The Graphene-Borophene

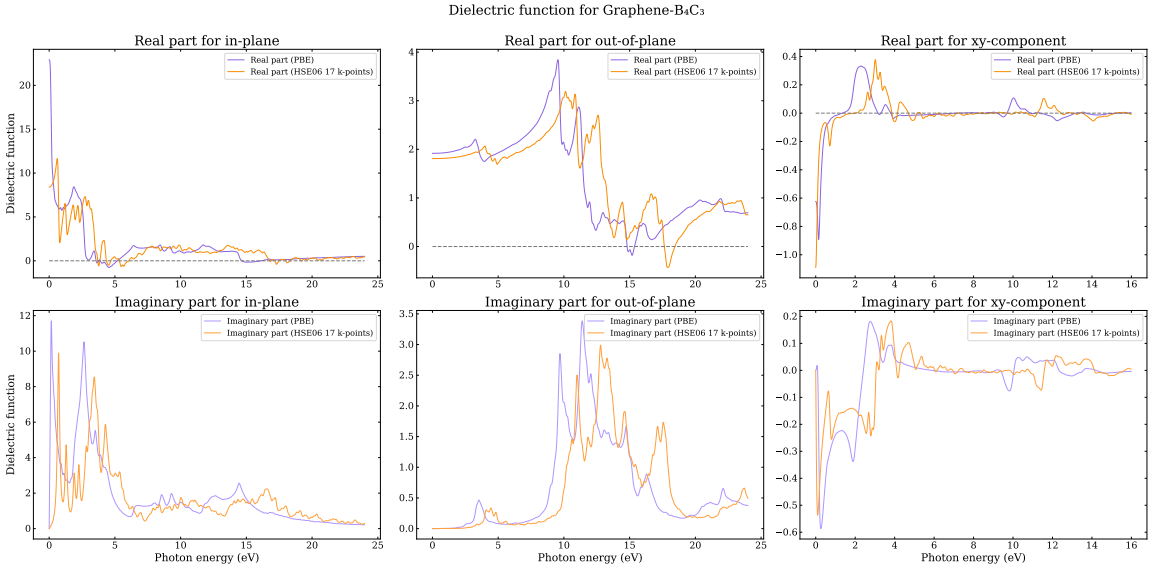

FIGURE 4.11. In-plane, out-of-plane, and off-diagonal ε_{xy} components of the dielectric function for the Graphene-B_4C_3 heterostructure.

heterostructure shows a significantly larger absorption coefficient than the other two systems, with a sharp rise around 1 eV. This feature can be attributed to transitions between the flat valence-band region around the Γ point and the low-lying conduction band in the same region of the Brillouin zone. The Graphene-BC_3 heterostructure shows a similar sharp rise at a slightly higher energy, around 2.5 eV, followed by Graphene-B_4C_3 at around 3.5 eV. These latter features arise from transitions between relatively flat valence and conduction bands along the Γ–M direction.

The energy-loss spectrum in figure 4.12 shows a sharp low-energy feature for Graphene-BC_3 and Graphene-B_4C_3, which is absent for Graphene-Borophene. This feature is related to Dirac-like plasmons. The broader features between 3 eV to 8 eV can be attributed to π-like plasmons. Individual interband transitions presumably coexist with these plasmons and form a broad background. When interband transitions and plasmons occur in the same energy range, the two types of excitations can couple, giving finite plasmon lifetimes as well as interband-transition character. The features around 15 eV, where a sharp increase begins, may be associated with $\pi - \sigma$-type plasmons. This assignment is consistent with the negative real part of the out-of-plane dielectric-function component, which indicates a plasmon around this energy for all three bilayer heterostructures.

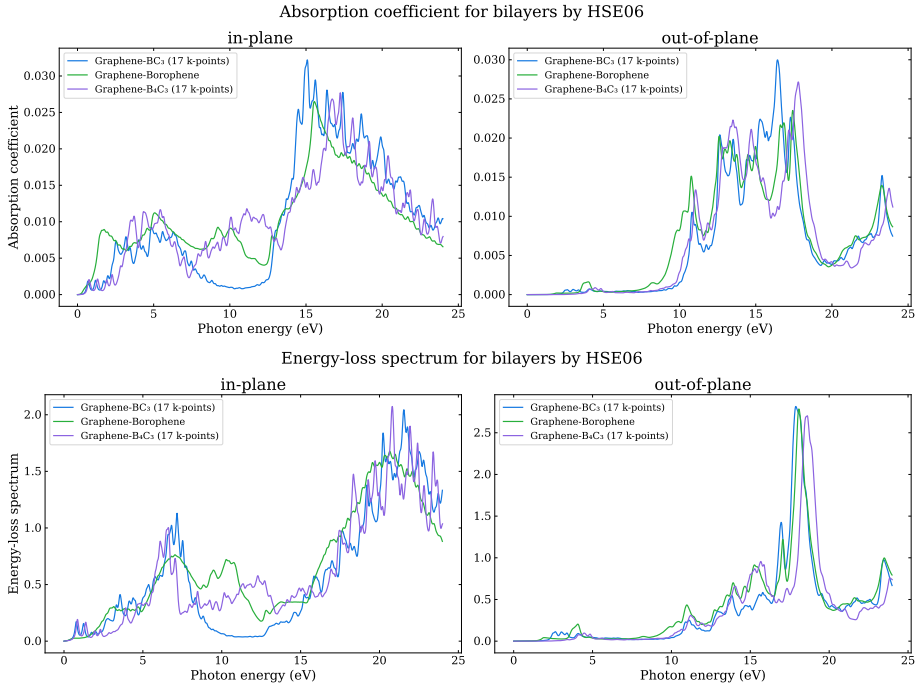

Figure 4.12. In-plane and out-of-plane absorption coefficients and energy-loss spectra of the three heterostructure bilayers.

Figure 4.13 shows the reflectivity and refractive index of the three heterostructures, as obtained using the HSE06 functional for in-plane and out-of-plane polarizations. The refractive index is related to the real and imaginary parts of the dielectric function. Its trend is similar to that of ε_1, indicating that the real part of the dielectric function plays the leading role. The refractive index determines how much the path of light is bent, or refracted, when light enters a material. For visible light, most transparent materials have refractive indices between 1 and 2. The absolute refractive index of an optical medium is defined as the ratio between the speed of light in vacuum and the phase velocity of light in the material, where the phase velocity is the speed at which the crests of the light wave propagate. Therefore, if the phase velocity is larger than the speed of light in vacuum, the refractive index can be smaller than 1. For frequencies near the plasmon resonance in 2D materials, the refractive index can drop below 1 due to the strong interaction between light and plasmonic oscillations.

The reflectivity spectra also show a clear polarization dependence. For in-plane polarization, Graphene-Borophene exhibits the strongest response below approximately 5 eV, while Graphene-BC_3 and Graphene-B_4C_3 show oscillatory but weaker reflection in the same range. All three systems then enter a low-reflectivity region from approximately

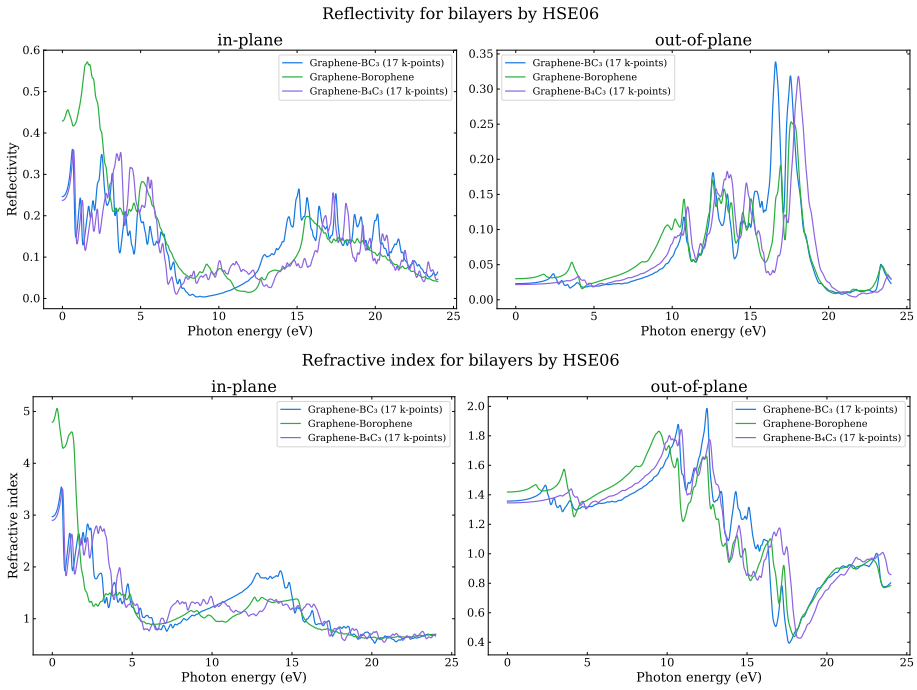

FIGURE 4.13. In-plane and out-of-plane reflectivity and refractive index of the three heterostructure bilayers.

7 eV to 13 eV. Above this range, Graphene-BC_3 and Graphene-B_4C_3 recover weak oscillatory features and gradually decay at higher photon energies, while Graphene-Borophene reaches a smaller peak around 16 eV before also decreasing. For out-of-plane polarization, the low-energy reflectivity remains small for all three systems, while pronounced peaks emerge around 16 eV to 18 eV, consistent with the plasmon-related structures discussed above.

For in-plane polarization, the refractive index is close to, or smaller than, 1 in the energy range of approximately 5 eV to 7 eV and above approximately 17 eV for all three systems. For out-of-plane polarization, the refractive index is close to, or smaller than 1 for energies above about 14 eV. These low-index regions occur near plasmon-related energies and indicate a strong dispersive dielectric response. Their optical transparency should be interpreted together with the extinction coefficient and energy-loss spectrum, because a refractive index close to or smaller than 1 does not by itself guarantee weak attenuation.

Figure 4.14 shows the HSE06 extinction coefficient for in-plane and out-of-plane polarizations. For in-plane polarization, the strongest response occurs in the low-energy region for all three heterostructures. Graphene-Borophene exhibits the largest

peak below approximately 2 eV and remains relatively strong up to about 6 eV, whereas Graphene-BC_3 and Graphene-B_4C_3 show weaker oscillatory features over approximately 1 eV to 6 eV. All three systems then enter a suppressed region around 7 eV to 13 eV, where the response of Graphene-BC_3 becomes particularly weak, followed by broader secondary features in the higher-energy range. For out-of-plane polarization, the extinction coefficient is small at low photon energies but rises strongly in the ultraviolet region, where pronounced peaks emerge mainly between approximately 10 eV to 18 eV. These energy regions therefore correspond to strong photon attenuation.

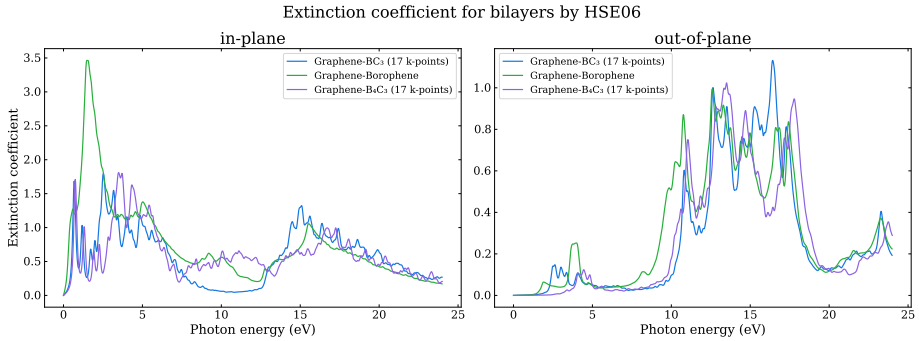

FIGURE 4.14. In-plane and out-of-plane extinction coefficients of the three heterostructure bilayers.

4.5 Conclusions

In the present work, we investigated the atomic, electronic, and optical properties of Graphene-Borophene, Graphene-BC_3, and Graphene-B_4C_3 bilayer heterostructure systems using first-principles calculations with the HSE06 and PBE functionals. We find that all systems have small binding energies between the layers, indicating that graphene interacts weakly with the monolayers through van der Waals interactions. The Graphene-Borophene bilayer is metallic, whereas the Graphene-BC_3 and Graphene-B_4C_3 structures maintain the characteristic linear bands of graphene around the K point, with only a very small band-gap opening of 20 meV to 50 meV. The calculated optical properties show absorption in the visible region of the spectrum for all heterostructures, with Graphene-Borophene exhibiting the strongest absorption. All heterostructures support plasmons. For Graphene-Borophene, two low-energy plasmons are observed, one emerging at 2 eV and the other around 5 eV. The Graphene-B_4C_3 bilayer heterostructure has non-negligible off-diagonal elements in the dielectric tensor over a range of energies, suggesting that it could exhibit optically active, possibly chiral, effects.

4.6 Data availability statement

The original data presented in this chapter are openly available from the GitHub repository: Photonico/Graphene-BC_20230622.

4.7 Supplementary information

The supplementary information presents the lattice-parameter determination for the graphene, borophene, BC_3, and B_4C_3 monolayers, as well as the corresponding results for the associated heterostructure bilayers under various relative lateral displacements. It also provides $\vec{k}$-point mesh convergence tests, projected densities of states, dielectric function convergence tests, and derived optical properties for the monolayer and bilayer systems investigated.

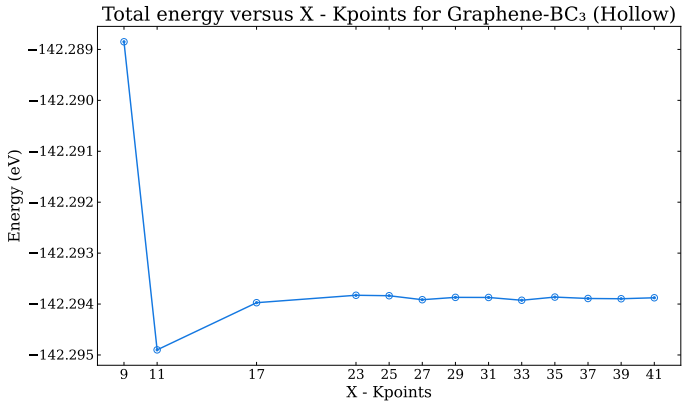

FIGURE 4.15. Total energy versus the $\vec{k}$-point mesh used to sample the Brillouin zone for the Graphene-BC_3 bilayer; here, X denotes an $X \times X \times 1$ $\vec{k}$-point mesh.

Figures 4.18–4.20 show the total energy versus the lattice constant and interlayer distance for the various lateral positions considered. In each figure, the equilibrium geometry is indicated by the "Extrema of fitted data" result.

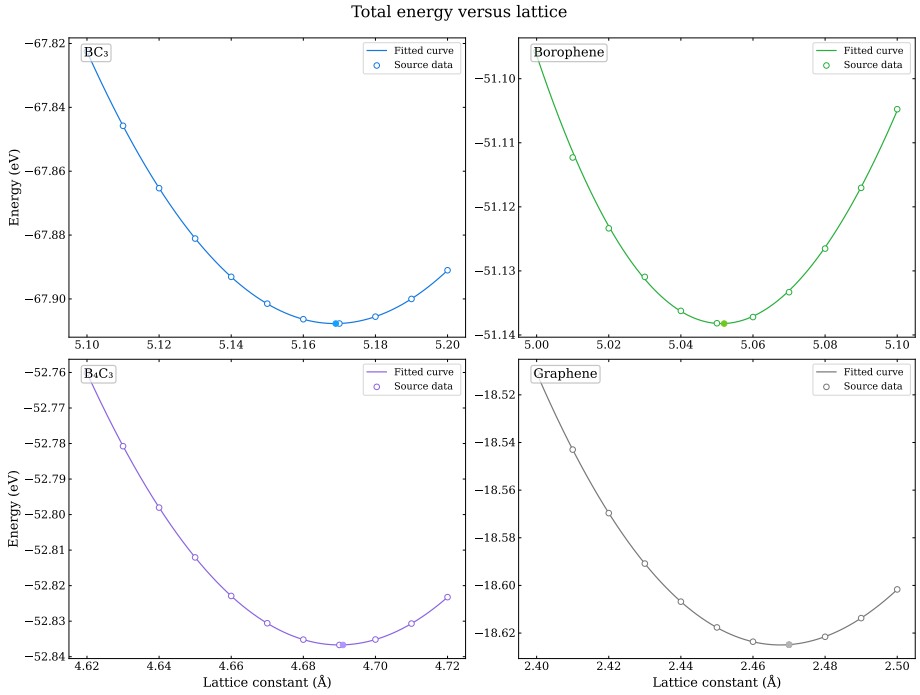

FIGURE 4.16. Total energy versus lattice constant for the four monolayer systems: graphene, borophene, BC_3, and B_4C_3. The lattice constant yielding the lowest total energy is shown by the full circle in each plot.

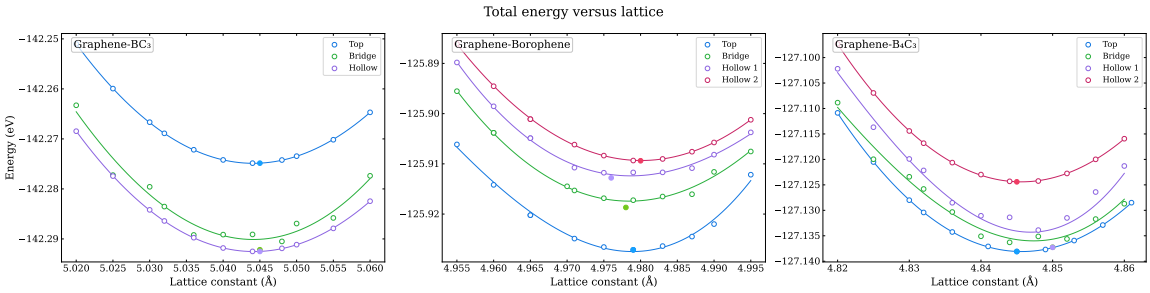

FIGURE 4.17. Total energy versus lattice constant for each bilayer system for the various stacking positions considered. The lines are fitted curves, with the full circle indicating the lowest-energy structure.

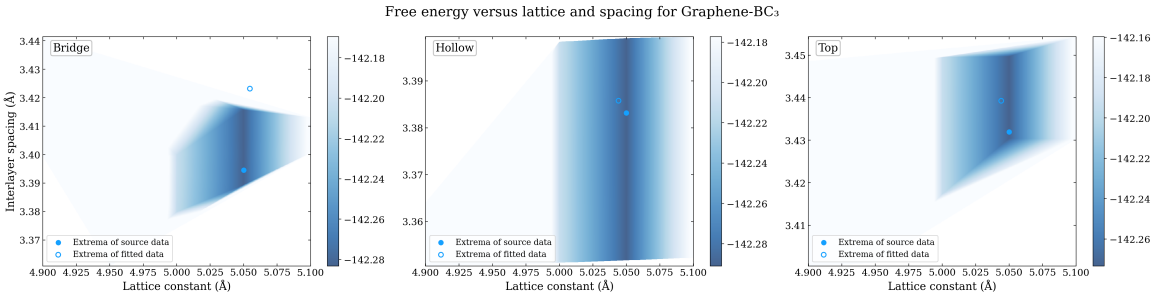

FIGURE 4.18. Total-energy landscape of the Graphene-BC_3 bilayer.

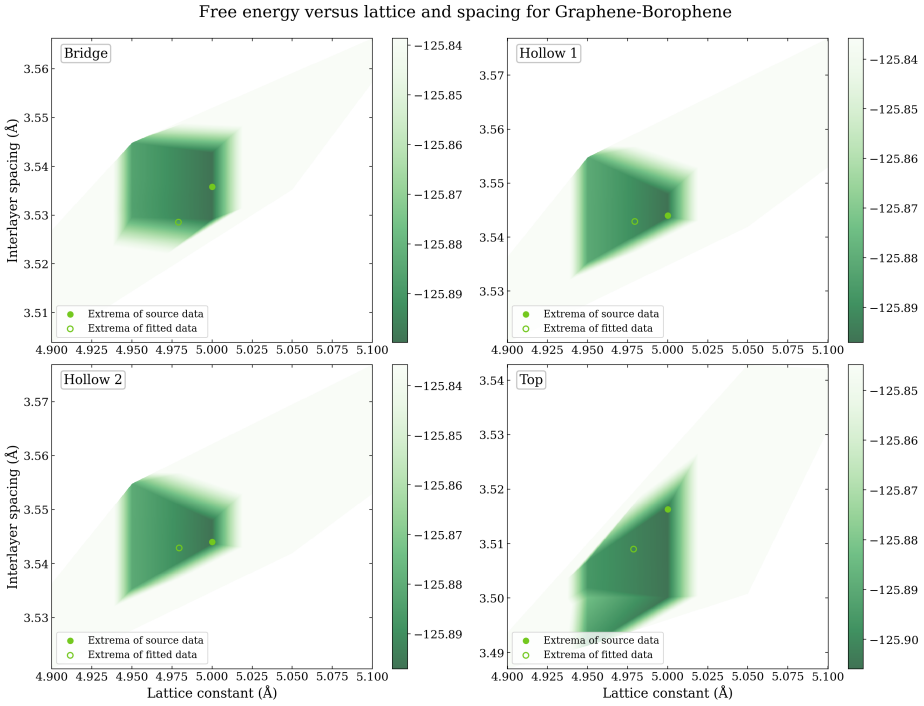

FIGURE 4.19. Total-energy landscape of the Graphene-Borophene bilayer.

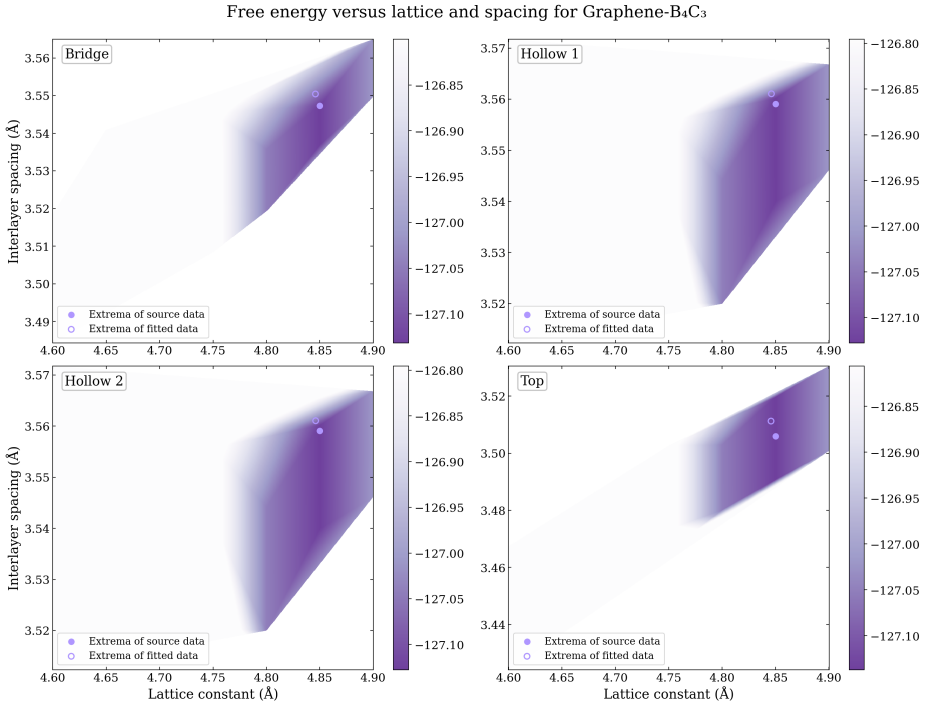

FIGURE 4.20. Total-energy landscape of the Graphene-B_4C_3 bilayer.

system	site	lattice constant (Å)	interlayer spacing (Å)	total energy (eV)	relative energy (eV)
Graphene-BC_3	Top	5.044	3.432	-142.275	0.018
	Bridge	5.044	3.384	-142.290	0.002
	Hollow	**5.044**	**3.375**	**-142.293**	**0.000**
Graphene-Borophene	**Top**	**4.979**	**3.496**	**-125.927**	**0.000**
	Bridge	4.979	3.505	-125.921	0.006
	Hollow 1	4.979	3.551	-125.914	0.013
	Hollow 2	4.979	3.548	-125.909	0.018
Graphene-B_4C_3	**Top**	**4.846**	**3.514**	**-127.138**	**0.000**
	Bridge	4.846	3.536	-127.135	0.003
	Hollow 1	4.846	3.545	-127.133	0.005
	Hollow 2	4.846	3.562	-127.124	0.014

TABLE 4.3. For each bilayer system and each lateral position considered, the optimized lattice constant, interlayer spacing, total energy, and relative energy are given. The relative energy is given with respect to the most favourable structure, which corresponds to the lowest total energy for each bilayer system.

In addition, figures 4.21–4.23 show the projected densities of states of the three heterostructure systems calculated using the HSE06 functional.

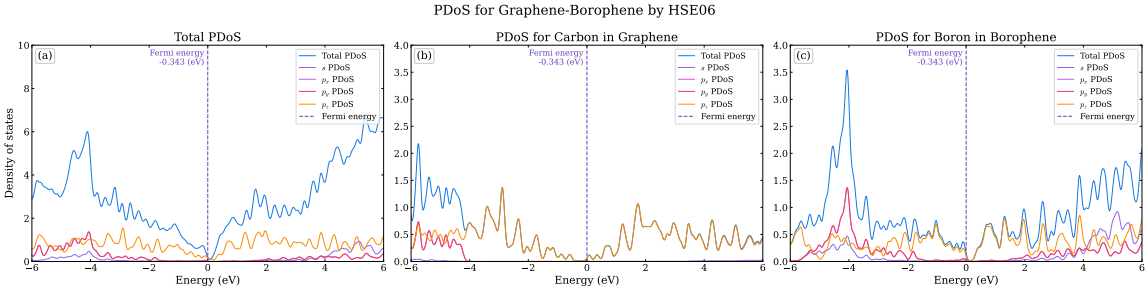

FIGURE 4.21. Projected density of states of Graphene-Borophene calculated using the HSE06 functional.

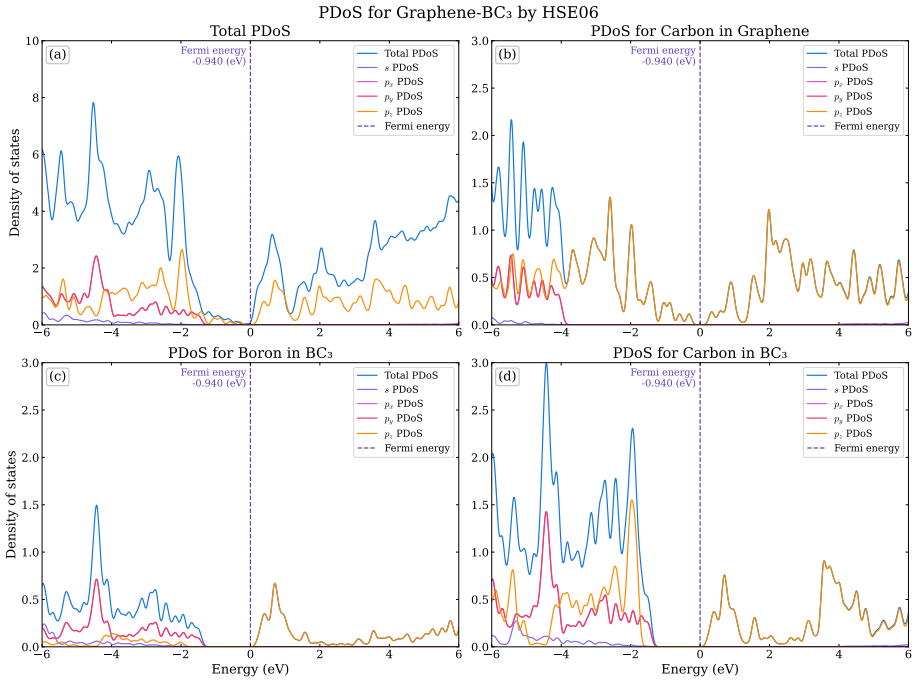

FIGURE 4.22. Projected density of states of the Graphene-BC_3 calculated using the HSE06 functional.

Figure 4.28 also shows the effect of the energy convergence criterion `EDIFF` on the calculated dielectric function, together with the effects of different $\vec{k}$-point meshes and the comparison between HSE06 and PBE.

Figures 4.29–4.31 show the full dielectric-function tensor components of the three bilayer systems, as calculated using both the PBE and HSE06 functionals. The diagonal components describe the dominant in-plane and out-of-plane optical responses,

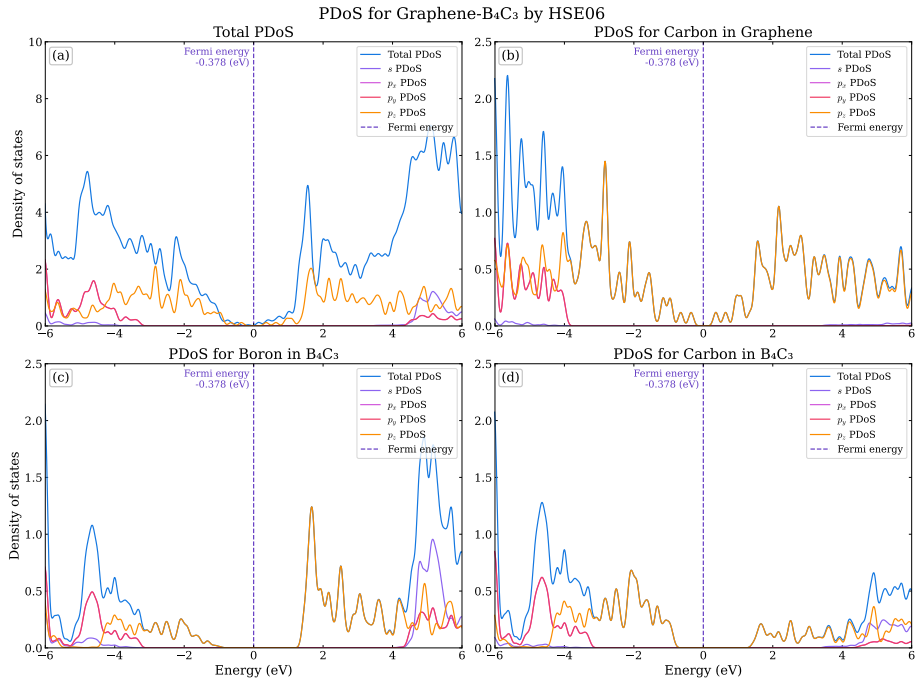

FIGURE 4.23. Projected density of states of the Graphene-B_4C_3 calculated using the HSE06 functional.

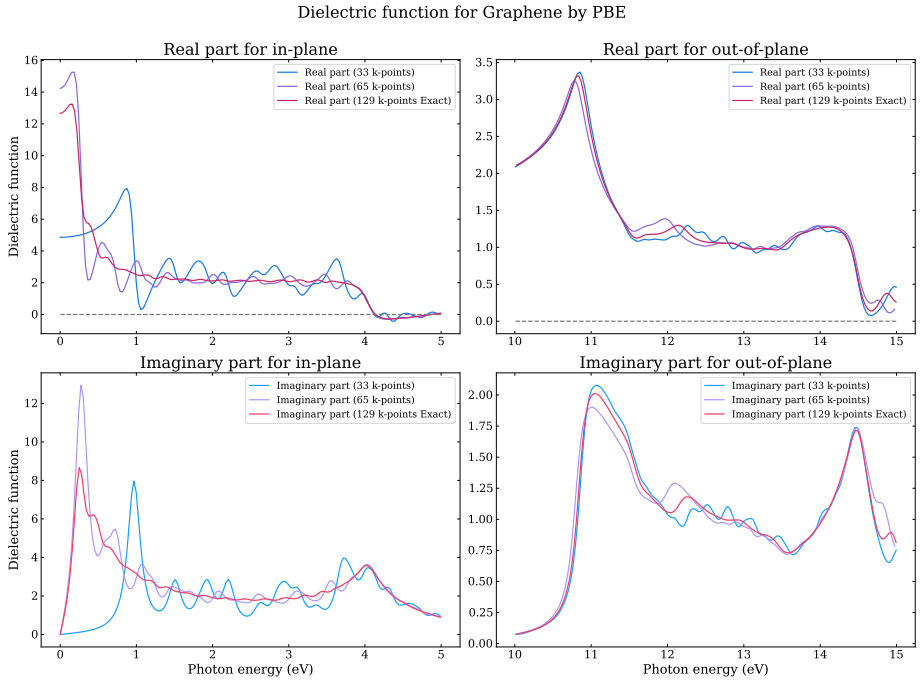

FIGURE 4.24. Dielectric function of graphene calculated using the PBE functional for different $\vec{k}$-point meshes; here, for example, 33 denotes a $33 \times 33 \times 1$ $\vec{k}$-point mesh.

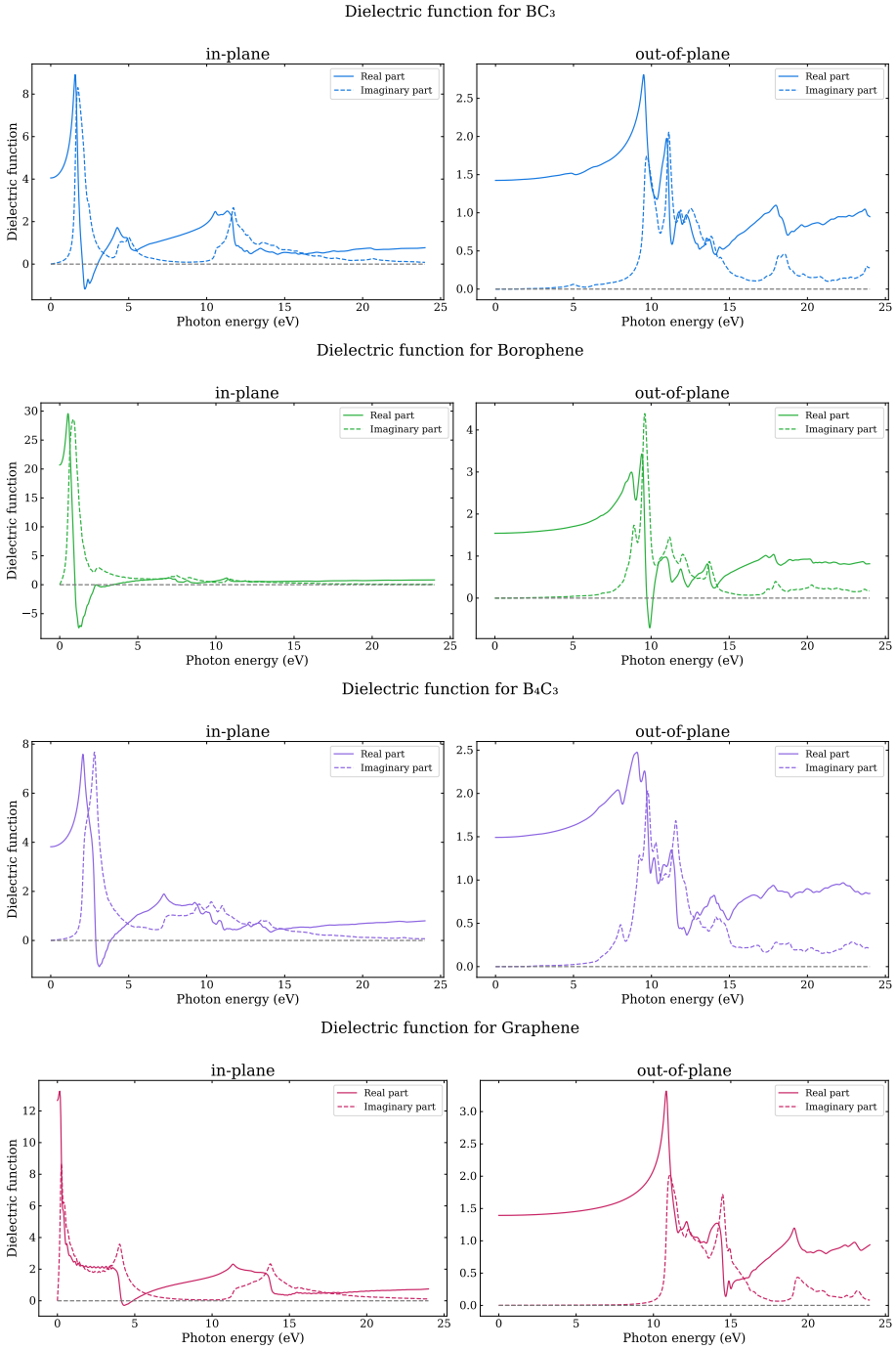

FIGURE 4.25. Real and imaginary parts of the dielectric function for the four monolayers as calculated using the PBE functional with a $129 \times 129 \times 1$ $\vec{k}$-point mesh for graphene, and a $65 \times 65 \times 1$ $\vec{k}$-point mesh for the borophene and boron-carbide monolayers..

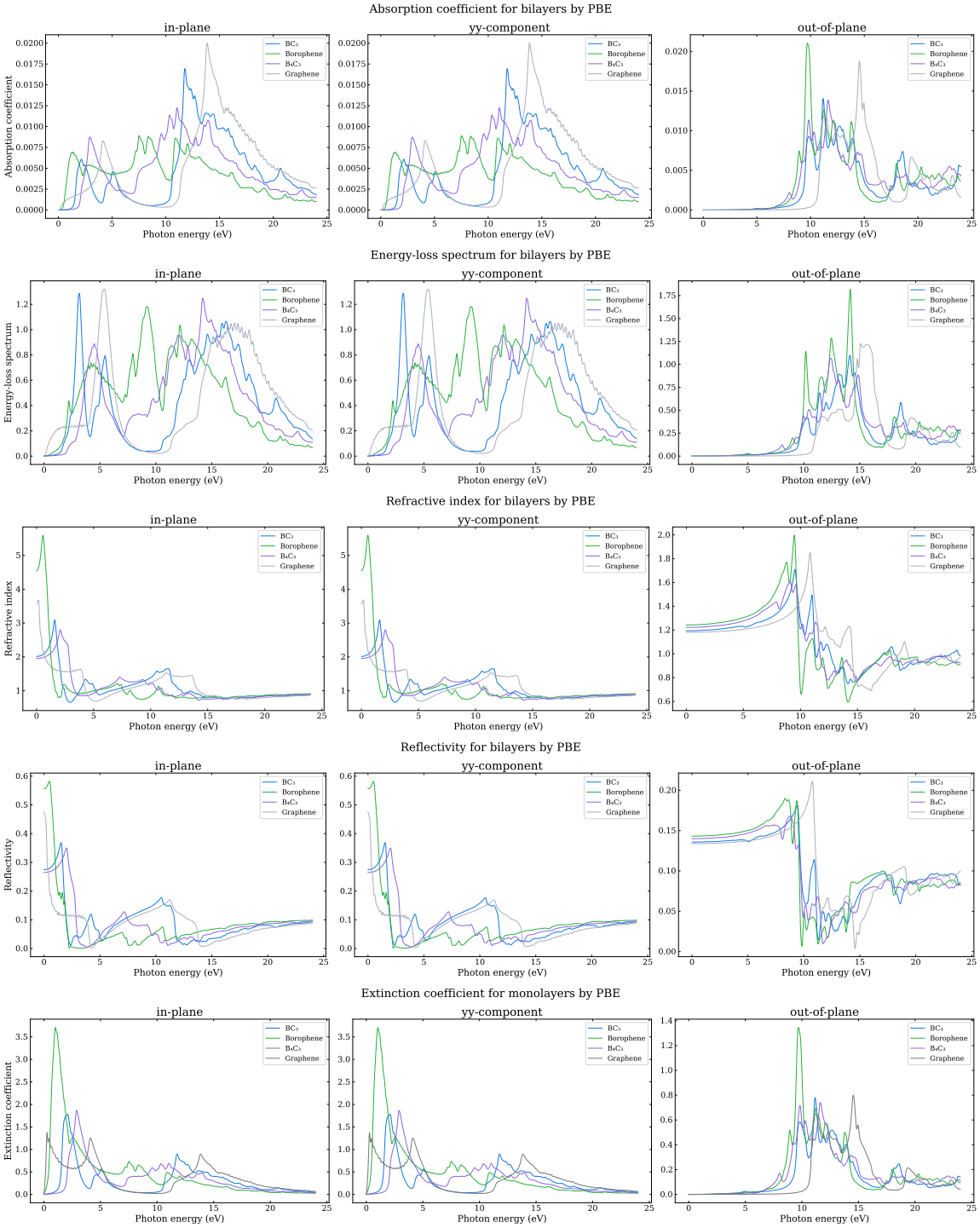

FIGURE 4.26. Optical properties for the four monolayers as calculated using the PBE functional with a $129 \times 129 \times 1$ $\vec{k}$-point mesh for graphene, and a $65 \times 65 \times 1$ $\vec{k}$-point mesh for the borophene and boron-carbide monolayers.

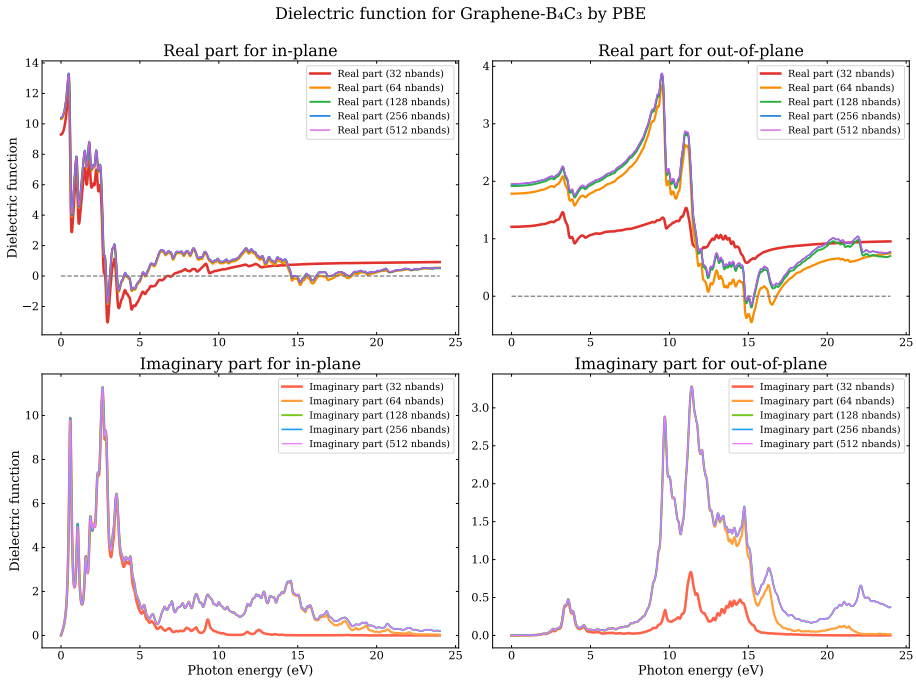

FIGURE 4.27. Convergence test of the dielectric function of Graphene-B_4C_3 with respect to the number of unoccupied bands, calculated using PBE.

whereas the off-diagonal components describe the coupling between different Cartesian directions. These off-diagonal components are negligible for Graphene-Borophene, but remain finite for Graphene-BC_3 and Graphene-B_4C_3. They satisfy the antisymmetric relations $\varepsilon_{yx} = -\varepsilon_{xy}$, $\varepsilon_{yz} = -\varepsilon_{zy}$, and $\varepsilon_{zx} = -\varepsilon_{xz}$.

In addition, figures 4.32–4.36 show the optical properties derived from the dielectric function for the three bilayer heterostructures, also calculated using both the PBE and HSE06 functionals. These properties include the absorption coefficient, energy-loss spectrum, reflectivity, refractive index, and extinction coefficient.

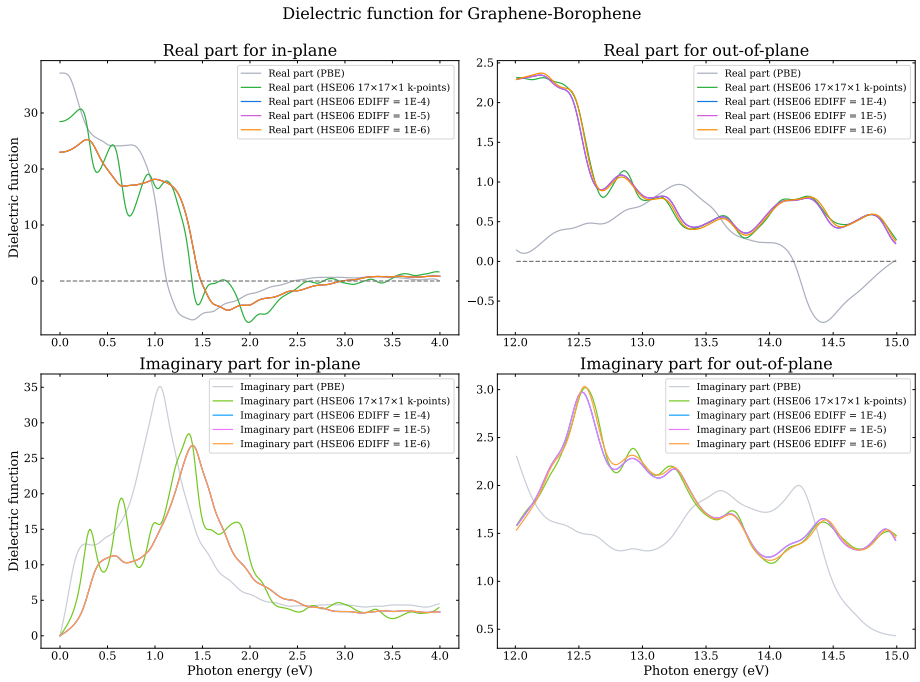

FIGURE 4.28. Dielectric function of the Graphene-Borophene bilayer comparing PBE and HSE06 results, HSE06 $\vec{k}$-point meshes of $17 \times 17 \times 1$ and $65 \times 65 \times 1$, and HSE06 energy convergence criteria `EDIFF` $= 10^{-4}$, 10^{-5}, and 10^{-6}.

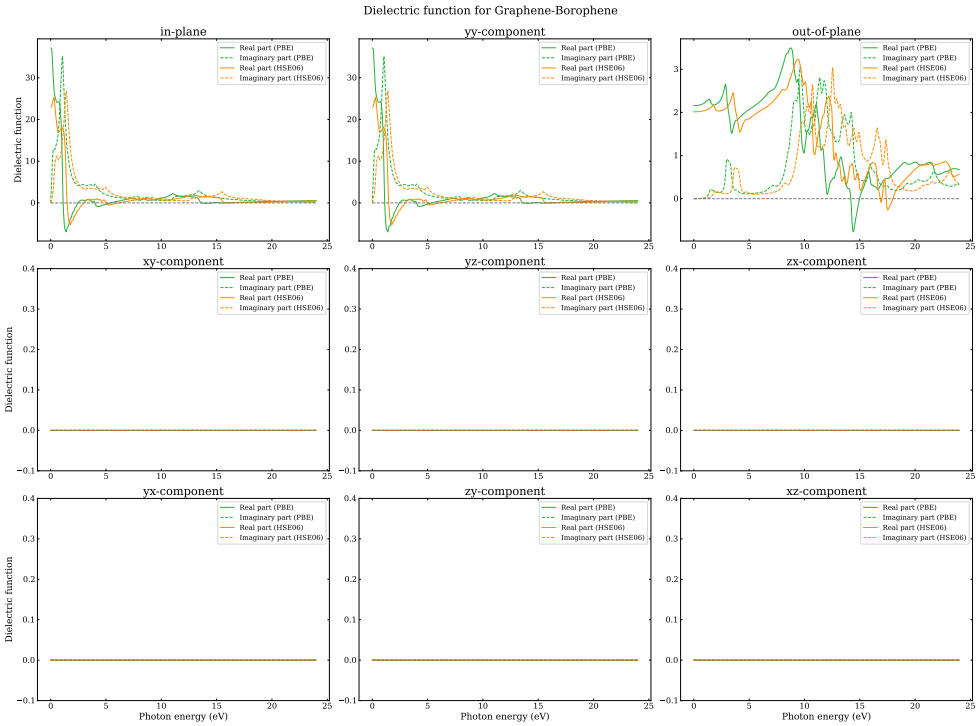

FIGURE 4.29. Dielectric function components of Graphene-Borophene.

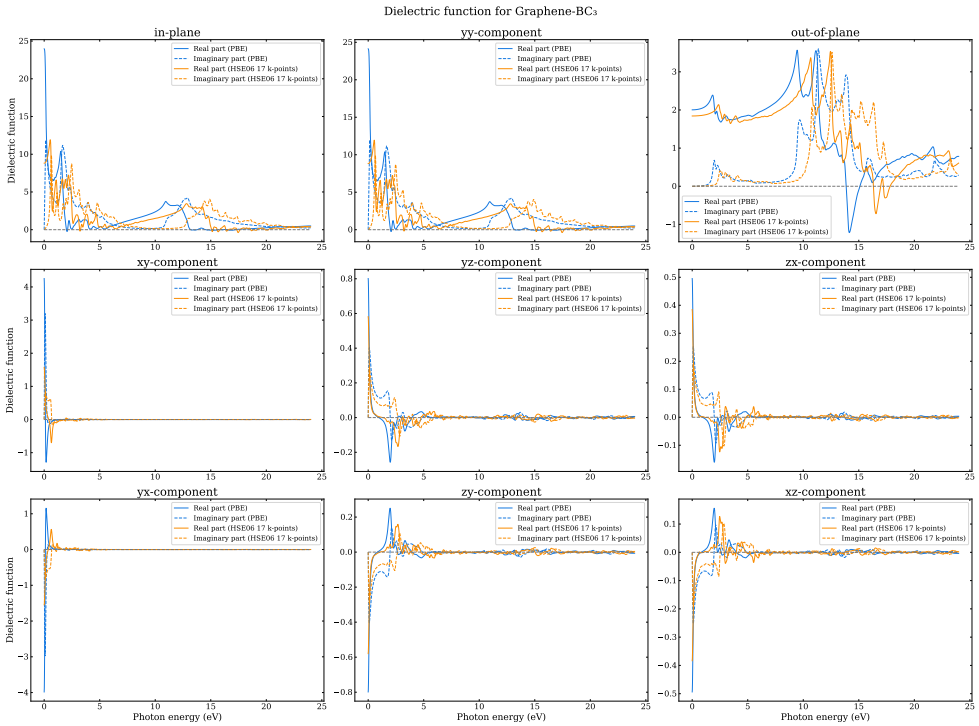

FIGURE 4.30. Dielectric function components of Graphene-BC_3.

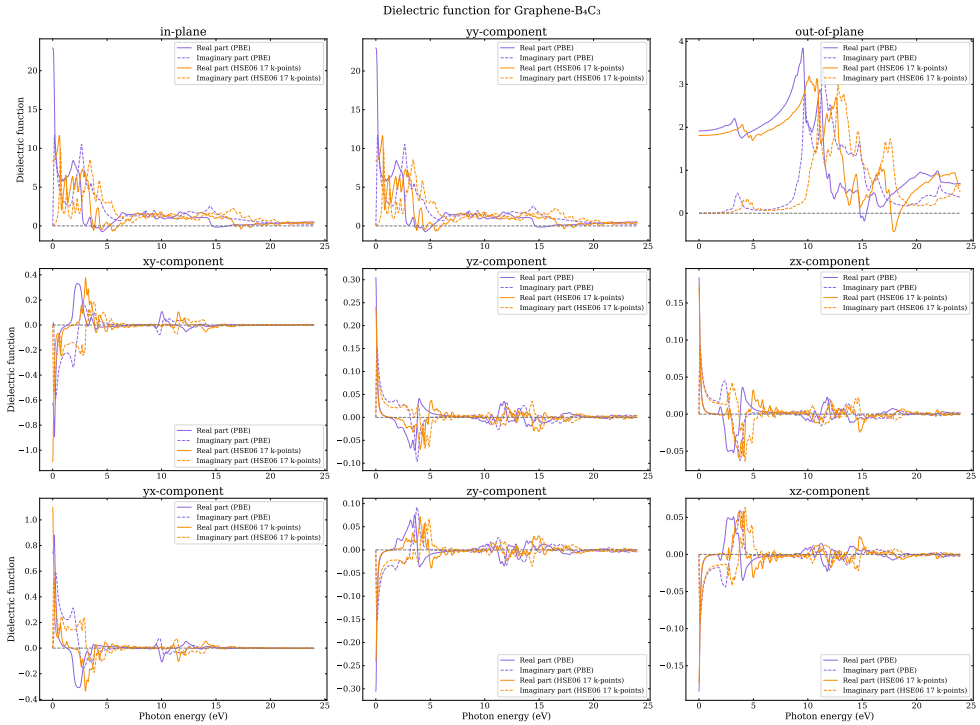

FIGURE 4.31. Dielectric function components of Graphene-B_4C_3.

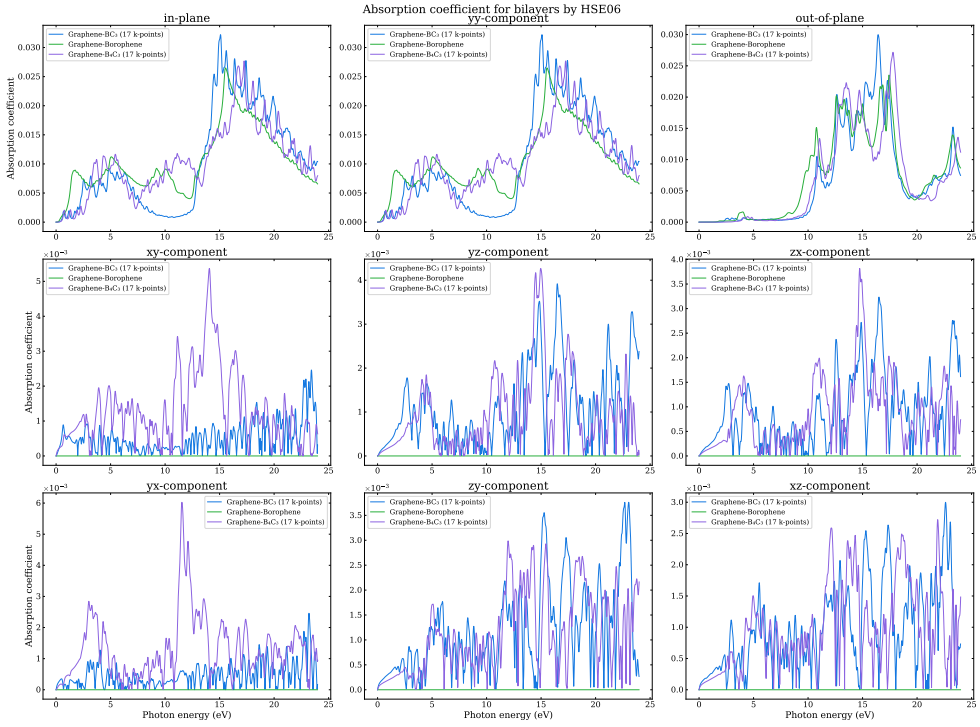

FIGURE 4.32. Absorption coefficient.

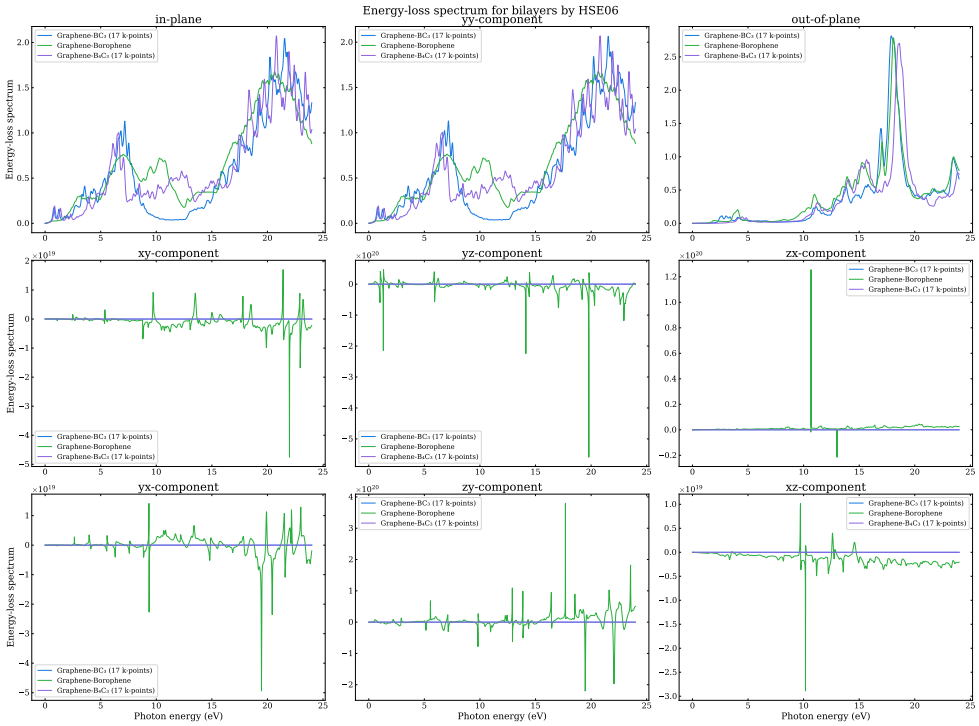

FIGURE 4.33. Energy-loss spectrum.

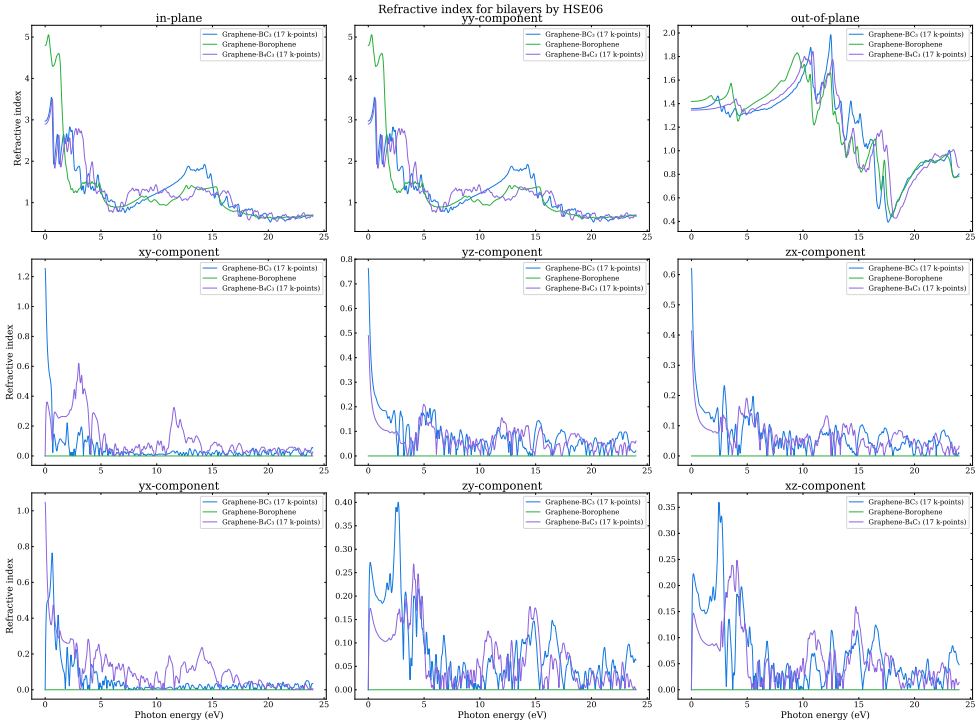

FIGURE 4.34. Refractive index.

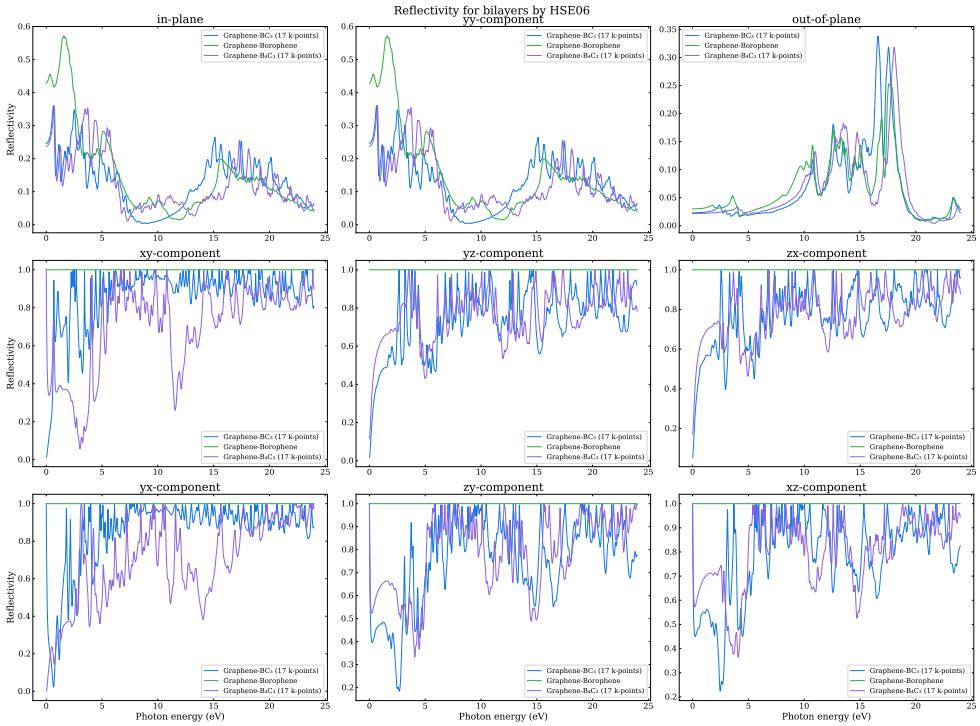

FIGURE 4.35. Reflectivity.

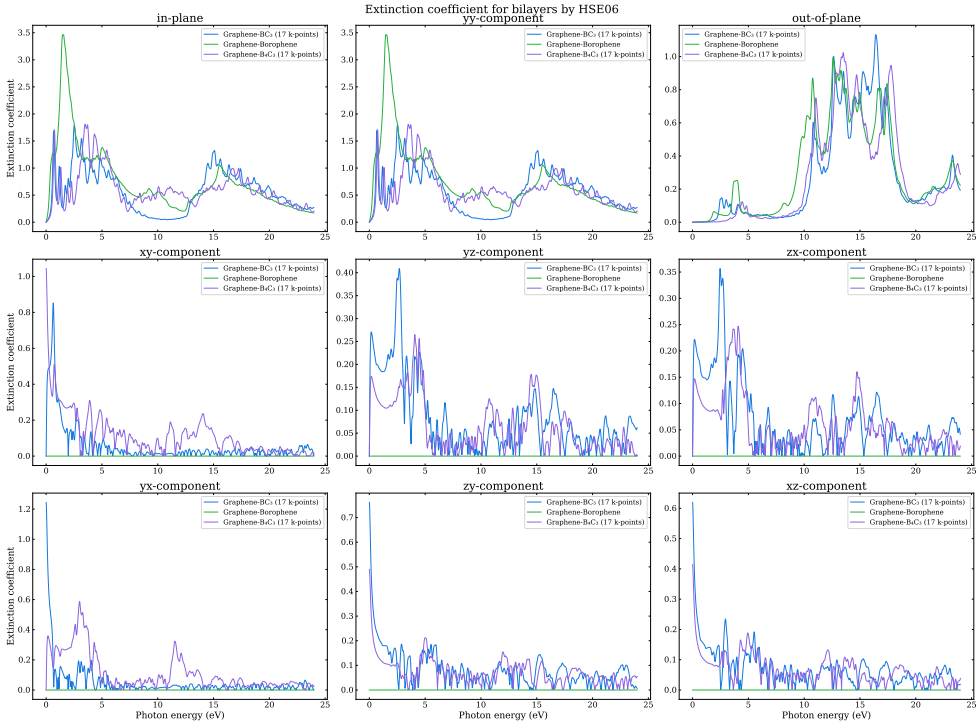

FIGURE 4.36. Extinction coefficient.

Chapter 5

Superconductivity, Magnetism, and Directional Plasmonic Modes in Two-Dimensional Penta-Bipyramid Boron Allotropes

5.1 Abstract

Bulk orthorhombic boron, o-B_{14}, represents an intriguing, recently predicted, three-dimensional (3D) boron allotrope. Its structure consists of edge-sharing pentagonal bipyramids with diverse bond lengths. These structural motifs facilitate the formation of unusual seven-centre-two-electron π-bonds, which were predicted for the first time in 3D boron allotropes. The presence of these bonds highlights the complexity and versatility of boron chemistry under ambient conditions. Moreover, this material is predicted to be a superconductor with a critical temperature T_c of 29.1 K [142].

In the present work, we investigate the electronic and optical properties of bulk o-B_{14}, its monolayer and bilayer forms, and their hydrogen-terminated counterparts using first-principles calculations. These two-dimensional structures are predicted to remain dynamically stable and to exhibit a remarkable variety of electronic behaviour, including semiconductivity with a band-gap of 0.67 eV in the bilayer, superconductivity with a T_c of 24.3 K in the H-terminated bilayer, and magnetism in the monolayer. The directional bonding network drives anisotropic plasmonic and optical responses, as well as absorption in the visible region of the spectrum. These properties suggest promising opportunities for applications in advanced optoelectronic, plasmonic and quantum devices.

5.2 Introduction

Elemental boron possesses a unique combination of light weight, exceptional hardness, and thermal and chemical stability. Due to these properties, it has potential for high-technology applications such as catalysis, sensors, energy storage, piezoelectrics, and nanodevices [143]. Boron exists in multiple allotropes with complex crystal structures arising from its electron-deficient bonding and small atomic size. The most well-characterized and stable crystalline forms are α-rhombohedral boron (α-B, 12 atoms per unit cell), β-rhombohedral boron (β-B, 105–108 atoms per unit cell) [144], and γ-orthorhombic boron (γ-B, 28 atoms per unit cell) [145]. α-B consists of B_{12} icosahedral building blocks and forms at low temperature, whereas β-B has the same structural motif, but also includes interstitial B atoms. β-B forms at room temperature and is the most stable structure. γ-B is also composed of B_{12} icosahedral units and forms at high pressure and temperature. The structures are predominantly characterized by covalent bonding, but the electron deficiency of boron gives rise to multi-centre bonding, such as three-centre-two-electron bonds. Additionally, boron-rich compounds exhibit promising semiconducting and thermoelectric properties, with potential use in next-generation electronics and energy-harvesting devices.

Generally, the superconducting critical temperature (T_c) of elemental superconductors is relatively low at ambient pressure. A representative elemental superconductor, Nb, has a T_c of about 9 K. Under high pressure, the T_c can be significantly increased compared with the ambient-pressure value, but high pressure hinders experimental studies and practical applications. Therefore, it is of high priority to explore elemental superconductors with a high T_c under low or ambient pressure. Interestingly, superconductivity with a T_c of 35 K has been reported in a three-dimensional (3D) kagome-like boron allotrope under 7 GPa on the basis of *ab initio* calculations [146], while earlier work predicts a T_c exceeding 15 K in 3D boron allotropes [147].

Boron can also form isolated clusters which exhibit many intriguing properties, including fluxionality [148]. Fluxionality describes the ability to undergo continuous structural rearrangements without breaking chemical bonds in a way that permanently alters the system. Recent studies have focused on two-dimensional (2D) forms of boron, including monolayer borophene, where many different structures have been observed. Recent review papers have summarized these developments: Wang *et al.* [149] have summarized the synthesis and properties of borophene, while Mishra *et al.* [150] have reviewed the potential of borophene in various technological fields such as batteries,

supercapacitors, and fuel cells, as well as the associated synthesis techniques. Other recent reviews of borophene focus on experimental aspects [151, 152, 153, 154], and provide a survey of structures starting from the bulk allotropes, boron clusters, potential building blocks for 2D boron crystals, and 2D few-layered boron sheets [155]. As a light 2D material with good flexibility, borophene has potential applications in flexible and wearable devices, biomedicines, and biosensors [156, 157].

From the theoretical side, 2D boron has also attracted significant recent interest due to its complex electronic structure, which can give rise to predicted semiconducting behaviour [158], including in twisted bilayer borophene [159], superconductivity [142, 160], magnetism [161], and topological states [162, 163, 164]. Furthermore, the β_{12} sheet has been found to host gapless Dirac cones [165], and a bilayer kagome structure has been predicted to host multiple van Hove singularities [166]. Half-metallic ferromagnetism has also been reported in a 2D kagome lattice composed of a triangular arrangement of boron atoms within triangles [167].

Synthesizing borophene is notably challenging due to several intrinsic and extrinsic factors. Unlike graphene, borophene lacks a naturally occurring layered bulk counterpart, which eliminates the possibility of exfoliation-based synthesis. Its high chemical reactivity, especially with oxygen and moisture, makes it unstable under ambient conditions, requiring ultra-high vacuum (UHV) environments for fabrication and handling. These factors collectively make the reproducible and scalable synthesis of monolayer borophene a significant materials science hurdle. To overcome the challenges of synthesizing monolayer borophene, researchers have focused on carefully controlled epitaxial growth techniques such as molecular beam epitaxy (MBE) on inert metal substrates [168]. Another study has achieved growth of β_{12}- and α-sheets of borophene on Ag(111) by direct evaporation of a pure boron source in an e-beam evaporator [169].

Substrates stabilize the 2D structure of borophene during growth by promoting specific polymorphs through surface templating. Maintaining UHV conditions during synthesis helps prevent oxidation and contamination. Another strategy to enhance stability uses hydrogen termination, which passivates the highly reactive surface boron atoms. This passivation can reduce the tendency to oxidize or react with ambient species, thereby improving chemical stability in air. Moreover, hydrogen termination can modify the electronic properties, potentially opening a band-gap or altering metallic behaviour, which may be advantageous for specific applications [170, 171]. Large-scale synthesis of freestanding atomic sheets of borophene has been reported through a novel

liquid-phase exfoliation approach, where electron microscopy confirmed the presence of the β_{12} and χ_3 phases, as well as their intermediate phases [172].

Different from flat borophenes, a novel highly stable 2D boron structure has been recently predicted on the basis of *ab initio* calculations. This structure is composed of B_{12} clusters linked by single B-B bonds and is a narrow band-gap semiconductor [173]. The optical properties of 2D allotropes have also been investigated. It has been reported that the plasmon frequencies reach the near infrared and even visible regions, making borophene among the first materials with 2D plasmons at such high energies without doping [140].

Recently, a bulk orthorhombic boron allotrope, o-B_{14}, with interesting properties was predicted [142]. Its structure is characterized by edge-sharing pentagonal bipyramids with varying B-B bond lengths ranging from 1.644 Å to 1.966 Å. These structural motifs facilitate the formation of peculiar seven-centre-two-electron π-bonds, which were predicted for the first time in 3D boron allotropes. Remarkably, this material is predicted to be a superconductor with a transition temperature of 29.1 K.

This orthorhombic o-B_{14} structure is a particularly promising candidate for further investigation, as it introduces the pentagonal B_7 bipyramidal motif as a fundamental structural unit in crystalline boron, which differs markedly from the B_{12} icosahedral units that dominate most known boron phases. This unusual topology leads to highly delocalized multicentre bonding, including the predicted seven-centre-two-electron π bonds, suggesting that the electronic structure may host unconventional features such as anisotropic transport, unusual phonon behaviour, and potentially strong electron-phonon coupling. Since the phase is predicted to be energetically competitive and dynamically stable at ambient pressure, *ab initio* studies of its electronic band structure, phonon spectrum, and superconducting properties would be valuable, as would studies that explore how the unique bipyramidal building blocks influence the broader physics of electron-deficient boron networks. Moreover, this also raises the question of whether it may be stable in 2D form, where the interlayer connectivity consists only of single B-B bonds.

In the present work, we use density-functional theory (DFT) calculations to investigate the electronic and optical properties of bulk o-B_{14} and its mono- and bilayers. Hydrogen termination of the 2D structures is also considered. Interestingly, bilayer o-B_{14} is found to be a semiconductor with a band-gap of around 0.69 eV as calculated using the HSE06 hybrid functional. The H-terminated bilayer, however, is predicted to be

superconducting with a T_c of 24.3 K, while monolayer o-B_{14} is found to be magnetic. The optical properties reveal distinctive anisotropic behaviour and the existence of plasmons in the near- to mid-infrared range for in-plane excitation, as well as the ultraviolet energy range for out-of-plane excitation, broadening the potential of boron for applications in optoelectronics, plasmonics, and catalysis.

This chapter is organised as follows: In Section 5.3 the computational methods are presented, followed by the results and discussion in Section 5.4. Herein, we describe the electronic, superconducting, and optical properties. This is followed by the conclusions in Section 5.5.

5.3 Computation methods

The DFT calculations are performed using the Vienna Ab initio Simulation Package (VASP) [123, 124, 125, 174]. The projector augmented wave (PAW) method pseudopotentials are used to describe the electron-ion interactions [126]. The generalized gradient approximation (GGA) of Perdew–Burke–Ernzerhof (PBE) for the exchange-correlation functional [48] is employed along with the Becke–Johnson damped D3(BJ) dispersion correction to account for long-range van der Waals (vdW) interactions [175, 176, 177]. For bulk o-B_{14}, a Γ-centered $\vec{k}$-point mesh of $34 \times 48 \times 31$ is used with a plane-wave energy cutoff of 600 eV. The electronic convergence criterion is set to 1×10^{-6} eV, and the structures are relaxed until all forces on each atom are below 0.01 eV Å^{-1}. For the 2D systems, a vacuum region of approximately 22 Å is included, and a $\vec{k}$-point mesh of $20 \times 18 \times 1$ is used. Convergence tests show that these settings yield accurate results, as shown in figures S1–S3 of the Supplementary Information.

For selected calculations, we also use the screened hybrid exchange-correlation functional of Heyd, Scuseria, and Ernzerhof (HSE06), which gives more accurate band-gap energies [50, 127]. Unless explicitly stated otherwise, all figures show results obtained using the PBE functional. For the density-of-states (DOS) calculations for bulk o-B_{14}, we use a Γ-centered $\vec{k}$-point mesh of $20 \times 28 \times 18$. For the 2D systems, we use a mesh of $20 \times 18 \times 1$. The smearing at the Fermi level is calculated using the Gaussian method with a broadening value of 0.1 eV.

The adsorption energy per atom, E_{ad}, of hydrogen atoms on the surface is calculated as:

$$E_{\mathrm{ad}} = \frac{E_{\mathrm{H/B}} - E_{\mathrm{B}} - E_{\mathrm{H}}}{n_{\mathrm{H}}} \tag{5.1}$$

here, $E_{\mathrm{H/B}}$ is the total energy of the adsorption system, E_{B} is the total energy of the pristine 2D boron structure, E_{H} is the total energy of a hydrogen atom in the H_2 molecule, and n_{H} is the number of H atoms in the unit cell.

The cohesive energy per atom, E_{c}, of the boron systems is calculated as:

$$E_{\mathrm{c}} = \frac{E_{\text{B-struc}} - n_{\mathrm{B}} E_{\text{B-atom}}}{n_{\mathrm{B}}} \tag{5.2}$$

here, $E_{\text{B-struc}}$ is the total energy of the boron structure of interest, $E_{\text{B-atom}}$ is the total energy of a free boron atom calculated in a large supercell, and n_{B} is the number of boron atoms in the unit cell of the structure.

The charge density difference, $\Delta\rho(\vec{r})$, for hydrogen adsorption on the surface is calculated as:

$$\Delta\rho(\vec{r}) = \rho_{\mathrm{H/B}}(\vec{r}) - \rho_{\mathrm{B}}(\vec{r}) - \rho_{\mathrm{H}}(\vec{r}) \tag{5.3}$$

here, $\rho_{\mathrm{H/B}}(\vec{r})$ is the total charge density of the adsorbed system, and $\rho_{\mathrm{B}}(\vec{r})$ and $\rho_{\mathrm{H}}(\vec{r})$ are the total charge densities of the clean boron structure and the hydrogen atom, respectively.

The superconducting behaviour of o-B_{14} is investigated systematically in three structural phases, namely bulk, monolayer, and hydrogen-terminated bilayer. Calculations are carried out using the PBE formulation of the generalized gradient approximation [48], as provided in Quantum ESPRESSO [178]. Norm-conserving pseudopotentials and a kinetic energy cutoff of 45 Ry are chosen [179], and a vacuum spacing of 15 Å is added for the 2D systems to suppress spurious periodic interactions. Monkhorst–Pack $\vec{k}$-point meshes of $9\times12\times8$ for the bulk, $10\times10\times1$ for the monolayer, and $9\times9\times1$ for the bilayer are employed to sample the Brillouin zone [180]. The phonon dispersions are calculated on $\vec{q}$-point grids of $3\times4\times3$, $5\times5\times1$, and $3\times3\times1$, respectively [181]. Atomic positions and lattice parameters are relaxed until residual forces are below 1×10^{-5} Ry/bohr, corresponding to 2.6×10^{-4} eV Å^{-1}.

Maximally localized Wannier functions are constructed in Wannier90 by initializing projections on B s and p orbitals [182]. Uniform $\vec{k}$-point meshes of $6 \times 8 \times 6$ for the bulk and $6 \times 6 \times 1$ for the monolayer and bilayer are used in Wannier interpolation. Superconductivity is subsequently analyzed through EPW [178, 183, 184, 185], which is employed to evaluate electron-phonon coupling strength and estimate Allen–Dynes transition temperatures, T_c [186]. The superconducting properties are also obtained by solving the fully anisotropic Migdal–Eliashberg equations [186, 187, 188], as implemented in the EPW code. The Morel–Anderson pseudopotential μ^* is chosen to be 0.13, consistent with the typical semi-empirical range of 0.10–0.16 used in Migdal–Eliashberg calculations for light-element and sp-bonded superconductors [189], and with the value adopted in the literature for boron-based 2D superconductors [190]. With this choice, the calculated electron-phonon coupling strengths, λ, and T_c indicate clear dimensionality effects, as discussed in Section 5.4.3.

For the calculation of the optical properties within VASP, in particular the frequency-dependent dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$, the $\vec{k}$-point meshes of $20 \times 28 \times 18$ and $20 \times 18 \times 1$ are found to yield converged results for the bulk and 2D systems, respectively, as shown in figure S4. A convergence check for the number of unoccupied bands included in the calculation of the dielectric function is also performed. Figure S5 shows the result for bulk o-B_{14}. It can be seen that 128 bands yield well-converged results, and this value is used for all systems. The definitions of the dielectric tensor and the derived optical quantities are given in Section 3.4.

The dielectric tensor contains the combined dielectric response of the 2D material and the vacuum, so the magnitude of $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$ depends linearly on the vacuum thickness [191, 192]. Therefore, to obtain the intrinsic dielectric function of the structures, the calculated dielectric functions are renormalized following Ref. [193]:

$$\varepsilon_2^{\mathrm{2D}}(\omega) = \frac{d_{\mathrm{SC}}}{d_{\mathrm{2D}}} \varepsilon_2^{\mathrm{SC}}(\omega) \tag{5.4}$$

$$\varepsilon_1^{\mathrm{2D}}(\omega) = 1 + \frac{d_{\mathrm{SC}}}{d_{\mathrm{2D}}} \left[\varepsilon_1^{\mathrm{SC}}(\omega) - 1\right] \tag{5.5}$$

here, $\varepsilon_i^{\mathrm{2D}}(\omega)$ and $\varepsilon_i^{\mathrm{SC}}(\omega)$, with $i = 1, 2$, are the dielectric functions of the studied 2D structure without the influence of the vacuum region and for the whole supercell, respectively, and d_{SC} is the length of the supercell. The thickness d_{2D} for the monolayer, bilayer, and H-terminated bilayer is 5.897 Å, 9.412 Å, and 11.209 Å, respectively. The

thickness is defined in figure S6. In the literature, this correction is often ignored in studies on 2D materials, which leads to errors in the reported dielectric functions and therefore the optical properties of the 2D materials. It is noted that interband transitions presumably coexist with the plasmons forming a broad background. The coexistence of individual transitions and plasmons in the same energy range leads to coupling of the two types of excitations, and hence finite plasmon lifetimes as well as interband transitions [141].

5.4 Results and discussion

5.4.1 Bulk o-B_{14}: atomic structure and electronic properties

5.4.2 Xene o-B_{14}: atomic structure and energetics

5.4.2.1 Structural properties

5.4.2.2 Dynamic stability: phonon dispersion

5.4.2.3 Electronic properties

5.4.3 Superconducting properties

5.4.4 Optical properties

5.5 Conclusions

5.6 Data availability statement

5.7 Supplementary information

Convergence tests are shown for the total energy, the plane-wave energy cutoff, and the $\vec{k}$-point set of bulk o-B_{14} in figure **??**. The lattice-constant determination is shown in the lower panel of figure **??**, and the cohesive-energy convergence is shown in figure **??**. The total energy as a function of vacuum-region thickness for the monolayer is given in figure **??**. Convergence tests for the calculation of the dielectric function are also shown for the $\vec{k}$-point set in figure **??** and for the number of unoccupied bands included in the calculation, the parameter `NBANDS`, in figure **??**. The definition

of d_{2D} used in the dielectric-function renormalization is displayed in figure **??**. The renormalization procedure is given in Section 5.3, and the general optical-property definitions are introduced in Chapter 3. Figure **??** shows the hollow-bridge site that was also considered for H adsorption on the monolayer and bilayer structures.

The calculated phonon dispersion curves for the two-dimensional systems, as obtained using (2×2) and (3×3) supercells, are shown in figures **??** and **??**. The results obtained using both supercells show qualitatively the same behaviour. The hydrogen-terminated monolayer is dynamically unstable due to imaginary frequencies across a wide range of the Brillouin zone.

Results of *ab initio* molecular dynamics (AIMD) calculations for the H-terminated bilayer are shown in figure **??** for a temperature of 400 K, displaying good structural stability.

Comparison of the spin-polarized FM band structures for the monolayer, as calculated using the HSE06 and GGA-PBE functionals, is shown in figure **??**. The corresponding total density of states (DOS) is shown in figure **??**. A comparison of the non-spin-polarized band structure of the monolayer, as calculated using the r^{2}SCAN, HSE06, and GGA-PBE functionals, is shown in figure **??**. In figure **??**, the non-spin-polarized band structure of the monolayer is shown as calculated in a (2×1) supercell without symmetry, which tests the effect of possible Jahn–Teller distortions. For the FM monolayer structure, the spin moment is shown in figure **??**, where a clear minimum can be seen near 2.0 μ_B. The projected DOS for the AFM and FM monolayers is shown in figure **??**. Figure **??** displays the spin density and the state distribution for the highest occupied and lowest unoccupied states of the FM monolayer midway along the T–Y direction. Finally, the frequency-dependent extinction coefficient for all structures considered is displayed in figure **??**.

Optical-property definitions

We obtain the optical observables used in this chapter from the frequency-dependent dielectric function. Chapter 3 introduced the general route from the dielectric function to optical properties in Section 3.4. Specifically, we defined the absorption coefficient $\alpha_\lambda(\omega)$ in equation (3.40), the refractive index $n_\lambda(\omega)$ in equation (3.45), the extinction coefficient $k_\lambda(\omega)$ in equation (3.46), the reflectivity $R_\lambda(\omega)$ in equation (3.50), and the energy-loss spectrum $L_\lambda(\omega)$ in equation (3.52). Therefore, we do not repeat these formula definitions here.

Renormalization of the dielectric function to account for the vacuum region

The effective thickness of the 2D systems, d_{2D}, is required for the dielectric-function renormalization introduced in Section 5.3. We define it from the difference in the z coordinate of the top and bottom atoms in the system, dz, together with one additional van der Waals radius on each side of the slab. Thus, $d_{2D} = dz + 2d_{vdW}$. For the bare monolayer and bilayer, the top and bottom atoms are the pyramidal apical B atoms. For the H-terminated bilayer, the top and bottom atoms are the H atoms. Here, we use $d_{vdW} = 1.91\,\text{Å}$ for boron and $d_{vdW} = 1.2\,\text{Å}$ for hydrogen. A schematic illustration of the total thickness, $D_{total} \equiv d_{2D}$, is shown in figure **??**.

Chapter 6

Emergent Topological and Optical Properties in Pristine and Hydrogen-Functionalized Two-Dimensional Beryllium Phases

6.1 Abstract

6.2 Introduction

6.3 Computation methods

6.4 Results and discussion

6.5 Conclusions

6.6 Data availability statement

6.7 Supplementary information

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Appendix

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